INCIDENT RESPONSE REPORT

Forensic analysis: test02

Case	test02
Main system	?
Verdict	Confirmed compromise (from memory evidence only) of DESKTOP-SDN1RPT (C137.LOCAL). Domain-level credentials were abused and there was hands-on-keyboard activity. Malware ran with Domain/Enterprise/Schema Admin rights and service persistence was installed. Code was injected into spoolsv.exe (holding a handle on lsass) and into powershell.exe, and live sessions to the suspected C2 203.78.103.109:443 were open when memory was captured. The whole domain should be treated as potentially compromised.
Trace period (UTC)	2019-10-15 13:51:00 → 2020-09-19 05:10:44 (99.8 % of the events; full range 2013-02-04 23:20:16 → 2020-09-19 05:11:17)
Incident window (UTC)	2020-09-18 21:44:13 → 2020-09-19 06:10:39
Findings	Critical: 1, High: 11, Medium: 6, Informational: 8
Report date	2026-10-20 16:01 UTC
Classification	CONFIDENTIAL: TLP:AMBER
Production	Report generated by Sleuthline (analyze.py) with the assistance of a language model. Model: anthropic / claude-opus-5-5.

Disclaimer: this report was produced automatically. Every finding is linked to verifiable records of the case database ("Reference" column) and to the original path of the artifact. It must be reviewed and validated by an analyst before any distribution or decision.

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1. Executive summary

Verdict: Confirmed compromise (from memory evidence only) of DESKTOP-SDN1RPT (C137.LOCAL). Domain-level credentials were abused and there was hands-on-keyboard activity. Malware ran with Domain/Enterprise/Schema Admin rights and service persistence was installed. Code was injected into spoolsv.exe (holding a handle on lsass) and into powershell.exe, and live sessions to the suspected C2 203.78.103.109:443 were open when memory was captured. The whole domain should be treated as potentially compromised.

On 2020-09-19 (UTC), someone logged on to workstation DESKTOP-SDN1RPT (10.42.85.115, Windows 10 2004, domain C137.LOCAL) over inbound RDP at 03:36:23 using the built-in domain Administrator account (C137\Administrator, RID 500, member of Domain/Enterprise/Schema Admins). This was the first time that account had ever logged on to the host. Within four minutes the operator:
- opened Windows Security, and Defender's network inspection driver (WdNisDrv.sys) was unloaded at 03:39:45;
- browsed with Microsoft Edge to a web host at the raw IP 194.61.24.x (03:39:26);
- received the file coreupdater.exe (written to disk at 03:40:00) and ran it elevated at 03:40:49 (PID 8324, SeDebugPrivilege enabled, exited 03:43:10).
An auto-start Windows service named "coreupdater" was registered during this window, which gives the attacker persistence across reboots. Between 03:45 and 03:47:39 the operator browsed the profile folders of every user (including ricksanchez and mortysmith) and opened an item called "loot" (loot.lnk).

When memory was captured (05:10:39), two pieces of injected code were active:
- a reflectively loaded PE in spoolsv.exe (PID 2188) with an open handle on lsass.exe, which points to credential theft;
- a reflectively loaded PE plus an RWX-started thread in a PowerShell chain (PID 508 -> 3316) that started at 05:08:37, 21 s after the Domain Admin ricksanchez logged on at the console.
Two TLS sessions to the external address 203.78.103.109:443 were ESTABLISHED; this is the most likely C2 channel. The host also held an SMB session and a mapped drive (Z:) to the domain controller CITADEL-DC01 (\\CITADEL-DC01\FileShare).

Impact:
- Two Domain Admin identities (Administrator and ricksanchez) were exposed or used. Credential material from lsass is probably compromised.
- Exfiltration is not proven: no archive or transfer tool was found, but the C2 sessions could have carried data.

Confidence: the order of events is solid, because process, MFT, driver and prefetch times agree to the second. However, every conclusion comes from a single source (the memory image). This is why the high/critical findings are rated "probable" rather than "confirmed".

Key figures

Findings	26 (critical: 1, high: 11, medium: 6, informational: 8)
Key timeline events	24
Indicators of compromise	17
Questions answered	0 / 0 asked in the guidance
First / last key event (UTC)	2020-09-18 22:44:13 → 2020-09-19 05:10:39

Main findings

ID	Severity	Confidence	Finding	Period (UTC)
F-003	Critical	probable	Code injection in spoolsv.exe (PID 2188): RWX private MZ region unlinked from loader lists + handle to lsass.exe with VM_READ access	2020-09-19 01:24:09 → 2020-09-19 01:24:09
F-018	High	probable	Administrator RDP session file/folder activity 03:45-03:47: profile folders enumerated, Portal_gun.png opened, "loot" item opened (loot.lnk) and Desktop folder opened (Desktop.lnk)	2019-12-07 09:14:52 → 2020-09-19 03:47:39
F-014	High	probable	Domain Administrator (RID 500) used via RDP (Remote	2020-09-19 01:24:09 → 2020-

ID	Severity	Confidence	Finding	Period (UTC)
			Interactive, logon type 10): first-ever logon on the host at 03:36:26, session 3, elevated Domain/Enterprise/Schema Admin token	09-19 03:39:37
F-016	High	probable	Inbound RDP session of C137\Administrator at 03:36:23 UTC: first-ever RDP device installation (rdpidd/termkbd PNF, RdpIdd, TerminalServices-PnPDevices logs) + RemoteInteractive tokens; source IP not recoverable	2020-09-19 01:24:09 → 2020-09-19 03:36:24
F-006	High	probable	coreupdater.exe execution details: launched interactively in session 3 by domain Administrator (High IL, Domain/Enterprise/Schema Admins) at 03:40:49, 49 s after being written to disk	2020-09-19 03:36:23 → 2020-09-19 03:46:18
F-021	High	probable	coreupdater.exe most probably downloaded with Microsoft Edge (Legacy) by C137\Administrator from a web host at 194.61.24.x (03:39:26-03:40:04 UTC)	2020-09-19 03:36:24 → 2020-09-19 03:40:49
F-009	High	probable	Microsoft Defender Network Inspection driver (WdNisDrv.sys) unloaded at 03:39:45 while domain Administrator had Windows Security open, 15 s before coreupdater.exe was dropped	2020-09-19 03:39:36 → 2020-09-19 03:40:00
F-005	High	probable	Suspicious auto-start service "coreupdater" and coreupdater.exe executed as domain Administrator (PID 8324, 03:40:49-03:43:10 UTC)	2020-09-19 03:40:00 → 2020-09-19 03:40:49
F-012	High	probable	Persistence: Windows service "coreupdater" (AUTO_START, own-process) registered at runtime during the Administrator session, after boot and before the 05:08 ricksanchez logon	2020-09-19 03:40:00 → 2020-09-19 03:40:49
F-010	High	probable	coreupdater.exe (PID 8324) token had SeDebugPrivilege explicitly enabled (non-default) - credential-access capability	2020-09-19 03:40:49 → 2020-09-19 03:40:49
F-007	High	probable	PowerShell executed at ricksanchez logon by a non-recoverable parent (PPID 1380); first PowerShell Prefetch on the host created 05:08:44; command line not recoverable	2020-09-19 05:08:16 → 2020-09-19 05:08:44
F-004	High	probable	Injected PE and RWX-started thread in powershell.exe PID 3316 (C137\ricksanchez, Domain Admin) spawned 21 s after logon by orphaned powershell.exe PID 508	2020-09-19 05:08:37 → 2020-09-19 05:08:43
F-011	Medium	possible	Network state at capture: two ESTABLISHED TLS sessions to external 203.78.103.109:443 (owner unresolved), SMB to DC, RDP listener; remaining external peers are Microsoft/CDN	2020-09-19 01:24:09 → 2020-09-19 01:24:09
F-023	Medium	probable	Microsoft Defender state at capture: WinDefend/WdFilter running, only Network Inspection (WdNisSvc/WdNisDrv) stopped; no MDE (Sense stopped)	2020-09-19 01:24:09 → 2020-09-19 03:39:45
F-013	Medium	possible	Possible logon autostart: PowerShell PID 508 started in the same second as the Run-key programs of the ricksanchez logon (05:08:37) by a transient parent (PPID 1380)	2020-09-19 03:36:26 → 2020-09-19 05:08:37
F-025	Medium	possible	Collection/exfiltration review: no archive, compression or transfer tool found; 203.78.103.109:443 sessions are the only candidate C2/exfil channel (ephemeral port ordering points to the 05:08 PowerShell chain and spoolsv injection)	2020-09-19 03:46:18 → 2020-09-19 03:46:18
F-019	Medium	probable	PowerShell custom jump list (590aee7bdd69b59b.customDestinations-ms) written at 05:08:37 - interactive PowerShell window at ricksanchez logon	2020-09-19 05:08:37 → 2020-09-19 05:08:37
F-017	Medium	possible	Mapped network drive Z: -> \\CITADEL-DC01\FileShare and ESTABLISHED SMB session to the DC (10.42.85.10:445) at capture	2020-09-19 05:09:52 → 2020-09-19 05:09:52

Technical assessment

Evidence base: one Volatility 3 analysis of DESKTOP-SDN1RPT.mem (2 GiB, SHA256 AC6EE0E8…67C3). Boot 01:24:07 UTC, capture 05:10:39 UTC, host time zone derived as UTC-7. There are no automatic detections, EVTX, registry hives or disk artefacts.

Chronology (UTC):
1. 2020-09-18 22:44:13 and 22:46:45: first logons of ricksanchez (RID 1106, Domain Admins) and mortysmith (RID 1108). Baseline activity: Thoughts.txt, Jessica.jpg and Portal_gun.png downloaded with Edge (F-015, F-020).
2. 2020-09-19 03:16:05-03:17:00: session 1, Edge PID 4592, user undetermined (T-021).
3. 03:36:23: inbound RDP session. First-ever RdpIdd/termkbd/rdpidd.PNF device installation and TerminalServices-PnPDevices logs created. S-1-5-14 (RemoteInteractive) is in the session 3 tokens of C137\Administrator (S-1-5-21-2232410529-1445159330-2725690660-500). Profile created at 03:36:26. Source IP not recoverable (F-014, F-016).

4. 03:39:26: Edge Legacy PID 860 opens 194.61.24.x (DOMStore 194.61.24[1].xml, MFT 25584).
5. 03:39:36: Windows Security opened. 03:39:45: WdNisDrv.sys unloaded (unloadedmodules). WdNisSvc and WdNisDrv were STOPPED at capture; WinDefend/WdFilter were running and Sense was stopped (F-009, F-023).
6. 03:39:57: DownloadHistory updated.
7. 03:40:00: coreupdater.exe written (MFT 26213, $SI=$FN, no timestomping; FN updated 03:40:04). 03:40:42: Downloads directory modified (F-021, F-024).
8. 03:40:49-03:43:10: coreupdater.exe PID 8324 (PPID 4008, Explorer) runs in session 3. Token: High IL, Domain/Enterprise/Schema Admins, SeDebugPrivilege explicitly enabled. It is the only process with SeDebugPrivilege enabled outside the default set (F-005, F-006, F-010).
9. Service "coreupdater": SERVICE_AUTO_START, WIN32_OWN_PROCESS, STOPPED, SCM Order 662. This sits between the boot-time services (up to 623) and the per-user services of the 05:08 logon (from 664), so the service was registered at runtime, most probably by coreupdater.exe. The binary path is empty in svcscan (F-012).
10. 03:44:52: Edge exits. 03:45:02: profile folders of all users enumerated. 03:45:53: Portal_gun.png opened. 03:46:18: loot.lnk created (MFT 87445). 03:47:39: Desktop.lnk created, the last activity of session 3 (F-018).
11. 05:08:15: ricksanchez console logon (session 2). 05:08:37: powershell.exe PID 508 starts (parent PPID 1380 had already exited), in the same second as the Run-key programs; the PowerShell jump list was committed at the same time. 05:08:43: child powershell.exe PID 3316. 05:08:44: first-ever POWERSHELL.EXE-CA1AE517.pf. PID 3316 contains an unlinked MZ RWX region at 0x10c6c070000 and thread TID 3892 started in RWX memory. No command line or console history was recoverable (F-004, F-007, F-013, F-019).
12. State at capture: spoolsv.exe PID 2188 holds a MZ RWX VadS region at 0x1840000 that is missing from the loader lists, a handle 0x1410 on lsass.exe PID 656 with VM_READ, and its CWD is C:\Users\mortysmith\Desktop. The injection time cannot be determined (F-003). Two ESTABLISHED sessions to 203.78.103.109:443 (local ports 50875 and 50972) have no owner in netscan/netstat; port ordering suggests PID 3316 and spoolsv (F-011, F-025). There was also an SMB session to the DC 10.42.85.10:445 and Z: mapped to \\CITADEL-DC01\FileShare (F-017). The mutant e2vgmtzqt3xcbq.mca has no resolved owner.

Ruled out with the available data:
- PsExec/WinRM/RMM tools, outbound RDP and pivots to other internal hosts.
- Log clearing (EventLog was running with its channels open) and wiping tools.
- Scheduled-task persistence: only Microsoft/GPO defaults found.
- Archivers and exfiltration tools.
- SSDT or syscall hooking, and DKOM-hidden processes.
The following were classed as noise: responder tools (FTK Imager from E:\, msinfo32, autorunsc) and the malfind/pebmasquerade false positives (MsMpEng, csrss, WmiPrvSE, audiodg).

Assessment: an operator with domain Administrator credentials (obtained by an undetermined method, probably elsewhere in the domain) used RDP to stage and run a payload. The likely result is a C2 implant (reflective loader / injection into spoolsv and PowerShell, consistent with Cobalt Strike/Meterpreter-style tradecraft; this is a hypothesis with no family attribution) plus service persistence. The PowerShell chain at the ricksanchez logon suggests a second user-context persistence (Run key or logon script; not verified). Credential theft from lsass is probable.

2. Scope, sources and methodology

2.1 Analyzed system

Observed host names	?
Observed accounts	ricksanchez (1)
Machine time zone	ActiveTimeBias = None min (UTC = local time + bias)
Time convention of the report	All times are UTC.
Available traces	177,718 timeline events, from 2013-02-04 23:20:16 to 2020-09-

	19 05:11:17

2.2 Collection and processing

KAPE collection (!SANS_Triage target); parsing by recognized tools orchestrated by orchestrator.py; normalization into DuckDB by normalize.py; analysis by analyze.py following the playbook. The raw tool outputs are kept in C:\Sleuthline\cases\test02\parsed and every cited piece of evidence points to them.

Job	Artifact / tool	Status	Tool and version	Duration
volatility	Volatility 3 - memory image (58 plugins, no secrets, no file extraction)	ok	vol volatility3 2.28.2	1633 s

Analysis database	C:\Sleuthline\cases\test02\db\case.duckdb
Pipeline version	commit 5bced44bc74f; content sha256 2186C3B0776C5224; deployed 2026-10-20 16:21
Collection integrity	not checked (run the inventory again with a recent version of the pipeline)
Normalization	normalize.py v1.1 on 2026-10-20T14:38:39Z; mappings v2.1 (sha256 64cf863da3e02e13…)
Normalization warnings	none
Normalization errors	none
Outputs without a validated mapping	none

2.3 Methodology

The analysis follows the playbook below (PLAYBOOK.md), one step at a time. At each step, the agent queries the read-only database, correlates the sources and records a finding only if all its evidence references exist in the database (automatic check). Artifact dates are interpreted according to their evidential value (e.g. ShimCache = presence, Prefetch = execution).

Step	Purpose	Status	Tool calls	Duration
1	System context and scope	done	28	681 s
2	Triage of automatic detections	done	40	754 s
3	Program execution	done	39	1428 s
4	Memory (RAM)	done	44	752 s
5	Persistence	done	35	146 s
6	Accounts, authentication and privileges	done	31	155 s
7	Remote access and lateral movement	done	31	160 s
8	User activity on files and folders	done	31	197 s
9	Web browsing, downloads and e-mail	done	23	123 s
10	Defense evasion and anti-forensics	done	34	196 s
11	Collection, exfiltration and communications	done	13	84 s
12	Incident timeline and correlation	done	17	76 s
13	Summary, answers and recommendations	done	3	60 s

2.4 Guidance provided by the analyst

Context	none
Period of interest	not specified
Questions	none
Provided IOCs	none
Suspects	none
Exclusions	none
Instructions	none

2.5 Limitations of the analysis

1. Single source. This is a memory-only case: no KAPE/disk collection, no EVTX, no registry hives, no Prefetch/Amcache files, no network or proxy logs. Under the evidence policy every critical/high finding therefore stays "probable".
2. No automatic detections exist and no hash sets were loaded.
3. Not recoverable from the image: the contents or SHA1/SHA256 of coreupdater.exe; the coreupdater service ImagePath (empty in svcscan); the command lines of powershell.exe PIDs 508 and 3316; the identity of parent PPID 1380; the exact last octet of 194.61.24.x; the RDP source IP; the target of loot.lnk; the processes owning the 203.78.103.109 sockets; the injection time and loader of the spoolsv.exe PE.
4. mftscan gives no parent paths, so the download location of coreupdater.exe (Administrator Downloads) is inferred from timing.
5. Attributions by timing are interpretations: RID 1108 as mortysmith, the socket owners by port ordering, and coreupdater.exe as the installer of its own service.
6. Initial access (how the domain Administrator credentials were obtained) is outside the visible scope.
7. Data exfiltration was neither confirmed nor excluded.
8. Time zone: meta gives no bias; UTC-7 is derived from an ETL file name.

- Step 1 (System context and scope): No disk artifacts (registry, EVTX, Prefetch, Amcache, MFT/USN from disk), no detections; time zone unknown; no Wi-Fi/USB/volume data; userassist/amcache plugins empty; hashdump/lsadump/dumpfiles not run so no binary hashes. All evidence is one source (memory) so high findings cannot reach 'confirmed' by independent corroboration.
- Step 2 (Triage of automatic detections): All evidence is from one memory image, so high/critical findings remain 'probable'. Command lines of PowerShell PIDs 508/3316 and coreupdater are not resident; service ImagePath and coreupdater.exe path/hash not recovered; netscan Owner/PID empty, so C2 attribution to a process is impossible; parent PIDs 1380 and 4008 absent from pslist/psscan.
- Step 3 (Program execution): No disk artefacts (Prefetch run counts, Amcache SHA1, registry ShimCache/BAM/UserAssist, SRUM, EVTX 4688/4104/Sysmon) - all conclusions rest on a single source (memory) and remain 'probable'. coreupdater.exe full path, hash and command line unrecoverable (exited); powershell command lines unreadable; parent processes 4008 and 1380 not carved. hash_hits empty.
- Step 4 (Memory (RAM)): There are no disk artifacts (EVTX, Prefetch files, Amcache, registry, SRUM, Defender logs), so nothing in memory could be checked against the disk. Every high finding is therefore single-source and stays "probable". I could not determine which process owns the 203.78.103.109 connections. The coreupdater service binary path is empty, and coreupdater.exe has no Zone.Identifier and no handle that would show its full path or how it was downloaded. The plugins hollowprocesses, processghosting, etwpatch, svcdiff, userassist, amcache and skeleton_key were not provided. I did not finish checking the callbacks, SSDT and driverirp tables for hooks within this step.
- Step 5 (Persistence): There are no disk artefacts, no EVTX and no parsed registry hives. As a result, these could not be checked: the ImagePath, ObjectName and LastWrite of Services\coreupdater (the svcscan "Binary (Registry)" column is empty everywhere), events 7045/4697/4698/4702/106/5861, Run/RunOnce, Winlogon Userinit/Shell, IFEO Debugger, AppInit_DLLs, HKCU CLSID COM hijacks, Active Setup, Print monitors/processors, BootExecute, WMI subscriptions (OBJECTS.DATA not parsed), the hosts file content, Office startup and PendingFileRenameOperations. The scheduled_tasks plugin returned many empty rows because of paged-out data. The creation time of the coreupdater service is inferred, not recorded.
- Step 6 (Accounts, authentication and privileges): There are no EVTX logs, so the Security/TS/RCM event IDs (4624, 4625, 4648, 4672, 4720, 4732, 1149, 21/25) and logon failures could not be checked. The SAM and SECURITY hives are in memory, but no registry, hashdump or cachedump output is available, so account creation, last logon, password changes and group membership changes in the SAM are undetermined. The RID 1108 to mortysmith mapping is inferred from timestamps only. The RDP client IP and the session 1 user could not be recovered.
- Step 7 (Remote access and lateral movement): There are no EVTX, so the following could not be checked: Security 4624 LT10/4648/4778/5140/5145, RdpCoreTS 131, TerminalServices-LSM 21/24/25, WinRM 6/91/168, PowerShell

400, System 7045. There are no registry hives (Terminal Server Client, MountPoints2, Network\Z, ShellBags), no RDP bitmap cache and no full $MFT. The RDP source IP and the user who owns LUID 0xb3ce1f cannot be determined. The 203.78.103.109:443 sessions still have no identified owner process.

- Step 8 (User activity on files and folders): There is no disk or KAPE collection, so none of these were available: LNK bodies (target, volume serial, MAC), Jump List contents, ShellBags, RecentDocs/OpenSave/LastVisited, WordWheelQuery, Windows Timeline, SIDR, Thumbcache, Notepad tabs, $MFT/$UsnJrnl/$LogFile (creations, renames and deletions beyond memory-resident records), Recycle Bin, or USB/USBSTOR/EMDMgmt/setupapi. The consequences:
  - The target, content and possible exfiltration of "loot" are unknown.
  - The parent directories of LNK files are inferred from timing.
  - MOTW absence on coreupdater.exe is not established.
  - Removable-media history, other than the responder E: drive, cannot be assessed.
  - Thoughts.txt and Plans contents are not in memory.
- Step 9 (Web browsing, downloads and e-mail): No browser databases were parsed (WebCacheV01.dat/spartan.edb of the Administrator profile are referenced in filescan but were not extracted). Because of that, the full URL, the last octet of 194.61.24.x, typed URLs and search queries, and the Edge download record cannot be recovered. coreupdater.exe has no Zone.Identifier in memory, so HostUrl/ReferrerUrl are unavailable. mftscan gives no parent directory, so the Downloads location is inferred from timing. All the evidence comes from the memory image (a single source), so corroborating with disk artefacts (MFT/$J, Zone.Identifier, WebCache) or proxy/firewall logs is required to confirm.
- Step 10 (Defense evasion and anti-forensics): No EVTX contents, so 1102/104/4616/Kernel-General 1/Defender 5001-5013 could not be checked. No registry hives (Defender exclusions, DisableRealtimeMonitoring, PowerShell logging policy). No full $MFT, $J or $LogFile (mass deletion, prefetch deletion and timestomp could only be checked on records cached in memory). vol_etwpatch not available. The system time change could not be assessed.
- Step 11 (Collection, exfiltration and communications): The case has no SRUM (network usage per application), no EVTX (5156, Sysmon 3/22, BITS 3/59/60, firewall 2004/2005), no $MFT/$UsnJrnl, no 7-Zip history, no browser upload history and no OneDrive ODL or SyncEngineDatabase parsing. Byte volumes to 203.78.103.109 are unknown. The socket owners are null in netscan, so attribution rests on inference. The target of "loot" and the contents of \\CITADEL-DC01\FileShare could not be recovered. RDP clipboard and drive-redirection transfers cannot be assessed.
- Step 12 (Incident timeline and correlation): Several things cannot be recovered from memory: the initial access vector (how the Administrator credentials were obtained), the RDP source IP, the last octet of 194.61.24.x, the coreupdater.exe hash and full path, the spoolsv injection time, the owner of each 203.78.103.109 connection, the target of loot.lnk, and the PowerShell command line. With memory as the only source, nothing can be corroborated independently. The disk image (EVTX 4624/4688/7045, Sysmon, Amcache, $MFT/$J, the SYSTEM and NTUSER hives) is needed to confirm the findings.
- Step 13 (Summary, answers and recommendations): Memory-only case: there are no EVTX, hives or disk artefacts, so every critical/high finding stays probable (single source). Not recoverable: the coreupdater.exe hash and ImagePath, the PowerShell command lines, the RDP source IP, the full 194.61.24.x address, the loot.lnk target, the C2 socket owners, initial credential access, and whether data was exfiltrated.

3. Incident timeline

Key events established by the analysis (UTC), with their evidence. The complete timeline of the incident window (all artifacts) is provided in incident_timeline.csv.

Timestamp (UTC)	Phase	Event	Description	MITRE	References
2020-09-18 22:44:13	context	First logon of C137\ricksanchez (RID 1106) on the host	Profile C:\Users\ricksanchez created at 22:44:11. SID-named directory and FontCache for -1106 created at 22:44:13. L.ricksanchez.cdp followed at 22:44:21.		volatility/windows.mftscan.MFTScan.jsonl#10758 volatility/windows.mftscan.MFTScan.jsonl#102070
2020-09-18 22:46:45	context	First logon of C137 RID 1108 (mortysmith); Thoughts.lnk created 22:47:34	The -1108 SID directory was created at 22:46:45 and L.mortysmith.cdp at 22:46:46. Thoughts.lnk (recent item) was created at 22:47:34. The injected spoolsv.exe later had C:\Users\mortysmith\Desktop as its CWD.		volatility/windows.mftscan.MFTScan.jsonl#28040 volatility/windows.mftscan.MFTScan.jsonl#26401 volatility/windows.mftscan.MFTScan.jsonl#79896
2020-09-19 01:24:07	context	System boot of DESKTOP-SDN1RPT	System (PID 4) / smss.exe (PID 312) created 2020-09-19 01:24:07 UTC: last boot before memory capture.		volatility/windows.pslist.PsList.jsonl#1 volatility/windows.pslist.PsList.jsonl#3
2020-09-19 03:16:05	context	Short interactive session 1 (user undetermined) runs MicrosoftEdge PID 4592 (03:16:05-03:17:00)	psscan: MicrosoftEdge.exe PID 4592 (PPID 764 svchost/DcomLaunch) in session 1, created 03:16:05, exited 03:17:00 UTC, i.e. ~19 min before the Administrator RDP logon. User of session 1 not recoverable; listed as context/possible precursor (undetermined relation).		volatility/windows.psscan.PsScan.jsonl#18
2020-09-19 03:36:23	lateral movement	First interactive logon of C137\Administrator (session 3, RDP indicators)	Administrator profile objects created 03:36:22-27 and RdpIdd/IddCx0102/termkbd.PNF created 03:36:23 (memory MFT); MicrosoftEdge PID 860 in session 3 from 03:36:40.	T1021.001 T1078.002	volatility/windows.mftscan.MFTScan.jsonl#23934 volatility/windows.psscan.PsScan.jsonl#158
2020-09-19 03:36:23	lateral movement	First-ever inbound RDP session established (rdpidd.PNF/termkbd.PNF/RdpIdd created)	RDP device artefacts were created on first use (rdpidd.PNF record 259, termkbd.PNF record 255, RdpIdd record 260, TerminalServices-PnPDevices evtx 03:36:24). This is session 3 of C137\Administrator (RemoteInteractive). The source IP is not recoverable.	T1021.001	volatility/windows.mftscan.MFTScan.jsonl#128240 volatility/windows.mftscan.MFTScan.jsonl#69180 volatility/windows.mftscan.MFTScan.jsonl#23939
2020-09-19 03:36:26	lateral movement	Domain Administrator RID 500 RDP (RemoteInteractive) logon - profile created	The -500 SID directory was created at 03:36:26, the first-ever profile artefact for this account on the host. Session 3 tokens (Edge PID 860, coreupdater PID 8324) contain S-1-5-14 Remote Interactive Logon.	T1021.001 T1078.002	volatility/windows.mftscan.MFTScan.jsonl#107218 volatility/windows.getsids.GetSIDs.jsonl#606 volatility/windows.getsids.GetSIDs.jsonl#625
2020-09-19 03:39:26	initial access / ingress	Administrator's Edge visits a host at IP 194.61.24.x (DOMStore 194.61.24[1].xml created)	Edge Legacy PID 860 (session 3, C137\Administrator) creates DOMStore item 194.61.24[1].xml (MFT 25584). This is the only raw-IP site in the browser artefacts and it comes 34 s before coreupdater.exe was written.	T1105	volatility/windows.mftscan.MFTScan.jsonl#51833 volatility/windows.psscan.PsScan.jsonl#158
2020-09-19 03:39:45	defense evasion	Defender Network Inspection driver WdNisDrv.sys unloaded (Windows Security UI open by Administrator)	WdNisDrv.sys unloaded 03:39:45; SecHealthUI package folder for RID 500 created 03:39:36; SECURITYHEALTHHOST pf 03:39:47; WdNisSvc/WdNisDrv STOPPED at capture. 15 s before coreupdater.exe drop. Probable disabling of Defender protection (F-009).	T1562.001	volatility/windows.unloadedmodules.UnloadedModules.jsonl#7 volatility/windows.mftscan.MFTScan.jsonl#111476 volatility/windows.mftscan.MFTScan.jsonl#13226
2020-09-19 03:39:57	initial access / ingress	Edge DownloadHistory updated: download started (Administrator profile)	$SI M/C/A of the Edge DownloadHistory directory (MFT 24179) set to 03:39:57. coreupdater.exe (MFT 26213) was created 3 s later (03:40:00, finalised 03:40:04). The Administrator Downloads directory (MFT 426) was modified at 03:40:42 and the file ran at 03:40:49.	T1105	volatility/windows.mftscan.MFTScan.jsonl#109281 volatility/windows.mftscan.MFTScan.jsonl#56378 volatility/windows.mftscan.MFTScan.jso

Timestamp (UTC)	Phase	Event	Description	MITRE	References
					nl#5594
2020-09-19 03:40:00	execution / persistence	coreupdater.exe written to disk (MFT record 26213)	$FILE_NAME creation of coreupdater.exe; executed at 03:40:49 as domain Administrator (PID 8324, exited 03:43:10); service "coreupdater" auto-start registered.	T1543.003	volatility/windows.mftscan.MFTScan.jsonl#56378 volatility/windows.pslist.PsList.jsonl#50 volatility/windows.svcscan.SvcScan.jsonl#20
2020-09-19 03:40:49	execution	coreupdater.exe executed (PID 8324) as elevated domain Administrator	PID 8324 PPID 4008 session 3, High IL, Domain/Enterprise Admins; $SI accessed 03:40:49 on MFT 26213; ChxSmartScreen prefetch created same second; exited 03:43:10.	T1204.002	volatility/windows.psscan.PsScan.jsonl#167 volatility/windows.mftscan.MFTScan.jsonl#56376 volatility/windows.mftscan.MFTScan.jsonl#105657
2020-09-19 03:40:49	persistence	Service "coreupdater" (AUTO_START) registered at runtime (SCM Order 662) - inferred during coreupdater.exe run 03:40:49-03:43:10	svclist shows coreupdater at Order 662, which is after the alphabetical boot enumeration (ends at Order 623) and before the per-user services of the 05:08 logon (Order 664+). The exact creation time is not recorded in memory; the timestamp shown is the start of the only coreupdater.exe execution (inferred).	T1543.003	volatility/windows.svclist.SvcList.jsonl#22 volatility/windows.psscan.PsScan.jsonl#167
2020-09-19 03:43:10	execution	coreupdater.exe PID 8324 exits after 2 min 21 s of runtime	psscan: coreupdater.exe PID 8324, PPID 4008 (explorer, session 3), CreateTime 03:40:49, ExitTime 03:43:10 UTC. During this window its token carried SeDebugPrivilege enabled (F-010) and the service "coreupdater" was most probably installed (F-012).	T1204.002 T1543.003	volatility/windows.psscan.PsScan.jsonl#167 volatility/windows.privileges.Privs.jsonl#1734
2020-09-19 03:44:52	execution	Administrator's Edge PID 860 exits (browser used for 194.61.24.x download)	psscan: MicrosoftEdge.exe PID 860 (session 3, C137\Administrator, RemoteInteractive token) created 03:36:40, exited 03:44:52 UTC. Session continued interactively until at least 03:47:39 (F-018).	T1105	volatility/windows.psscan.PsScan.jsonl#158
2020-09-19 03:45:02	collection	Administrator session enumerates all user profile folders (desktop.ini/shell folders accessed)	At 03:45:02, $SI Accessed is updated in the same second on the shell folders and desktop.ini files of the ricksanchez, mortysmith and OOBE local profiles. This is consistent with browsing or searching across C:\Users during the RDP session of C137\Administrator.	T1083	volatility/windows.mftscan.MFTScan.jsonl#36769 volatility/windows.mftscan.MFTScan.jsonl#97439 volatility/windows.mftscan.MFTScan.jsonl#126444
2020-09-19 03:45:53	collection	Portal_gun.png (Edge download of 23:07:44) opened in Photos during Administrator session	Portal_gun.png (MFT rec 96326) has $SI Accessed = 03:45:53. The Photos app resources PhtoMDL2.ttf and Blank_PhotosSplashWideTile.png were accessed at 03:45:53-03:46:02.		volatility/windows.mftscan.MFTScan.jsonl#49436 volatility/windows.mftscan.MFTScan.jsonl#84896
2020-09-19 03:46:18	collection	Shortcut loot.lnk created during Administrator session	MFT record 87445 loot.lnk created 03:46:18, 3 min after coreupdater exit; suggests a file/folder named "loot" was opened (possible data staging).	T1074.001	volatility/windows.mftscan.MFTScan.jsonl#78568
2020-09-19 03:47:39	collection	Desktop.lnk created in Administrator Recent folder (last observed activity of session 3)	Desktop.lnk was created (rec 87493). The Administrator Recent directory (rec 520, created 03:36:24) and the Explorer jump lists 5f7b5f1e01b83767 and f01b4d95cf55d32a were modified at 03:47:39. A mortysmith-profile desktop.ini (rec 89652) was accessed at 03:47:34.	T1005	volatility/windows.mftscan.MFTScan.jsonl#82925 volatility/windows.mftscan.MFTScan.jsonl#64444 volatility/windows.mftscan.MFTScan.jsonl#89564
2020-09-19 05:08:15	context	Console logon of C137\ricksanchez (session 2)	sihost.exe and the user svchost started at 05:08:15, then explorer.exe (SESSIONNAME=Console) at 05:08:16. LOGONSERVER is \\CITADEL-DC01. This logon was followed by the PowerShell chain at 05:08:37.		volatility/windows.sessions.Sessions.jsonl#53 volatility/windows.sessions.Sessions.jsonl#57 volatility/windows.envars.Envars.jsonl#602
2020-09-19 05:08:37	execution	powershell.exe chain at ricksanchez logon (PID 508 -> 3316) with injected PE	powershell.exe PID 508 (PPID 1380 absent) started 21 s after logon in session 2, spawned PID 3316 at 05:08:43, which contains an unlinked RWX MZ region and an RWX-started thread.	T1059.001 T1055	volatility/windows.pslist.PsList.jsonl#78 volatility/windows.pslist.PsList.jsonl#81 volatility/windows.malware.malfind.Malfind.jsonl#6
2020-09-19 05:08:44	execution	First PowerShell prefetch on host (POWERSHELL.EXE-CA1AE517.pf)	Prefetch created 1 s after powershell PID 3316 start (parent 508, grandparent 1380 not recoverable); ricksanchez, Medium IL, no command line recoverable.	T1059.001	volatility/windows.mftscan.MFTScan.jsonl#16979 volatility/windows.psscan.PsScan.jsonl#101
2020-09-19 05:09:56	collection (responder)	Memory acquisition with FTK Imager (collection)	FTK Imager.exe PID 6544 started from E:\FTK Imager\ by explorer.exe PID 5896 (ricksanchez session 2); image SystemTime 05:10:39 UTC.		volatility/windows.pslist.PsList.jsonl#87 volatility/windows.cmdline.CmdLine.jso

Timestamp (UTC)	Phase	Event	Description	MITRE	References
					nl#87 volatility/windows.info.Info.jsonl#12
2020-09-19 05:10:39	credential access / command and control	State at capture: reflective PE in spoolsv.exe PID 2188 with lsass handle; 2 ESTABLISHED sessions to 203.78.103.109:443	At memory capture (05:10:39 UTC): spoolsv.exe PID 2188 (created 01:24:09, session 0) holds a private RWX region at 0x1840000 starting with MZ (malfind), unlinked from loader lists (ldrmodules), and a handle on lsass.exe PID 656 (process VM_READ access). netscan/netstat: 10.42.85.115:50875 and :50972 -> 203.78.103.109:443 ESTABLISHED, owner unresolved. Injection time inside spoolsv is not determinable (process start is boot time, not injection time).	T1055.001 T1620 T1003.001 T1071.001	volatility/ windows.malware.malfind.Malfind.jsonl #1 volatility/windows.malware.ldrmodules. LdrModules.jsonl#32 volatility/windows.handles.Handles.jsonl #14206 volatility/windows.netscan.NetScan.json l#110 volatility/windows.netscan.NetScan.json l#114 volatility/windows.psscan.PsScan.jsonl# 106

Detailed evidence timeline

All the evidence cited by the findings and the timeline, sorted by time.

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
	raw.vol_image	{"image": "C:\\Sleuthline\\cases\\test02\\memory\\DESKTOP-SDN1RPT.mem", "name": "DESKTOP-SDN1RPT.mem", "bytes": "2147483648", "sha256": "AC6EE0E8C273103B94D776F54192D2C6ACF8303AABBD6A199254030C69F667C3"}	volatility/memory_image.jsonl#1		F-001, F-002
	raw.vol_info	{"TreeDepth": "0", "Value": "15.19041", "Variable": "Major/Minor"}	volatility/windows.info.Info.jsonl#9		F-001, F-026
	raw.vol_info	{"TreeDepth": "0", "Value": "2020-09-19 05:10:39+00:00", "Variable": "SystemTime"}	volatility/windows.info.Info.jsonl#12		F-001, F-002, T-002
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x146a4e03120", "PID": "616", "Process": "services.exe", "Value": "DESKTOP-SDN1RPT", "Variable": "COMPUTERNAME"}	volatility/ windows.envars.Envars.jsonl#5		F-001
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x215a22b1bd0", "PID": "3232", "Process": "sihost.exe", "Value": "\\\\CITADEL-DC01", "Variable": "LOGONSERVER"}	volatility/ windows.envars.Envars.jsonl#565		F-001
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x215a22b1bd0", "PID": "3232", "Process": "sihost.exe", "Value": "C137.LOCAL", "Variable": "USERDNSDOMAIN"}	volatility/ windows.envars.Envars.jsonl#585		F-001
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x215a22b1bd0", "PID": "3232", "Process": "sihost.exe", "Value": "ricksanchez", "Variable": "USERNAME"}	volatility/ windows.envars.Envars.jsonl#588		F-001
	raw.vol_getsids	{"TreeDepth": "0", "PID": "3232", "Process": "sihost.exe", "SID": "S-1-5-21-2232410529-1445159330-2725690660-1106"}	volatility/ windows.getsids.GetSIDs.jsonl#638		F-001
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/ windows.getsids.GetSIDs.jsonl#613		F-001
	raw.vol_hivelist	{"TreeDepth": "0", "File output": "Disabled", "FileFullPath": "\\??\\C:\\Users\\Admin\\ ntuser.dat", "Offset": "227618253631488"}	volatility/ windows.registry.hivelist.HiveList.jsonl #13		F-001, F-015
	raw.vol_hivelist	{"TreeDepth": "0", "File output": "Disabled", "FileFullPath": "\\??\\C:\\Users\\Administrator\\ ntuser.dat", "Offset": "227618280730624"}	volatility/ windows.registry.hivelist.HiveList.jsonl #14		F-001, F-014
	raw.vol_hivelist	{"TreeDepth": "0", "File output": "Disabled", "FileFullPath": "\\??\\C:\\Users\\ricksanchez\\ ntuser.dat", "Offset": "227618280112128"}	volatility/ windows.registry.hivelist.HiveList.jsonl #16		F-001
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "10.42.85.10", "ForeignPort": "445", "LocalAddr": "10.42.85.115", "LocalPort": "50957", "Offset": "209519101090720", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#88		F-001, F-017, F-025
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "3232", "Process": "sihost.exe", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/ windows.getsids.GetSIDs.jsonl#650		F-001

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Administrator", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-21-2232410529-1445159330-2725690660-500"}	volatility/ windows.getsids.GetSIDs.jsonl#600		F-001, F-014
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "36", "Disasm": "\"\n0x1840000:\tpop\tr10\n0x1840002:\ tnop\t\n0x1840003:\tadd\tbyte ptr [rbx], al\n0x1840005:\tadd\tbyte ptr [rax], al\ n0x1840007:\tadd\tbyte ptr [rax + rax], al\n0x184000a:\tadd\tbyte ptr [rax], al\"", "End VPN": "25575423", "File output": "Disabled", "Hexdump": "4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 b8 00 00 00 00 00 00 00 40 00 00 00	volatility/ windows.malware.malfind.Malfind.jsonl#1		F-003, I-012, T-024
	raw.vol_ldrmodules	{"TreeDepth": "0", "Base": "25427968", "InInit": "false", "InLoad": "false", "InMem": "false", "Pid": "2188", "Process": "spoolsv.exe"}	volatility/ windows.malware.ldrmodules.LdrModules.jsonl#32		F-003, I-012, T-024
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "5136", "HandleValue": "1964", "Name": "lsass.exe Pid 656", "Offset": "209519057219712", "PID": "2188", "Process": "spoolsv.exe", "Type": "Process"}	volatility/ windows.handles.Handles.jsonl#14206		F-003, I-012, T-024
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1048608", "HandleValue": "1912", "Name": "\\Device\\ HarddiskVolume3\\Users\\mortysmith\\Desktop", "Offset": "209519093754240", "PID": "2188", "Process": "spoolsv.exe", "Type": "File"}	volatility/ windows.handles.Handles.jsonl#14195		F-003, F-015, I-008
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "36", "Disasm": "\"\n0x10c6c070000:\tpop\tr10\ n0x10c6c070002:\tnop\t\n0x10c6c070003:\tadd\tbyte ptr [rbx], al\n0x10c6c070005:\tadd\ tbyte ptr [rax], al\n0x10c6c070007:\tadd\tbyte ptr [rax + rax], al\n0x10c6c07000a:\tadd\tbyte ptr [rax], al\"", "End VPN": "1152863780863", "File output": "Disabled", "Hexdump": "4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 b8 00 0	volatility/ windows.malware.malfind.Malfind.jsonl#6		F-004, I-013, T-004
	raw.vol_ldrmodules	{"TreeDepth": "0", "Base": "1152863633408", "InInit": "false", "InLoad": "false", "InMem": "false", "Pid": "3316", "Process": "powershell.exe"}	volatility/ windows.malware.ldrmodules.LdrModules.jsonl#176		F-004
	raw.vol_suspicious_threads	{"TreeDepth": "0", "Address": "1152857931776", "Context": "Win32Start", "Note": "VAD at base address (0x10c6bb00000) hosting this thread has an unexpected starting protection PAGE_EXECUTE_READWRITE", "PID": "3316", "Process": "powershell.exe", "TID": "3892", "VAD Path": "<Non-File Backed Region>"}	volatility/ windows.malware.suspicious_threads. SuspiciousThreads.jsonl#44		F-004, I-013
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "coreupdater", "Name": "coreupdater", "Offset": "1402927956048", "Order": "662", "Start": "SERVICE_AUTO_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_WIN32_OWN_PROCESS"}	volatility/ windows.svcscan.SvcScan.jsonl#20		F-005, F-006, F-012, I-001, T-003
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Administrator", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-500"}	volatility/ windows.getsids.GetSIDs.jsonl#619		F-006, I-015
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/ windows.getsids.GetSIDs.jsonl#632		F-006
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Enterprise Admins", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-519"}	volatility/ windows.getsids.GetSIDs.jsonl#634		F-006
	raw.vol_getsids	{"TreeDepth": "0", "Name": "High Mandatory Level", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-16-12288"}	volatility/ windows.getsids.GetSIDs.jsonl#637		F-006, F-014
	raw.vol_cmdline	{"TreeDepth": "0", "PID": "508", "Process": "powershell.exe"}	volatility/ windows.cmdline.CmdLine.jsonl#78		F-007, F-013
	raw.vol_cmdline	{"TreeDepth": "0", "PID": "3316", "Process": "powershell.exe"}	volatility/ windows.cmdline.CmdLine.jsonl#81		F-007
	raw.vol_cmdscan	{"TreeDepth": "1", "ConsoleInfo": "138492641775368", "Data": "0", "PID": "728", "Process": "conhost.exe", "Property": "_COMMAND_HISTORY.CommandCount"}	volatility/ windows.cmdscan.CmdScan.jsonl#4		F-007
	raw.vol_getsids	{"TreeDepth": "0", "PID": "3316", "Process": "powershell.exe", "SID": "S-1-5-21-2232410529-1445159330-2725690660-1106"}	volatility/ windows.getsids.GetSIDs.jsonl#1098		F-007
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Medium Mandatory Level", "PID": "3316", "Process": "powershell.exe", "SID": "S-1-16-8192"}	volatility/ windows.getsids.GetSIDs.jsonl#1113		F-007
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1442207", "HandleValue": "2368", "Name": "\\Device\\ Afd\\Endpoint", "Offset": "209519091203248", "PID": "3316", "Process": "powershell.exe", "Type": "File"}	volatility/ windows.handles.Handles.jsonl#28137		F-007, F-025
	raw.vol_ads	{"TreeDepth": "0", "ADS Filename": "Zone.Identifier", "Filename": "autorunsc64.exe", "Hexdump": "5b 5a 6f 6e 65 54 72 61 6e 73 66 65 72 5d 0d 0a 5a 6f 6e 65 49 64 3d 33 0d 0a 52 65 66 65 72 72 65 72 55 72 6c 3d 43 3a 5c 55 73 65 72 73 5c 4a 61 6d 65 73 5c 44 6f 77 6e 6c 6f 61 64 73 5c 53 79 73 69 6e 74 65 72 6e 61 6c 73 53 75 69 74 65 2e 7a 69 70 0d 0a", "MFT Type": "DATA", "Offset": "345084272",	volatility/ windows.mftscan.ADS.jsonl#13		F-008

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "Microsoft Defender Antivirus Network Inspection System Driver", "Name": "WdNisDrv", "Offset": "1402926010544", "Order": "574", "Start": "SERVICE_DEMAND_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_KERNEL_DRIVER"}	volatility/ windows.svcscan.SvcScan.jsonl#70		F-009, F-023, I-005
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "獀撈昮獀撈昮蔉案傑桂助醝狵捩獟牴慲扭晉瑙粵挤慨麗徾牴檇獀瑙獀撈昮㘝殼睜 Z 瓤甒醝狵捩潟瑳敲涛瑙粵挤慨麗徾牴檇獀瑙獀撈昮獀撈昮蔉案址弹逮湩瑩悚楬毆瞙眸也婀璩囧慢楳姣獯牴慲瞭坟粵挤慨麗牴檇獀瑙瞶瑳瞙昮瑳瞙鐽桂晾ι蔉啜囧昮 Z 瓤甒醝狵捩潟瑳敲涛瑙档牡瑟慵瑩瞬坟獀撈昮獀撈昮蔉案傑桂助醝狵捩獟牴慲扭晉瑙啗囧档牡瑟慵瑩瞬坟獀撈昮㘝殼睜 Z 瓤甒醝狵捩潟瑳敲涛瑙啗囧档牡瑟慵瑩瞬坟獀撈昮獀撈昮蔉案址弹逮湩瑩悚楬毆瞙眸也婀璩囧慢楳姣獯敲涛瑙粵瞻啄囧档牡瑟慵瑩瞬眸瑳瞙昮瑳瞙鐽桂晾棄噂囧昮 Z 瓤甒醝狵捩獟牴慲扭晉臏粵挤慨麗牴檇獀瞬獀撈昮", "Name": "WdNisSvc", "Offset": "1402926015184", "Order": "575", "Start": "SERVICE_DEMAND_START", "State": "SERV	volatility/ windows.svcscan.SvcScan.jsonl#69		F-009, F-023
	raw.vol_privileges	{"TreeDepth": "0", "Attributes": "Present,Enabled", "Description": "Debug programs", "PID": "8324", "Privilege": "SeDebugPrivilege", "Process": "coreupdater.ex", "Value": "20"}	volatility/ windows.privileges.Privs.jsonl#1734		F-010, T-022
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#110		F-011, F-025, I-003, T-024
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#114		F-011, F-025, I-003, T-024
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netstat.NetStat.jsonl#12		F-011, F-025
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netstat.NetStat.jsonl#15		F-011, F-025
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "10.42.85.10", "ForeignPort": "445", "LocalAddr": "10.42.85.115", "LocalPort": "50957", "Offset": "209519101090720", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netstat.NetStat.jsonl#2		F-011
	raw.vol_svclist	{"TreeDepth": "0", "Display": "coreupdater", "Name": "coreupdater", "Offset": "1402927956048", "Order": "662", "Start": "SERVICE_AUTO_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_WIN32_OWN_PROCESS"}	volatility/ windows.svclist.SvcList.jsonl#22		F-012, T-010
	raw.vol_svclist	{"TreeDepth": "0", "Name": "xinputhid", "Offset": "1402926051120", "Order": "623", "Start": "SERVICE_DEMAND_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_KERNEL_DRIVER"}	volatility/ windows.svclist.SvcList.jsonl#23		F-012
	raw.vol_svclist	{"TreeDepth": "0", "Binary": "C:\\Windows\\system32\\svchost.exe -k LocalServiceNoNetwork -p", "Display": "CoreMessaging", "Name": "CoreMessagingRegistrar", "Offset": "1402925696448", "Order": "93", "PID": "904", "Start": "SERVICE_AUTO_START", "State": "SERVICE_RUNNING", "Type": "SERVICE_WIN32_SHARE_PROCESS"}	volatility/ windows.svclist.SvcList.jsonl#553		F-012
	raw.vol_svclist	{"TreeDepth": "0", "Display": "Agent Activation Runtime_b3e3d7", "Name": "AarSvc_b3e3d7", "Offset": "1402927959024", "Order": "664", "Start": "SERVICE_DEMAND_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_WIN32_SHARE_PROCESS"}	volatility/ windows.svclist.SvcList.jsonl#21		F-012
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Remote Interactive Logon", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-14"}	volatility/ windows.getsids.GetSIDs.jsonl#606		F-014, F-016, I-015, T-013
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Remote Interactive Logon", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-14"}	volatility/ windows.getsids.GetSIDs.jsonl#625		F-014, F-016, I-007, T-013
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Schema Admins", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-21-2232410529-1445159330-2725690660-518"}	volatility/ windows.getsids.GetSIDs.jsonl#614		F-014
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Enterprise Admins", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-21-2232410529-1445159330-2725690660-519"}	volatility/ windows.getsids.GetSIDs.jsonl#615		F-014
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x761c00", "PID": "5896", "Process": "explorer.exe", "Value": "Console", "Variable": "SESSIONNAME"}	volatility/ windows.envars.Envars.jsonl#657		F-014
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x1b9f82033f0", "PID": "1172", "Process": "svchost.exe", "Value": "\\\\CITADEL-DC01", "Variable": "LOGONSERVER"}	volatility/ windows.envars.Envars.jsonl#602		F-015, T-014
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\Users\\mortysmith\\Desktop\\Thoughts.txt", "Offset": "209519130648096"}	volatility/ windows.filescan.FileScan.jsonl#5309		F-015, F-020

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
	raw.vol_hivelist	{"TreeDepth": "0", "File output": "Disabled", "FileFullPath": "\\SystemRoot\\System32\\Config\\SAM", "Offset": "227618137903104"}	volatility/windows.registry.hivelist.HiveList.jsonl#8		F-015
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "Remote Desktop Services", "Name": "TermService", "Offset": "1402925935040", "Order": "457", "PID": "200", "Start": "SERVICE_DEMAND_START", "State": "SERVICE_RUNNING", "Type": "SERVICE_WIN32_SHARE_PROCESS"}	volatility/windows.svcscan.SvcScan.jsonl#187		F-016
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1048576", "HandleValue": "2480", "Name": "\\Device\\Mup\\;Z:0000000000b3ce1f\\CITADEL-DC01\\FileShare", "Offset": "209519096092816", "PID": "1096", "Process": "svchost.exe", "Type": "File"}	volatility/windows.handles.Handles.jsonl#11901		F-017, I-009
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\;Z:0000000000b3ce1f\\CITADEL-DC01\\FileShare", "Offset": "209519096092816"}	volatility/windows.filescan.FileScan.jsonl#3433		F-017, I-009
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x761c00", "PID": "5896", "Process": "explorer.exe", "Value": "\\\\CITADEL-DC01", "Variable": "LOGONSERVER"}	volatility/windows.envars.Envars.jsonl#641		F-017
	raw.vol_ads	{"TreeDepth": "0", "ADS Filename": "Zone.Identifier", "Filename": "Portal_gun.png", "Hexdump": "5b 5a 6f 6e 65 54 72 61 6e 73 66 65 72 5d 0d 0a 4c 61 73 74 57 72 69 74 65 72 50 61 63 6b 61 67 65 46 61 6d 69 6c 79 4e 61 6d 65 3d 4d 69 63 72 6f 73 6f 66 74 2e 4d 69 63 72 6f 73 6f 66 74 45 64 67 65 5f 38 77 65 6b 79 62 33 64 38 62 62 77 65 0d 0a 5a 6f 6e 65 49 64 3d 33 0d 0a", "MFT Type": "DATA", "Of	volatility/windows.mftscan.ADS.jsonl#12		F-018
	raw.vol_ads	{"TreeDepth": "0", "ADS Filename": "Zone.Identifier", "Filename": "Jessica.jpg", "Hexdump": "5b 5a 6f 6e 65 54 72 61 6e 73 66 65 72 5d 0d 0a 4c 61 73 74 57 72 69 74 65 72 50 61 63 6b 61 67 65 46 61 6d 69 6c 79 4e 61 6d 65 3d 4d 69 63 72 6f 73 6f 66 74 2e 4d 69 63 72 6f 73 6f 66 74 45 64 67 65 5f 38 77 65 6b 79 62 33 64 38 62 62 77 65 0d 0a 5a 6f 6e 65 49 64 3d 33 0d 0a", "MFT Type": "DATA", "Offse	volatility/windows.mftscan.ADS.jsonl#1		F-020
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\Users\\Administrator\\AppData\\Local\\Microsoft\\Windows\\WebCache\\WebCacheV01.dat", "Offset": "209519123245328"}	volatility/windows.filescan.FileScan.jsonl#4452		F-021
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1180063", "HandleValue": "1328", "Name": "\\Device\\HarddiskVolume3\\Windows\\System32\\winevt\\Logs\\Security.evtx", "Offset": "209519066688144", "PID": "988", "Process": "svchost.exe", "Type": "File"}	volatility/windows.handles.Handles.jsonl#10927		F-022
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1180063", "HandleValue": "1316", "Name": "\\Device\\HarddiskVolume3\\Windows\\System32\\winevt\\Logs\\System.evtx", "Offset": "209519066674944", "PID": "988", "Process": "svchost.exe", "Type": "File"}	volatility/windows.handles.Handles.jsonl#10924		F-022
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1180063", "HandleValue": "1348", "Name": "\\Device\\HarddiskVolume3\\Windows\\System32\\winevt\\Logs\\Windows PowerShell.evtx", "Offset": "209519066691744", "PID": "988", "Process": "svchost.exe", "Type": "File"}	volatility/windows.handles.Handles.jsonl#10932		F-022
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "Windows Event Log", "Name": "EventLog", "Offset": "1402925743360", "Order": "136", "PID": "988", "Start": "SERVICE_AUTO_START", "State": "SERVICE_RUNNING", "Type": "SERVICE_WIN32_SHARE_PROCESS"}	volatility/windows.svcscan.SvcScan.jsonl#508		F-022
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\Windows\\System32\\winevt\\Logs\\Security.evtx", "Offset": "209519066688144"}	volatility/windows.filescan.FileScan.jsonl#1143		F-022
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\$Recycle.Bin\\S-1-5-21-2232410529-1445159330-2725690660-1106\\desktop.ini", "Offset": "209519130828096"}	volatility/windows.filescan.FileScan.jsonl#5448		F-022
	raw.vol_svcscan	{"TreeDepth": "0", "Binary": "\"C:\\ProgramData\\Microsoft\\Windows Defender\\platform\\4.18.2009.2-0\\MsMpEng.exe\"", "Display": "Microsoft Defender Antivirus Service", "Name": "WinDefend", "Offset": "1402926031872", "Order": "585", "PID": "2404", "Start": "SERVICE_AUTO_START", "State": "SERVICE_RUNNING", "Type": "SERVICE_WIN32_OWN_PROCESS"}	volatility/windows.svcscan.SvcScan.jsonl#59		F-023
	raw.vol_svcscan	{"TreeDepth": "0", "Binary": "\\FileSystem\\WdFilter", "Name": "WdFilter", "Offset": "1402926014720", "Order": "569", "Start": "SERVICE_BOOT_START", "State": "SERVICE_RUNNING", "Type": "SERVICE_FILE_SYSTEM_DRIVER"}	volatility/windows.svcscan.SvcScan.jsonl#75		F-023
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "瑶瑠啄對獢慴整瞳鼻瑳瞗兀秸晾葉嚐鯷捯湩潦㹈彀瞵 Z 虠瓡挤摯捥瑶蹓啄對獢慴整瞳鼻瑳瞗兀秸晾饈婐捯湩潦㹈彀瞵 Z 虠瓡挤摯捥瑶蹓啄對獢慴整瞳鼻瑳瞗兀秸晾饈婐捯湩潦㹈彀瞵 Z 虠瓡挤摯捥瑶蹓啄對獢慴整瞳鼻瑳瞗兀秸晾葉嚐鯷捯湩潦㹈彀瞵 Z 虠瓡挤摯捥瑶蹓啄對獢慴整瞳鼻瑳瞗兀秸晾葉嚐鯷捯湩潦㹈彀瞵 Z 虠瓡挤摯捥瑶蹓啄對獢慴整瞳鼻瑳瞗兀秸晾饈婐捯湩潦㹈彀瞵 Z 虠瓡挤摯捥瑶蹓啄對獢慴整瞳鼻瑳瞗兀秸晾葉嚐鯷捯湩潦㹈彀瞵 Z 虠瓡挤摯捥瑶蹓啄對獢慴整瞳鼻瑳瞗兀秸晾葉嚐鯷捯湩潦㹈彀瞵 Z 虠瓡挤摯捥瑶蹓啄對獢慴整瞳鼻瑳瞗兀秸晾饈婐捯湩潦㹈彀瞵 Z 虠瓡挤摯捥瑶蹓啄對獢慴整瞳鼻瑳瞗兀秸晾葉嚐鯷捯湩潦㹈彀瞵 Z", "Name": "Sense", "Offset": "1402925916784", "Order": "397", "Start": "SERVICE_DEMAND_START",	volatility/windows.svcscan.SvcScan.jsonl#247		F-023

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		"State": "SERVICE			
	raw.vol_callbacks	{"TreeDepth": "0", "Callback": "272684865443760", "Module": "WdFilter", "Type": "PspLoadImageNotifyRoutine"}	volatility/ windows.callbacks.Callbacks.jsonl#1		F-023
	raw.vol_ssdt	{"TreeDepth": "0", "Address": "272684835679072", "Index": "0", "Module": "ntoskrnl", "Symbol": "NtAccessCheck"}	volatility/windows.ssdt.SSDT.jsonl#1		F-024
	raw.vol_direct_system_calls	{"TreeDepth": "0", "Address": "140726349991939", "Disasm": "0x7ffd681a1003: add eax, 0xc32ecdc3; 0x7ffd681a1008: nop dword ptr [rax + rax]; 0x7ffd681a1010: mov r10, rcx; 0x7ffd681a1013: mov eax, 0x13ff; 0x7ffd681a1018: test byte ptr [0x7ffe0308], 1; 0x7ffd681a1020: jne 0x7ffd681a1025; 0x7ffd681a1022: syscall ; 0x7ffd681a1024: ret ; ", "PID": "3708", "Process": "SearchIndexer.", "Range": "\\Windows	volatility/ windows.malware.direct_system_calls .DirectSystemCalls.jsonl#4		F-024
	raw.vol_direct_system_calls	{"TreeDepth": "0", "Address": "140726497199904", "Disasm": "0x7ffd70e04720: mov r10, rcx; 0x7ffd70e04723: mov eax, 0; 0x7ffd70e04728: test byte ptr [0x7ffe0308], 1; 0x7ffd70e04730: jne 0x7ffd70e04735; 0x7ffd70e04732: syscall ; 0x7ffd70e04734: ret ; ", "PID": "4312", "Process": "svchost.exe", "Range": "\\Windows\\System32\\FirewallAPI.dll"}	volatility/ windows.malware.direct_system_calls .DirectSystemCalls.jsonl#258		F-024
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1442207", "HandleValue": "572", "Name": "\\Device\\ Afd", "Offset": "209519069517776", "PID": "2188", "Process": "spoolsv.exe", "Type": "File"}	volatility/ windows.handles.Handles.jsonl#13880		F-025
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1442207", "HandleValue": "600", "Name": "\\Device\\ Afd\\Endpoint", "Offset": "209519069533376", "PID": "2188", "Process": "spoolsv.exe", "Type": "File"}	volatility/ windows.handles.Handles.jsonl#13887		F-025
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\Users\\ricksanchez\\AppData\\Local\\Microsoft\\OneDrive\\ logs\\Personal\\SyncEngine-2020-09-19.0508.8904.1.aodl", "Offset": "209519128394784"}	volatility/ windows.filescan.FileScan.jsonl#5010		F-025
	raw.vol_cmdline	{"TreeDepth": "0", "Args": "\"E:\\FTK Imager\\FTK Imager.exe\" ", "PID": "6544", "Process": "FTK Imager.exe"}	volatility/ windows.cmdline.CmdLine.jsonl#87		T-002
	raw.vol_mutantscan	{"TreeDepth": "0", "Name": "e2vgmtzqt3xcbq.mca", "Offset": "209519120171152"}	volatility/ windows.mutantscan.MutantScan.jsonl#440		I-006
2018-04-02 18:27:58	Memory: ShimCache	ShimCache: file modified (memory) SIGN.MEDIA=11692A28 FTK Imager\FTK Imager.exe Order=4	volatility/ windows.shimcachemem.ShimcacheMem.jsonl#5	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-008
2019-07-16 17:11:36	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=24160 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#31242	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-024
2019-12-07 09:14:52	Memory: MFT records in memory	file created (memory, MFT record) Record Number=1527 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#36769	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018, T-016
2020-09-13 03:22:55	Memory: ShimCache	ShimCache: file modified (memory) SIGN.MEDIA=3ACFB0 autorunsc64.exe Order=8	volatility/ windows.shimcachemem.ShimcacheMem.jsonl#9	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-008
2020-09-18 05:41:34	Memory: MFT records in memory	file created (memory, MFT record) Record Number=27975 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#143841	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-022
2020-09-18 05:41:39	Memory: MFT records in memory	file created (memory, MFT record) Record Number=81794 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#136684	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-022
2020-09-18 05:47:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-41211245-796119838-3940169921-1000.pckgdep Record Number=83741 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#103228	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015
2020-09-18 05:50:04	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-41211245-796119838-3940169921-1001 Record Number=84914 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#117835	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015
2020-09-18 06:35:22	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Users Record Number=1475 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#140290	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-001
2020-09-18 21:41:48	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-41211245-796119838-3940169921-1001.pckgdep Record Number=87445 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#15020	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018
2020-09-18 21:41:48	Memory: MFT records in	file created (memory, MFT record) ; file modified (memory, MFT record)	volatility/	C:\Sleuthline\cases\test02\memory\	F-024

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
	memory	S-1-5-~1.PCK Record Number=87445 | Attribute Type=FILE_NAME | MFT Type=File	windows.mftscan.MFTScan.jsonl#15019	DESKTOP-SDN1RPT.mem	
2020-09-18 21:41:53	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-41211245-796119838-3940169921-1001.pckgdep Record Number=87543 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#104464	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015
2020-09-18 22:44:11	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) ricksanchez Record Number=87848 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#10758	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-001, F-015, T-011
2020-09-18 22:44:11	Memory: MFT records in memory	file created (memory, MFT record) Record Number=87863 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#126444	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018, T-016
2020-09-18 22:46:43	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=89652 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#89564	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018, T-018
2020-09-18 22:46:44	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=89698 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#97439	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018, T-016
2020-09-18 22:46:45	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-2232410529-1445159330-2725690660-1108 Record Number=90542 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#28040	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015, T-012
2020-09-18 22:46:45	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-2232410529-1445159330-2725690660-1108.pckgdep Record Number=90533 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#100724	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015
2020-09-18 22:46:46	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) L.mortysmith.cdp Record Number=90592 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#26401	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015, I-008, T-012
2020-09-18 22:47:34	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Thoughts.lnk Record Number=90986 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#79896	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015, F-020, T-012
2020-09-18 22:48:06	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Plans.lnk Record Number=91266 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#42892	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-020
2020-09-18 22:53:08	Memory: MFT records in memory	file created (memory, MFT record) Record Number=89157 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#141846	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-022
2020-09-18 22:53:47	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-2232410529-1445159330-2725690660-1106 Record Number=92893 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#102070	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015, T-011
2020-09-18 23:01:11	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Jessica.jpg Record Number=94877 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#541	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-020
2020-09-18 23:07:44	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=96326 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#49436	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018, T-017
2020-09-18 23:07:44	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Portal_gun.png Record Number=96326 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#49438	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-020
2020-09-18 23:07:44	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Portal_gun.lnk Record Number=96327 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#49441	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-020
2020-09-19 01:24:07	Memory: process list (pslist)	process created (memory, pslist) System PID=4 | PPID=0 | ImageFileName=System | Threads=151	volatility/windows.pslist.PsList.jsonl#1	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	T-001
2020-09-19 01:24:07	Memory: process list (pslist)	process created (memory, pslist) smss.exe PID=312 | PPID=4 | ImageFileName=smss.exe | Threads=2	volatility/windows.pslist.PsList.jsonl#3	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	T-001
2020-09-19 01:24:09	Memory: process list (pslist)	process created (memory, pslist)	volatility/	C:\Sleuthline\cases\test02\memory\	F-003

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		spoolsv.exe PID=2188 | PPID=616 | ImageFileName=spoolsv.exe | SessionId=0 | Threads=10	windows.pslist.PsList.jsonl#27	DESKTOP-SDN1RPT.mem	
2020-09-19 01:24:09	Memory: network connections (netstat)	network endpoint created (memory, netstat) svchost.exe Proto=TCPv4 | LocalAddr=0.0.0.0 | LocalPort=3389 | ForeignAddr=0.0.0.0 | ForeignPort=0 | State=LISTENING | PID=520	volatility/ windows.netstat.NetStat.jsonl#23	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-011
2020-09-19 01:24:09	Memory: network connections (netscan)	network endpoint created (memory) svchost.exe Proto=TCPv4 | LocalAddr=0.0.0.0 | LocalPort=3389 | ForeignAddr=0.0.0.0 | ForeignPort=0 | State=LISTENING | PID=520	volatility/ windows.netscan.NetScan.jsonl#5	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-014, F-016
2020-09-19 01:24:09	Memory: process list (pslist)	process created (memory, pslist) MsMpEng.exe PID=2404 | PPID=616 | ImageFileName=MsMpEng.exe | SessionId=0 | Threads=9	volatility/ windows.pslist.PsList.jsonl#32	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-023
2020-09-19 01:24:09	Memory: process scan (psscan)	process created (memory, psscan) spoolsv.exe PID=2188 | PPID=616 | ImageFileName=spoolsv.exe | SessionId=0	volatility/ windows.psscan.PsScan.jsonl#106	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	T-024
2020-09-19 01:31:22	Memory: MFT records in memory	file created (memory, MFT record) Record Number=102649 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#84896	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018, T-017
2020-09-19 03:16:05	Memory: process scan (psscan)	process created (memory, psscan) MicrosoftEdge. PID=4592 | PPID=764 | ImageFileName=MicrosoftEdge. | SessionId=1	volatility/ windows.psscan.PsScan.jsonl#71	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015
2020-09-19 03:16:05	Memory: process scan (psscan)	process created (memory, psscan) MicrosoftEdge. PID=4592 | PPID=764 | ImageFileName=MicrosoftEdge. | SessionId=1	volatility/ windows.psscan.PsScan.jsonl#18	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	T-021
2020-09-19 03:17:00	Memory: process scan (psscan)	process created (memory, psscan) winlogon.exe PID=6456 | PPID=4084 | ImageFileName=winlogon.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#132	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015
2020-09-19 03:36:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) RdpIdd Record Number=260 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#23934	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-006, F-016, F-024, T-005
2020-09-19 03:36:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) rdpidd.PNF Record Number=259 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#128240	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-016, T-015
2020-09-19 03:36:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) termkbd.PNF Record Number=255 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#69180	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-016, T-015
2020-09-19 03:36:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) user-not-present-trace-2020-09-18-20-36-23.etl Record Number=256 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#128233	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-016
2020-09-19 03:36:24	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Microsoft-Windows-TerminalServices-PnPDevices%4Admin.evtx Record Number=262 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#23939	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-016, T-015
2020-09-19 03:36:24	Memory: MFT records in memory	file created (memory, MFT record) Record Number=520 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#64444	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018, T-018
2020-09-19 03:36:24	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Recent Record Number=520 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#64445	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018
2020-09-19 03:36:24	Memory: MFT records in memory	file created (memory, MFT record) Record Number=426 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#5594	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-021, T-020
2020-09-19 03:36:26	Memory: scheduled tasks (registry)	scheduled task created (memory) ; scheduled task last run (memory) C:\Windows\Explorer.EXE Action Arguments=/NOUACCHECK | Principal ID=Author | Enabled=true	volatility/ windows.registry.scheduled_tasks.ScheduledTasks.jsonl#32	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-013
2020-09-19 03:36:30	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-2232410529-1445159330-2725690660-500.pckgdep Record Number=22129 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#102754	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-014

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
2020-09-19 03:36:40	Memory: process scan (psscan)	process created (memory, psscan) MicrosoftEdge. PID=860 | PPID=764 | ImageFileName=MicrosoftEdge. | SessionId=3	volatility/ windows.psscan.PsScan.jsonl#158	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-021, F-026, T-005, T-019, T-023
2020-09-19 03:36:41	Memory: MFT records in memory	file created (memory, MFT record) Record Number=24179 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#109281	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-021, T-020
2020-09-19 03:36:43	Memory: MFT records in memory	file created (memory, MFT record) Record Number=23462 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#117560	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018
2020-09-19 03:39:04	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) www.msn[1].xml Record Number=24781 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#29019	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-021
2020-09-19 03:39:26	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) 194.61.24[1].xml Record Number=25584 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#51833	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-021, F-026, I-011, I-014, T-019
2020-09-19 03:39:26	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) 194612~1.XML Record Number=25584 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#112169	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-021
2020-09-19 03:39:26	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) ntp[1].htm Record Number=25585 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#51835	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-021
2020-09-19 03:39:36	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Microsoft.Windows.SecHealthUI_cw5n1h2txyewy Record Number=82883 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#111476	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-009, T-009
2020-09-19 03:39:37	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-2232410529-1445159330-2725690660-500 Record Number=82944 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#107218	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-014, I-007, T-013
2020-09-19 03:39:45	Memory: unloaded kernel modules	kernel module unloaded (memory) WdNisDrv.sys 272684914507776	volatility/ windows.unloadedmodules.UnloadedModules.jsonl#7	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-009, F-023, F-026, I-005, T-009
2020-09-19 03:39:47	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) SECURITYHEALTHHOST.EXE-06344EE9.pf Record Number=24491 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#13226	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-009, T-009
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) coreupdater.exe Record Number=26213 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#56378	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-005, F-006, F-009, F-021, F-024, F-026, I-002, I-017, T-003, T-020
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=26213 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#56376	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-006, F-012, F-024, T-006
2020-09-19 03:40:49	Memory: process list (pslist)	process created (memory, pslist) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3 | Threads=0	volatility/ windows.pslist.PsList.jsonl#50	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-005, I-001, T-003
2020-09-19 03:40:49	Memory: process scan (psscan)	process created (memory, psscan) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3	volatility/ windows.psscan.PsScan.jsonl#167	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-006, F-010, F-012, F-021, F-026, T-006, T-010, T-022
2020-09-19 03:40:49	Memory: logon sessions	process in session (memory) coreupdater.ex Session ID=3 | Process ID=8324	volatility/ windows.sessions.Sessions.jsonl#93	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-006
2020-09-19 03:40:49	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) CHXSMARTSCREEN.EXE-D6E6DEB7.pf Record Number=83165 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#105657	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-006, T-006
2020-09-19 03:46:18	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) loot.lnk Record Number=87445 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#78568	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-006, F-018, F-025, I-004, T-007

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
2020-09-19 03:46:18	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=87445 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#7856 7	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018
2020-09-19 03:47:39	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=87493 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#8292 4	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018
2020-09-19 03:47:39	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Desktop.lnk Record Number=87493 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#8292 5	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-018, I-010, T-018
2020-09-19 05:03:02	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) sysinfo.txt Record Number=260 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#1143 58	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-024
2020-09-19 05:08:15	Memory: logon sessions	process in session (memory) sihost.exe Session ID=2 | Process ID=3232 | User Name=C137/ricksanchez	volatility/ windows.sessions.Sessions.jsonl#53	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-015, T-014
2020-09-19 05:08:16	Memory: logon sessions	process in session (memory) explorer.exe Session ID=2 | Session Type=Console | Process ID=5896 | User Name=C137/ricksanchez	volatility/ windows.sessions.Sessions.jsonl#57	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-001, F-015, T-014
2020-09-19 05:08:16	Memory: process scan (psscan)	process created (memory, psscan) userinit.exe PID=5204 | PPID=6456 | ImageFileName=userinit.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#161	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-007, F-013
2020-09-19 05:08:33	Memory: process scan (psscan)	process created (memory, psscan) SecurityHealth PID=356 | PPID=5896 | ImageFileName=SecurityHealth | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#121	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-013
2020-09-19 05:08:34	Memory: process scan (psscan)	process created (memory, psscan) vmtoolsd.exe PID=7252 | PPID=5896 | ImageFileName=vmtoolsd.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#156	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-013
2020-09-19 05:08:37	Memory: process list (pslist)	process created (memory, pslist) powershell.exe PID=508 | PPID=1380 | ImageFileName=powershell.exe | SessionId=2 | Threads=0	volatility/ windows.pslist.PsList.jsonl#78	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-004, T-004
2020-09-19 05:08:37	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe PID=508 | PPID=1380 | ImageFileName=powershell.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#101	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-007, F-013, F-019, F-026, I-013, I-016, T-008
2020-09-19 05:08:37	Memory: process scan (psscan)	process created (memory, psscan) OneDrive.exe PID=8904 | PPID=5896 | ImageFileName=OneDrive.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#118	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-013
2020-09-19 05:08:37	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) JNKXKDDMCSDYIYL32PRB.temp Record Number=91449 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#7487 5	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-019, F-026
2020-09-19 05:08:37	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=91449 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#1337 83	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-019
2020-09-19 05:08:37	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) 590aee7bdd69b59b.customDestinations-ms Record Number=91449 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#1337 85	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-019
2020-09-19 05:08:40	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Microsoft-Windows-PowerShell%4Operational.evtx Record Number=93203 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#1311 49	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-022
2020-09-19 05:08:43	Memory: process list (pslist)	process created (memory, pslist) powershell.exe PID=3316 | PPID=508 | ImageFileName=powershell.exe | SessionId=2 | Threads=11	volatility/ windows.pslist.PsList.jsonl#81	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-004, I-016, T-004
2020-09-19 05:08:43	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe PID=3316 | PPID=508 | ImageFileName=powershell.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#154	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-007, I-013
2020-09-19 05:08:43	Memory: process scan (psscan)	process created (memory, psscan) conhost.exe	volatility/ windows.psscan.PsScan.jsonl#145	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-007

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		PID=728 | PPID=3316 | ImageFileName=conhost.exe | SessionId=2			
2020-09-19 05:08:44	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) POWERSHELL.EXE-CA1AE517.pf Record Number=93244 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#16979	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-007, T-008
2020-09-19 05:09:13	Memory: process scan (psscan)	process created (memory, psscan) msinfo32.exe PID=7328 | PPID=5896 | ImageFileName=msinfo32.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#152	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-008
2020-09-19 05:09:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) MSINFO32.EXE-C3C668DA.pf Record Number=104666 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#78621	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-008
2020-09-19 05:09:26	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) DESKTOP-SDN1RPT Record Number=262 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#114362	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-024
2020-09-19 05:09:52	Memory: loaded DLLs	DLL loaded (memory) C:\Windows\System32\ntlanman.dll PID=5896 | Process=explorer.exe	volatility/ windows.dllist.DllList.jsonl#936	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-017
2020-09-19 05:09:56	Memory: process scan (psscan)	process created (memory, psscan) FTK Imager.exe PID=6544 | PPID=5896 | ImageFileName=FTK Imager.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#126	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	F-008
2020-09-19 05:09:56	Memory: process list (pslist)	process created (memory, pslist) FTK Imager.exe PID=6544 | PPID=5896 | ImageFileName=FTK Imager.exe | SessionId=2 | Threads=17	volatility/ windows.pslist.PsList.jsonl#87	C:\Sleuthline\cases\test02\memory\ DESKTOP-SDN1RPT.mem	T-002

4. Detailed findings

4.1 System context and scope

Case = single memory image DESKTOP-SDN1RPT.mem (2 GiB, SHA256 AC6EE0E8...67C3), Volatility 3 only. Host DESKTOP-SDN1RPT, Windows 10 x64 build 19041, domain C137.LOCAL (DC CITADEL-DC01, SMB session to 10.42.85.10), IP 10.42.85.115, boot 2020-09-19 01:24:07 UTC, capture 05:10:39 UTC. Users: C137\ricksanchez (RID 1106, Domain Admins, console session 2 from 05:08:15), domain Administrator (RID 500) in session 3 (03:36-03:44) running MicrosoftEdge and suspicious coreupdater.exe (PID 8324, PPID 4008, 03:40:49-03:43:10), Admin profile hive loaded. OS install approx 2020-09-18 06:35 UTC (C:\Users $FN creation). Collection tool FTK Imager (E:\, 05:09:56) noted as noise. Findings F-001 (system description) and F-002 (scope limitations) recorded; timeline T-001, T-002.

F-002: Scope limitation: memory image only - no disk/KAPE artifacts, several Volatility plugins empty or not run

Severity	**Informational**
Confidence	confirmed
Independent sources	memory
Category	context
MITRE ATT&CK	n/a
Period (UTC)	? → ?

Observed facts

The `sources` table contains only Volatility 3 outputs (job 'volatility', status ok, 2026-10-20 14:23:50-14:34:29 UTC). No KAPE/EZTools/EVTX/registry/Prefetch/Amcache/$MFT/$J disk sources exist, and the detections table is empty (no Hayabusa/Chainsaw run). Plugins returning 0-byte output (status 'empty'): windows.registry.userassist, windows.registry.amcache, windows.getservicesids, windows.malware.svcdiff, windows.malware.hollowprocesses, windows.malware.processghosting, windows.malware.indirect_system_calls, windows.malware.skeleton_key_check, windows.orphan_kernel_threads, windows.suspended_threads, windows.symlinkscan, windows.etwpatch, windows.debugregisters, windows.registry.getcellroutine. Not run by design: hashdump, lsadump, cachedump, truecrypt.Passphrase, dumpfiles, pedump (no binaries extracted, no hashes available). Memory timeline covers 2020-09-19 01:24:04 (boot) to 05:10:51 UTC; memory MFT records reach back to 2013 (file system metadata).

Analysis

All conclusions will rest on a single source (the memory image), so the program's two-independent-source rule cannot be satisfied for high/critical findings: they will remain 'probable' unless strictly memory-internal. Execution history prior to boot (01:24 UTC) and process creation events for exited processes not in psscan are unavailable; no file hashes can be computed (dumpfiles/pedump not run). Empty userassist/amcache output may be a plugin limitation (hive pages paged out) rather than absence of data.

Evidence (2)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_image	{"image": "C:\\Sleuthline\\cases\\test02\\memory\\DESKTOP-SDN1RPT.mem", "name": "DESKTOP-SDN1RPT.mem", "bytes": "2147483648", "sha256": "AC6EE0E8C273103B94D776F54192D2C6ACF8303AABBD6A199254030C69F667C3"}	volatility/memory_image.jsonl#1	
	raw.vol_info	{"TreeDepth": "0", "Value": "2020-09-19 05:10:39+00:00", "Variable": "SystemTime"}	volatility/windows.info.Info.jsonl#12	

F-001: System description: DESKTOP-SDN1RPT (Windows 10 build 19041, domain C137.LOCAL) - memory-only case

Severity	**Informational**

Confidence	confirmed
Independent sources	memory
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2020-09-18 06:35:22 → 2020-09-19 05:08:16

Observed facts

Evidence base: a single 2 GiB RAM image DESKTOP-SDN1RPT.mem (2147483648 bytes, SHA256 AC6EE0E8C273103B94D776F54192D2C6ACF8303AABBD6A199254030C69F667C3) processed by Volatility 3 (volatility3 2.28.2, 58 plugins). windows.info: x64 kernel, NtProductWinNt, NtMajor/Minor 10.0, Major/Minor 15.19041 (Windows 10 version 2004 / 20H2 family, build 19041), 2 processors, SystemTime at capture 2020-09-19 05:10:39 UTC. Host name DESKTOP-SDN1RPT (envar COMPUTERNAME), domain member: USERDOMAIN=C137, USERDNSDOMAIN=C137.LOCAL, LOGONSERVER=\\CITADEL-DC01; machine account C137/DESKTOP-SDN1RPT$. Interactive user: C137\ricksanchez (USERPROFILE C:\Users\ricksanchez, console session 2, logon processes from 2020-09-19 05:08:15 UTC); domain SID S-1-5-21-2232410529-1445159330-2725690660, user token RID -1106 observed in 38 processes and Domain Admins (-512) group membership in tokens. Loaded NTUSER hives in memory: C:\Users\Admin\ntuser.dat, C:\Users\Administrator\ntuser.dat (+UsrClass.dat), C:\Users\ricksanchez\ntuser.dat. Profile folder 'ricksanchez' $FN created 2020-09-18 22:44:11 UTC; C:\Users created 2020-09-18 06:35:22 UTC (approximate OS install date). Network: single IPv4 10.42.85.115; an established SMB session to 10.42.85.10:445 (probably the DC) and multiple HTTPS connections to Microsoft/CDN ranges. Last boot: System/smss created 2020-09-19 01:24:07 UTC. Additional session 3 (03:36:40-03:44:52 UTC) containing coreupdater.exe (PID 8324, PPID 4008, 03:40:49-03:43:10) – analysed in later steps. Time zone of the machine not available (meta machine_active_time_bias_min = None; no SYSTEM hive parsing); all times UTC.

Analysis

The machine is a domain-joined Windows 10 workstation of domain C137.LOCAL, primary user ricksanchez (member of Domain Admins according to the token SIDs - an elevated account used interactively). The OS appears to have been installed on 2020-09-18 (lab-like / recently built host). The presence of 'Admin' and 'Administrator' profiles loaded in memory indicates other logons since boot (session 1/3 activity) that must be examined. Identity/version information is consistent across windows.info, envars, getsids and hivelist (one source: the memory image).

Evidence (18)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_image	{"image": "C:\\Sleuthline\\cases\\test02\\memory\\DESKTOP-SDN1RPT.mem", "name": "DESKTOP-SDN1RPT.mem", "bytes": "2147483648", "sha256": "AC6EE0E8C273103B94D776F54192D2C6ACF8303AABBD6A199254030C69F667C3"}	volatility/memory_image.jsonl#1	
	raw.vol_info	{"TreeDepth": "0", "Value": "15.19041", "Variable": "Major/Minor"}	volatility/windows.info.Info.jsonl#9	
	raw.vol_info	{"TreeDepth": "0", "Value": "2020-09-19 05:10:39+00:00", "Variable": "SystemTime"}	volatility/windows.info.Info.jsonl#12	
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x146a4e03120", "PID": "616", "Process": "services.exe", "Value": "DESKTOP-SDN1RPT", "Variable": "COMPUTERNAME"}	volatility/windows.envars.Envars.jsonl#5	
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x215a22b1bd0", "PID": "3232", "Process": "sihost.exe", "Value": "\\\\CITADEL-DC01", "Variable": "LOGONSERVER"}	volatility/windows.envars.Envars.jsonl#565	
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x215a22b1bd0", "PID": "3232", "Process": "sihost.exe", "Value": "C137.LOCAL", "Variable": "USERDNSDOMAIN"}	volatility/windows.envars.Envars.jsonl#585	
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x215a22b1bd0", "PID": "3232", "Process": "sihost.exe", "Value": "ricksanchez", "Variable": "USERNAME"}	volatility/windows.envars.Envars.jsonl#588	
	raw.vol_getsids	{"TreeDepth": "0", "PID": "3232", "Process": "sihost.exe", "SID": "S-1-5-21-2232410529-1445159330-2725690660-1106"}	volatility/windows.getsids.GetSIDs.jsonl#638	

UTC	Artifact	Detail	Reference	Original source
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/windows.getsids.GetSIDs.jsonl#613	
	raw.vol_hivelist	{"TreeDepth": "0", "File output": "Disabled", "FileFullPath": "\\??\\C:\\Users\\Admin\\ntuser.dat", "Offset": "227618253631488"}	volatility/windows.registry.hivelist.HiveList.jsonl#13	
	raw.vol_hivelist	{"TreeDepth": "0", "File output": "Disabled", "FileFullPath": "\\??\\C:\\Users\\Administrator\\ntuser.dat", "Offset": "227618280730624"}	volatility/windows.registry.hivelist.HiveList.jsonl#14	
	raw.vol_hivelist	{"TreeDepth": "0", "File output": "Disabled", "FileFullPath": "\\??\\C:\\Users\\ricksanchez\\ntuser.dat", "Offset": "227618280112128"}	volatility/windows.registry.hivelist.HiveList.jsonl#16	
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "10.42.85.10", "ForeignPort": "445", "LocalAddr": "10.42.85.115", "LocalPort": "50957", "Offset": "209519101090720", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/windows.netscan.NetScan.jsonl#88	
2020-09-19 05:08:16	Memory: logon sessions	process in session (memory) explorer.exe Session ID=2 | Session Type=Console | Process ID=5896 | User Name=C137/ricksanchez	volatility/windows.sessions.Sessions.jsonl#57	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-18 22:44:11	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) ricksanchez Record Number=87848 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/windows.mftscan.MFTScan.jsonl#10758	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-18 06:35:22	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Users Record Number=1475 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/windows.mftscan.MFTScan.jsonl#140290	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "3232", "Process": "sihost.exe", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/windows.getsids.GetSIDs.jsonl#650	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Administrator", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-21-2232410529-1445159330-2725690660-500"}	volatility/windows.getsids.GetSIDs.jsonl#600	

4.2 Triage of automatic detections

No automatic detections exist (detections=0, hash_hits/hashset=0; no EVTX in this memory-only case). Memory anomaly plugins were triaged as substitute alerts. True positives: F-003 (critical, probable) reflective PE in spoolsv.exe PID 2188 (malfind MZ RWX 0x1840000, ldrmodules unlinked) holding a 0x1410 handle on lsass.exe PID 656 and cwd C:\Users\mortysmith\Desktop -> injection + credential access; F-004 (high, probable) powershell.exe 508->3316 (ricksanchez, Domain Admin) at 05:08:37-43 after logon, unlinked MZ RWX region 0x10c6c070000 and RWX-started thread TID 3892; F-005 (high, probable) service "coreupdater" AUTO_START + coreupdater.exe (MFT 26213 created 03:40:00) run as domain Administrator PID 8324 03:40:49-03:43:10. IOCs: coreupdater service/file, 203.78.103.109:443 (2 established sessions, owner unresolved). FPs listed in note (MsMpEng malfind, csrss/WmiPrvSE/audiodg pebmasquerade, generic suspicious_threads, FTK/autorunsc responder tools).

F-003: Code injection in spoolsv.exe (PID 2188): RWX private MZ region unlinked from loader lists + handle to lsass.exe with VM_READ access

Severity	**Critical**
Confidence	probable
Independent sources	memory
Category	execution
MITRE ATT&CK	T1055.001, T1620, T1003.001
Period (UTC)	2020-09-19 01:24:09 → 2020-09-19 01:24:09

Observed facts

Facts (memory image DESKTOP-SDN1RPT.mem, Volatility 3):
- windows.malfind: spoolsv.exe PID 2188 (PPID 616 services.exe, created 2020-09-19 01:24:09 UTC, session 0, S-1-5-18 / System Mandatory Level) has a VadS private region at 0x1840000 (Start VPN 25427968) with PAGE_EXECUTE_READWRITE whose content starts with a PE header "4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 b8 00..." (MZ).
- windows.ldrmodules: the same base 25427968 (0x1840000) is InLoad=false / InInit=false / InMem=false with no mapped

path, i.e. a PE image in memory not registered by the Windows loader (reflectively loaded).
- windows.handles: spoolsv.exe PID 2188 holds a Process handle to lsass.exe PID 656 with GrantedAccess 5136 = 0x1410 (PROCESS_QUERY_INFORMATION | PROCESS_VM_READ | PROCESS_QUERY_LIMITED_INFORMATION) (row 14206), a File handle (access 0x100020, typical current-directory handle) to \Device\HarddiskVolume3\Users\mortysmith\Desktop (row 14195), plus \Device\Afd\Endpoint socket handles.
- No automatic detections (Hayabusa/Chainsaw/Zircolite/Defender) exist in this case: detections table empty, no EVTX collected.

Analysis

A full PE in RWX private memory, invisible to the PEB loader lists, inside a SYSTEM service process is the classic signature of reflective DLL injection / process migration (Meterpreter "migrate", Cobalt Strike inject, etc.). spoolsv.exe is a common migration target for SYSTEM-level implants. The 0x1410 handle on lsass.exe is exactly the access mask used by Mimikatz sekurlsa / credential-dumping tooling; spoolsv has no legitimate need to read lsass memory. A working directory under C:\Users\mortysmith\Desktop for a service started at boot is abnormal and suggests the injected payload/operator shell changed directory into the mortysmith profile (to pivot). Classified TRUE POSITIVE. All evidence comes from the single memory image (one source), hence 'probable' under the evidence policy, although three independent plugins agree. Network attribution of the injected code is not possible because netscan/netstat Owner/PID are empty; established TCP sessions to 203.78.103.109:443 (local ports 50875, 50972) are C2 candidates to be pivoted later.

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "36", "Disasm": "\"\n0x1840000:\tpop\tr10\n0x1840002:\tnop\t\n0x1840003:\tadd\tbyte ptr [rbx], al\n0x1840005:\tadd\tbyte ptr [rax], al\n0x1840007:\tadd\tbyte ptr [rax + rax], al\n0x184000a:\tadd\tbyte ptr [rax], al\"", "End VPN": "25575423", "File output": "Disabled", "Hexdump": "4d 5a 90 00 03 00 00 00 04 00 00 0	volatility/windows.malware.malfind.Malfind.jsonl#1	
	raw.vol_ldrmodules	{"TreeDepth": "0", "Base": "25427968", "InInit": "false", "InLoad": "false", "InMem": "false", "Pid": "2188", "Process": "spoolsv.exe"}	volatility/windows.malware.ldrmodules.LdrModules.jsonl#32	
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "5136", "HandleValue": "1964", "Name": "lsass.exe Pid 656", "Offset": "209519057219712", "PID": "2188", "Process": "spoolsv.exe", "Type": "Process"}	volatility/windows.handles.Handles.jsonl#14206	
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1048608", "HandleValue": "1912", "Name": "\\Device\\HarddiskVolume3\\Users\\mortysmith\\Desktop", "Offset": "209519093754240", "PID": "2188", "Process": "spoolsv.exe", "Type": "File"}	volatility/windows.handles.Handles.jsonl#14195	
2020-09-19 01:24:09	Memory: process list (pslist)	process created (memory, pslist) spoolsv.exe PID=2188 | PPID=616 | ImageFileName=spoolsv.exe | SessionId=0 | Threads=10	volatility/windows.pslist.PsList.jsonl#27	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem

F-005: Suspicious auto-start service "coreupdater" and coreupdater.exe executed as domain Administrator (PID 8324, 03:40:49-03:43:10 UTC)

Severity	**High**
Confidence	probable
Independent sources	memory
Category	persistence
MITRE ATT&CK	T1543.003, T1036.005, T1204.002
Period (UTC)	2020-09-19 03:40:00 → 2020-09-19 03:40:49

Observed facts

Facts (Volatility 3):
- svcscan: service Name/Display "coreupdater", SERVICE_WIN32_OWN_PROCESS, SERVICE_AUTO_START, SERVICE_STOPPED at capture, binary path not recovered (rows 20, 679, 2906, 3548).
- pslist/psscan: coreupdater.exe PID 8324, PPID 4008 (parent absent), session 3, created 2020-09-19 03:40:49, exited 03:43:10 UTC.
- getsids PID 8324: S-1-5-21-2232410529-1445159330-2725690660-500 (domain Administrator) with Domain Admins, Schema Admins, Enterprise Admins, Group Policy Creator Owners, High Mandatory Level.
- mftscan: coreupdater.exe MFT record 26213, $FILE_NAME Created/Modified 2020-09-19 03:40:00, Accessed 03:40:01, Updated 03:40:04 UTC, link count 2 (parent directory not resolved in the plugin output).

Analysis

A binary named to imitate an updater, written to disk 49 s before being run interactively by the domain Administrator in a short session (03:36-03:44), and registered as an auto-start own-process service, is consistent with malware drop + service persistence (later executed as SYSTEM at boot would explain a SYSTEM implant such as the one injected into spoolsv.exe, F-003 - hypothesis). Not a known Windows service. Classified TRUE POSITIVE / to dig into (path, hash and ImagePath to be recovered from filescan/registry in later steps). Single source (memory) => probable.

Evidence (3)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "coreupdater", "Name": "coreupdater", "Offset": "1402927956048", "Order": "662", "Start": "SERVICE_AUTO_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_WIN32_OWN_PROCESS"}	volatility/ windows.svcscan.SvcScan .jsonl#20	
2020-09-19 03:40:49	Memory: process list (pslist)	process created (memory, pslist) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3 | Threads=0	volatility/ windows.pslist.PsList.json l#50	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) coreupdater.exe Record Number=26213 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTSca n.jsonl#56378	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

F-004: Injected PE and RWX-started thread in powershell.exe PID 3316 (C137\ricksanchez, Domain Admin) spawned 21 s after logon by orphaned powershell.exe PID 508

Severity	**High**
Confidence	probable
Independent sources	memory
Category	execution
MITRE ATT&CK	T1059.001, T1055, T1620
Period (UTC)	2020-09-19 05:08:37 → 2020-09-19 05:08:43

Observed facts

Facts (Volatility 3):
- pslist/psscan: powershell.exe PID 508 (PPID 1380, parent not present in pslist/psscan) created 2020-09-19 05:08:37 UTC in session 2, exited 05:08:43; it spawned powershell.exe PID 3316 at 05:08:43 (still running at capture, child conhost.exe PID 728). User logon of session 2 (userinit 5204 / explorer 5896) at 05:08:16.
- getsids: PIDs 508 and 3316 run as S-1-5-21-2232410529-1445159330-2725690660-1106 (ricksanchez) with Domain Admins (-512).
- cmdline: Args empty for both PIDs (PEB command line not resident); cmdscan of conhost 728: CommandCount 0.
- malfind PID 3316: VadS RWX private region at 0x10c6c070000 starting with MZ header (row 6); further RWX regions 0x10c6c000000, 0x10c6bfc0000 (zero-filled), 0x7df4f5a80000/0x7df4f5a90000 (data).

- ldrmodules: base 0x10c6c070000 (1152863633408) InLoad/InInit/InMem=false, no mapped path (row 176).
- suspicious_threads: TID 3892 starts at 0x10c6bb00000 in a non-file-backed VAD with unexpected PAGE_EXECUTE_READWRITE initial protection (rows 43-44).

Analysis

PowerShell hosts .NET JIT code which produces some RWX false positives, but an unlinked MZ image plus a thread whose start address lies in an RWX non-image allocation is consistent with in-memory payload staging (e.g. PowerShell download cradle / reflective loader, Invoke-ReflectivePEInjection, Meterpreter/Empire stager). The execution chain (powershell launching a second powershell 21 s after logon, parent PID 1380 no longer existing) is consistent with a logon-triggered persistence mechanism (Run key, startup folder, WMI or scheduled task) - to be pivoted in the persistence step. Classified TRUE POSITIVE / to dig into. Single source (memory) => probable. Note: the capture was taken at 05:10:39, about 2 min later; analyst activity cannot be fully excluded but FTK Imager runs from E:\ and is not the parent.

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "36", "Disasm": "\"\n0x10c6c070000:\tpop\tr10\n0x10c6c070002:\tnop\t\n0x10c6c070003:\tadd\tbyte ptr [rbx], al\n0x10c6c070005:\tadd\tbyte ptr [rax], al\n0x10c6c070007:\tadd\tbyte ptr [rax + rax], al\n0x10c6c07000a:\tadd\tbyte ptr [rax], al\"", "End VPN": "1152863780863", "File output": "Disabled", "Hexdump": "4d 5a	volatility/windows.malware.malfind.Malfind.jsonl#6	
	raw.vol_ldrmodules	{"TreeDepth": "0", "Base": "1152863633408", "InInit": "false", "InLoad": "false", "InMem": "false", "Pid": "3316", "Process": "powershell.exe"}	volatility/windows.malware.ldrmodules.LdrModules.jsonl#176	
	raw.vol_suspicious_threads	{"TreeDepth": "0", "Address": "1152857931776", "Context": "Win32Start", "Note": "VAD at base address (0x10c6bb00000) hosting this thread has an unexpected starting protection PAGE_EXECUTE_READWRITE", "PID": "3316", "Process": "powershell.exe", "TID": "3892", "VAD Path": "<Non-File Backed Region>"}	volatility/windows.malware.suspicious_threads.SuspiciousThreads.jsonl#44	
2020-09-19 05:08:43	Memory: process list (pslist)	process created (memory, pslist) powershell.exe PID=3316 | PPID=508 | ImageFileName=powershell.exe | SessionId=2 | Threads=11	volatility/windows.pslist.PsList.jsonl#81	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-19 05:08:37	Memory: process list (pslist)	process created (memory, pslist) powershell.exe PID=508 | PPID=1380 | ImageFileName=powershell.exe | SessionId=2 | Threads=0	volatility/windows.pslist.PsList.jsonl#78	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem

4.3 Program execution

Program execution reconstructed from memory only. Malicious/suspicious executions: (1) coreupdater.exe PID 8324 (F-006, high/probable) dropped 03:40:00 (MFT 26213, SI=FN, no timestomp), executed 03:40:49 ($SI access + ChxSmartScreen pf same second -> Explorer launch) in session 3 by C137\Administrator RID 500 with High IL and Domain/Enterprise/Schema Admins, exited 03:43:10; auto-start service "coreupdater" (F-005). Administrator session 3 started 03:36:23 with RDP display driver artefacts (RDP likely). loot.lnk created 03:46:18 (IOC). (2) PowerShell chain 508->3316 for ricksanchez at 05:08:37-43 (F-007, high/probable): parent PPID 1380 missing, no command line / console history, first-ever POWERSHELL pf created 05:08:44, socket handle; complements F-004 (injected PE). (3) spoolsv injection already F-003. Context F-008: msinfo32, FTK Imager (E:), autorunsc (Zone.Identifier referrer James\Downloads\SysinternalsSuite.zip) = responder tools. Timeline T-005..T-008 added.

F-006: coreupdater.exe execution details: launched interactively in session 3 by domain Administrator (High IL, Domain/Enterprise/Schema Admins) at 03:40:49, 49 s after being written to disk

Severity	**High**
Confidence	probable
Independent sources	memory
Category	execution
MITRE ATT&CK	T1204.002, T1543.003, T1078.002, T1021.001

Period (UTC)	2020-09-19 03:36:23 → 2020-09-19 03:46:18

Observed facts

Facts (all UTC, all from memory image DESKTOP-SDN1RPT.mem):
- psscan: coreupdater.exe PID 8324, PPID 4008, SessionId 3, CreateTime 2020-09-19 03:40:49, ExitTime 03:43:10 (runtime ~2 min 21 s). Parent PID 4008 is not present in pslist/psscan (exited / not recoverable). Command line not recoverable (process exited; cmdline plugin has no entry).
- getsids PID 8324: S-1-5-21-2232410529-1445159330-2725690660-500 (C137\Administrator), groups Domain Admins (-512), Enterprise Admins (-519), Schema Admins (-518), Group Policy Creator Owners (-520), integrity S-1-16-12288 (High Mandatory Level) -> elevated token.
- sessions plugin: PID 8324 in session 3 (same session as MicrosoftEdge PID 860, 03:36:40-03:44:52).
- memory-resident $MFT record 26213 (coreupdater.exe / 8.3 COREUP~1.EXE): $FN Created/Modified 03:40:00, $FN Accessed 03:40:01, $FN MFT-changed 03:40:04; $SI Created/Modified 03:40:00, $SI Accessed 03:40:49 and $SI Updated 03:40:49 = exact process creation time (consistent with the image being opened for execution). $SI and $FN creation identical -> no sign of timestomping.
- Prefetch file CHXSMARTSCREEN.EXE-D6E6DEB7.pf (MFT 83165) created at 03:40:49, i.e. SmartScreen (Explorer shell check) ran at the same second as coreupdater.exe start -> consistent with a user double-click launch from Explorer in session 3.
- Service "coreupdater" (svcscan, SERVICE_WIN32_OWN_PROCESS, SERVICE_AUTO_START, SERVICE_STOPPED at capture; binary path not resolved by svcscan) - see F-005.
- No COREUPDATER.EXE-*.pf entry was found in the memory-resident MFT records and no Zone.Identifier ADS for record 26213 in vol_ads (only partial MFT in memory: absence not conclusive). Full path of the file not resolvable from mftscan (no parent reference exported) and not present in filescan.
- Session 3 context: Administrator profile objects (UsrClass.dat transaction files, Desktop/Documents folders, Administrator SID crypto folders) were created 03:36:22-03:36:27, together with RdpIdd / IddCx0102 / termkbd.PNF / rdpidd.PNF at 03:36:23 (first interactive logon of Administrator on this host, Remote Desktop indirect display driver artefacts).
- 03:46:18: shortcut loot.lnk created (MFT 87445), 3 min after coreupdater exit, during the Administrator session.

Analysis

coreupdater.exe was dropped at 03:40:00, ~3.5 min after the first logon of the domain Administrator (very likely an RDP session, based on RdpIdd/termkbd artefacts created at logon time - hypothesis, no TerminalServices EVTX available), then executed at 03:40:49 with an elevated Domain/Enterprise Admin token. The ChxSmartScreen prefetch at the same second suggests a manual launch from Explorer (parent PID 4008, likely explorer.exe of session 3, now gone). It ran 2 min 21 s and a service of the same name set to AUTO_START exists, so coreupdater.exe very probably installed itself (or was installed) as a persistence service. The creation of "loot.lnk" at 03:46:18 suggests the attacker then opened a file/folder named "loot" (staging of collected data) - to be examined in the collection step. All evidence comes from the memory image (one source); disk Prefetch/Amcache/EVTX would be needed to confirm, hence 'probable'. Hash of the binary unavailable (no Amcache, no dumped file).

Evidence (12)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:40:49	Memory: process scan (psscan)	process created (memory, psscan) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3	volatility/ windows.psscan.PsScan.jsonl#167	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Administrator", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-500"}	volatility/ windows.getsids.GetSIDs.jsonl#619	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID":	volatility/	

UTC	Artifact	Detail	Reference	Original source
		"8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	windows.getsids.GetSIDs.jsonl#632	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Enterprise Admins", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-519"}	volatility/windows.getsids.GetSIDs.jsonl#634	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "High Mandatory Level", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-16-12288"}	volatility/windows.getsids.GetSIDs.jsonl#637	
2020-09-19 03:40:49	Memory: logon sessions	process in session (memory) coreupdater.ex Session ID=3 | Process ID=8324	volatility/windows.sessions.Sessions.jsonl#93	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=26213 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/windows.mftscan.MFTScan.jsonl#56376	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) coreupdater.exe Record Number=26213 | Attribute Type=FILE_NAME | MFT Type=File	volatility/windows.mftscan.MFTScan.jsonl#56378	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-19 03:40:49	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) CHXSMARTSCREEN.EXE-D6E6DEB7.pf Record Number=83165 | Attribute Type=FILE_NAME | MFT Type=File	volatility/windows.mftscan.MFTScan.jsonl#105657	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-19 03:36:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) RdpIdd Record Number=260 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/windows.mftscan.MFTScan.jsonl#23934	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-19 03:46:18	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) loot.lnk Record Number=87445 | Attribute Type=FILE_NAME | MFT Type=File	volatility/windows.mftscan.MFTScan.jsonl#78568	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "coreupdater", "Name": "coreupdater", "Offset": "1402927956048", "Order": "662", "Start": "SERVICE_AUTO_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_WIN32_OWN_PROCESS"}	volatility/windows.svcscan.SvcScan.jsonl#20	

F-007: PowerShell executed at ricksanchez logon by a non-recoverable parent (PPID 1380); first PowerShell Prefetch on the host created 05:08:44; command line not recoverable

Severity	**High**
Confidence	probable
Independent sources	memory
Category	execution
MITRE ATT&CK	T1059.001, T1055, T1547.001, T1546.003
Period (UTC)	2020-09-19 05:08:16 → 2020-09-19 05:08:44

Observed facts

Facts (UTC, memory image):
- psscan: powershell.exe PID 508, PPID 1380, session 2, created 05:08:37, exited 05:08:43; powershell.exe PID 3316, PPID 508, session 2, created 05:08:43, still running at capture (05:10:39); conhost.exe PID 728 (PPID 3316) created 05:08:43. PPID 1380 is absent from pslist/psscan/psxview (exited before capture, not carved). Logon chain: userinit.exe 5204 05:08:16-05:08:42, explorer.exe 5896 05:08:16.
- pstree Audit path for PID 3316: \Device\HarddiskVolume3\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (legitimate binary location). cmdline plugin returns empty Args for PIDs 508 and 3316 (PEB not readable/paged); cmdscan for conhost 728: CommandCount 0, consoles: no information -> no interactive commands typed.
- getsids: PIDs 508 and 3316 run as S-1-5-21-2232410529-1445159330-2725690660-1106 (C137\ricksanchez), Domain Admins (-512), Medium Mandatory Level (S-1-16-8192).

- Memory-resident $MFT: POWERSHELL.EXE-CA1AE517.pf (record 93244) created 05:08:44 - the only PowerShell prefetch file observed, i.e. first recorded execution of powershell.exe on this host (prefetch timestamps in mftscan for the whole period 2020-09-18 04:58 -> 2020-09-19 05:10 show no earlier POWERSHELL pf).
- handles PID 3316: \Device\Afd\Endpoint (network socket), Microsoft.PowerShell.Commands.Utility.dll, Windows Defender policy keys (MACHINE\SOFTWARE\POLICIES\MICROSOFT\WINDOWS DEFENDER\POLICY MANAGER). Injected MZ region and RWX thread recorded in F-004.

Analysis

A two-stage powershell.exe chain (stager 508 exits after spawning 3316, the classic pattern of encoded/download-cradle launchers such as Empire/Cobalt Strike PowerShell stagers) started 21 s after the interactive logon of a Domain Admin, without user interaction (no console history) and from a parent that is not userinit/explorer. Likely launch mechanisms (hypotheses, not provable from memory alone): logon-triggered persistence (Run key, WMI subscription, logon script/GPO, scheduled task not seen in the registry task cache) or remote execution (WMI/WinRM). The injected PE and socket handle in PID 3316 indicate an in-memory implant (C2). Corroboration from disk (Prefetch content, PowerShell 4104/400 EVTX, Sysmon) is not available in this case -> 'probable'.

Evidence (11)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 05:08:37	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe PID=508 | PPID=1380 | ImageFileName=powershell.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#101	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:43	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe PID=3316 | PPID=508 | ImageFileName=powershell.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#154	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:43	Memory: process scan (psscan)	process created (memory, psscan) conhost.exe PID=728 | PPID=3316 | ImageFileName=conhost.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#145	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_cmdline	{"TreeDepth": "0", "PID": "508", "Process": "powershell.exe"}	volatility/ windows.cmdline.CmdLine.jsonl#78	
	raw.vol_cmdline	{"TreeDepth": "0", "PID": "3316", "Process": "powershell.exe"}	volatility/ windows.cmdline.CmdLine.jsonl#81	
	raw.vol_cmdscan	{"TreeDepth": "1", "ConsoleInfo": "138492641775368", "Data": "0", "PID": "728", "Process": "conhost.exe", "Property": "_COMMAND_HISTORY.CommandCount"}	volatility/ windows.cmdscan.CmdScan.jsonl#4	
	raw.vol_getsids	{"TreeDepth": "0", "PID": "3316", "Process": "powershell.exe", "SID": "S-1-5-21-2232410529-1445159330-2725690660-1106"}	volatility/ windows.getsids.GetSIDs.jsonl#1098	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Medium Mandatory Level", "PID": "3316", "Process": "powershell.exe", "SID": "S-1-16-8192"}	volatility/ windows.getsids.GetSIDs.jsonl#1113	
2020-09-19 05:08:44	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) POWERSHELL.EXE-CA1AE517.pf Record Number=93244 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#16979	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1442207", "HandleValue": "2368", "Name": "\\Device\\Afd\\Endpoint", "Offset": "209519091203248", "PID": "3316", "Process": "powershell.exe", "Type": "File"}	volatility/ windows.handles.Handles.jsonl#28137	
2020-09-19 05:08:16	Memory: process scan (psscan)	process created (memory, psscan) userinit.exe PID=5204 | PPID=6456 | ImageFileName=userinit.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#161	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

F-008: Context: responder/collection tool executions (msinfo32, FTK Imager from E:\, Sysinternals autorunsc) and benign program execution baseline

Severity	Informational
Confidence	confirmed

Independent sources	memory
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2018-04-02 18:27:58 → 2020-09-19 05:09:56

Observed facts

- msinfo32.exe PID 7328 (PPID explorer 5896, session 2, ricksanchez) created 05:09:13; MSINFO32.EXE-C3C668DA.pf created 05:09:23.
- FTK Imager.exe PID 6544 "E:\FTK Imager\FTK Imager.exe" created 05:09:56 (\Device\HarddiskVolume5), prefetch FTK IMAGER.EXE-E67CC266.pf 05:10:06; ShimCache (memory) entries SIGN.MEDIA=11692A28 FTK Imager\FTK Imager.exe / ADIso.exe / adencrypt_gui.exe (removable media).
- ShimCache (memory) SIGN.MEDIA=3ACFB0 autorunsc64.exe (LastModified 2020-09-13 03:22:55); ADS records 224/225 autorunsc64.exe/autorunsc.exe Zone.Identifier ZoneId=3 ReferrerUrl=C:\Users\James\Downloads\SysinternalsSuite.zip (responder media, user "James" is not a local profile).
- Baseline prefetch creations in memory MFT: OneDriveSetup/FileSyncConfig 03:37, SecurityHealthHost 03:39:47, RuntimeBroker, BackgroundTransferHost, dllhost - standard Windows.
- Running processes outside C:\Windows: vmtoolsd.exe (VMware Tools), OneDrive (ricksanchez AppData), FTK Imager. No executables observed running from Temp/Downloads/Public/ProgramData other than Windows Defender platform binaries.

Analysis

msinfo32 (05:09:13) immediately precedes FTK Imager (05:09:56) and is most likely part of the responder's triage; FTK Imager and autorunsc are incident-response tools run from removable media. Excluded from malicious activity. The responder logged on with the Domain Admin account ricksanchez, which coincided with the PowerShell chain of F-007.

Evidence (6)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 05:09:13	Memory: process scan (psscan)	process created (memory, psscan) msinfo32.exe PID=7328 | PPID=5896 | ImageFileName=msinfo32.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#152	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:09:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) MSINFO32.EXE-C3C668DA.pf Record Number=104666 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#78621	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:09:56	Memory: process scan (psscan)	process created (memory, psscan) FTK Imager.exe PID=6544 | PPID=5896 | ImageFileName=FTK Imager.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#126	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2018-04-02 18:27:58	Memory: ShimCache	ShimCache: file modified (memory) SIGN.MEDIA=11692A28 FTK Imager\FTK Imager.exe Order=4	volatility/ windows.shimcachemem.ShimcacheMem.jsonl#5	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-13 03:22:55	Memory: ShimCache	ShimCache: file modified (memory) SIGN.MEDIA=3ACFB0 autorunsc64.exe Order=8	volatility/ windows.shimcachemem.ShimcacheMem.jsonl#9	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_ads	{"TreeDepth": "0", "ADS Filename": "Zone.Identifier", "Filename": "autorunsc64.exe", "Hexdump": "5b 5a 6f 6e 65 54 72 61 6e 73 66 65 72 5d 0d 0a 5a 6f 6e 65 49 64 3d 33 0d 0a 52 65 66 65 72 72 65 72 55 72 6c 3d 43 3a 5c 55 73 65 72 73 5c 4a 61 6d 65 73 5c 44 6f 77 6e 6c 6f 61 64 73 5c 53 79 73 69 6e 74 65 72 6e 61 6c 73 53 75 69 74 65 2e 7a 69 70 0	volatility/ windows.mftscan.ADS.jsonl#13	

4.4 Memory (RAM)

I reviewed the memory image (Volatility 3 only) for this step. It covered network, services, scheduled tasks, drivers and unloaded modules, DLLs, privileges, mutants, ADS, console history, psxview, suspicious threads and direct syscalls. New findings: F-009

(high/probable, defense evasion): Defender's network inspection driver WdNisDrv.sys was unloaded at 03:39:45 UTC. Nine seconds earlier, at 03:39:36, the Administrator (RID 500) opened the Windows Security app; the SECURITYHEALTHHOST prefetch file appeared at 03:39:47. At capture time WdNisSvc and WdNisDrv were STOPPED. coreupdater.exe was written to disk 15 s after the unload, at 03:40:00. F-010 (high/probable): coreupdater.exe (PID 8324) was the only process with SeDebugPrivilege enabled at runtime (Present,Enabled without the Default flag). F-011 (medium/possible): there were two ESTABLISHED sessions to 203.78.103.109:443 from local ports 50875 and 50972; netscan and netstat give no owning process. The host also had an ESTABLISHED SMB session to the DC on 10.42.85.10:445 and an RDP listener on 3389 (svchost PID 520). The other external peers are Microsoft/CDN. Added timeline event T-009, IOC I-005 (WdNisDrv stopped) and IOC I-006 (mutant e2vgmtzqt3xcbq.mca, no owner found). Checked and ruled out: scheduled tasks are benign (sc config upnphost, usoclient, Explorer /NOUACCHECK, gpupdate). No service runs a binary from a user or Temp folder; the coreupdater service has an empty binary path in svcscan. ad_driver.sys in ricksanchez\AppData\Local\Temp is most likely the FTK Imager driver (responder tool). The odd psxview rows (pslist false / psscan true) are stale duplicate copies of live processes. The direct-syscall and non-file-backed thread hits are in system and browser processes and look benign. No cmd history was recovered (conhost 728 has CommandCount 0). Zone.Identifier ADS exist only for Edge downloads and the Sysinternals tools; none for coreupdater.exe. The DLL lists contain no unexpected DLL from a user folder (only OneDrive and FTK Imager from E:).

F-009: Microsoft Defender Network Inspection driver (WdNisDrv.sys) unloaded at 03:39:45 while domain Administrator had Windows Security open, 15 s before coreupdater.exe was dropped

Severity	**High**
Confidence	probable
Independent sources	memory
Category	defense evasion
MITRE ATT&CK	T1562.001
Period (UTC)	2020-09-19 03:39:36 → 2020-09-19 03:40:00

Observed facts

Facts (memory image DESKTOP-SDN1RPT.mem, Volatility 3):
- windows.unloadedmodules: WdNisDrv.sys (Microsoft Defender Antivirus Network Inspection System driver), range 0xF8...(272684914507776-272684914601984), unloaded at 2020-09-19 03:39:45 UTC. It is the only unload in session time apart from boot-time drivers (01:24:07), WudfRd/IndirectKmd at 03:52:13 (RDP display/UMDF teardown) and EhStorClass at 05:08:58.
- windows.svcscan: WdNisDrv = SERVICE_DEMAND_START / SERVICE_STOPPED and WdNisSvc (Network Inspection Service) = SERVICE_STOPPED at capture time, whereas WinDefend (MsMpEng.exe PID 2404, started 01:24:09) and WdFilter are still RUNNING.
- windows.mftscan, in the profile of C137\Administrator (SID S-1-5-21-2232410529-1445159330-2725690660-500, session 3): package folder Microsoft.Windows.SecHealthUI_cw5n1h2txyewy (MFT 82883) created 03:39:36 (first launch of the Windows Security app by this user), Prefetch SECURITYHEALTHHOST.EXE-06344EE9.pf (MFT 24491) created 03:39:47.
- Sequence: SecHealthUI opened 03:39:36 -> WdNisDrv unloaded 03:39:45 -> SecurityHealthHost pf 03:39:47 -> coreupdater.exe written 03:40:00 (MFT 26213) -> executed 03:40:49 (PID 8324).

Analysis

Interpretation: the tight sequence (Windows Security UI opened by the domain Administrator, NIS driver unloaded 9 s later, payload dropped 15 s after) is consistent with an operator interactively turning off Defender real-time protection (which stops WdNisSvc/WdNisDrv) before downloading/dropping the coreupdater.exe payload so that it would not be detected. Alternatives: WdNisDrv can also be unloaded by Defender platform/engine updates or idle stop of the demand-start NIS service; this cannot be excluded without the Microsoft-Windows-Windows Defender/Operational log (EID 5001/5004/5007) or the registry value HKLM\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection\DisableRealtimeMonitoring, none of which is available (memory-only case). All evidence comes from a single source (memory image), hence 'probable'.

Evidence (6)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:39:45	Memory: unloaded kernel modules	kernel module unloaded (memory) WdNisDrv.sys 272684914507776	volatility/ windows.unloadedmodules.UnloadedModules.jsonl#7	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "Microsoft Defender Antivirus Network Inspection System Driver", "Name": "WdNisDrv", "Offset": "1402926010544", "Order": "574", "Start": "SERVICE_DEMAND_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_KERNEL_DRIVER"}	volatility/ windows.svcscan.SvcScan.jsonl#70	
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "[illegible]", "Name": "WdNisSvc", "Offset": "1402926015184", "Order": "575	volatility/ windows.svcscan.SvcScan.jsonl#69	
2020-09-19 03:39:36	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Microsoft.Windows.SecHealthUI_cw5n1h2txyewy Record Number=82883 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#111476	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:39:47	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) SECURITYHEALTHHOST.EXE-06344EE9.pf Record Number=24491 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#13226	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) coreupdater.exe Record Number=26213 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#56378	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

F-010: coreupdater.exe (PID 8324) token had SeDebugPrivilege explicitly enabled (non-default) - credential-access capability

Severity	**High**
Confidence	probable
Independent sources	memory
Category	credential access
MITRE ATT&CK	T1134, T1003.001
Period (UTC)	2020-09-19 03:40:49 → 2020-09-19 03:40:49

Observed facts

Facts: windows.privileges for PID 8324 coreupdater.exe (exited 03:43:10 UTC, EPROCESS still recoverable): SeDebugPrivilege = "Present,Enabled" (without the Default flag, i.e. enabled at runtime through AdjustTokenPrivileges). Other enabled privileges are only defaults (SeChangeNotify, SeImpersonate, SeCreateGlobal). Across all 95+ processes in the image, coreupdater.exe is the ONLY process whose SeDebugPrivilege is enabled non-by-default. Token belongs to C137\Administrator (RID 500, High IL, Domain/Enterprise/Schema Admins - see F-006).

Analysis

A legitimate updater has no reason to enable SeDebugPrivilege; this is the typical behaviour of credential dumpers (mimikatz 'privilege::debug', procdump-like LSASS readers) or injectors that open protected/system processes. Combined with the reflective PE in spoolsv.exe holding a VM_READ handle to lsass.exe (F-003), this supports a credential-access phase. The precise action taken with the privilege (which process was opened) is not recoverable because the process has exited and its handle table is gone. Single source (memory) -> probable.

Evidence (2)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_privileges	{"TreeDepth": "0", "Attributes": "Present,Enabled", "Description": "Debug programs", "PID": "8324", "Privilege": "SeDebugPrivilege", "Process": "coreupdater.ex", "Value": "20"}	volatility/ windows.privileges.Privs.jsonl#1734	
2020-09-19 03:40:49	Memory: process scan (psscan)	process created (memory, psscan) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3	volatility/ windows.psscan.PsScan.jsonl#167	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

F-011: Network state at capture: two ESTABLISHED TLS sessions to external 203.78.103.109:443 (owner unresolved), SMB to DC, RDP listener; remaining external peers are Microsoft/CDN

Severity	**Medium**
Confidence	possible
Independent sources	memory
Category	command and control
MITRE ATT&CK	T1071.001, T1021.001
Period (UTC)	2020-09-19 01:24:09 → 2020-09-19 01:24:09

Observed facts

Facts (windows.netscan / windows.netstat):
- TCPv4 10.42.85.115:50875 -> 203.78.103.109:443 ESTABLISHED and 10.42.85.115:50972 -> 203.78.103.109:443 ESTABLISHED (netscan rows 110, 114; netstat rows 12, 15). PID/Owner/Created fields are empty in both plugins (TCB not linked to an EPROCESS - owner process exited or structure partially paged).
- 10.42.85.115:50957 -> 10.42.85.10:445 ESTABLISHED (SMB to DC CITADEL-DC01).
- TCP 0.0.0.0:3389 and [::]:3389 LISTENING, svchost.exe PID 520 (TermService), created 01:24:09.
- Other foreign peers: 13.107.x, 204.79.197.200/222, 131.253.33.254, 52.242.211.89, 13.78.149.173, 52.167.250.154, 23.x/104.92.247.90 (Akamai), 72.21.91.29:80 (Verizon/Edgecast OCSP) - Microsoft/Bing/MSN/CDN endpoints consistent with Edge/OneDrive/Windows traffic.
- Ephemeral port order: 50875 is lower than the SMB session (50957) and the Edge/Microsoft sessions created after the ricksanchez logon (~50965-51003), so the 203.78.103.109 session on 50875 was opened earlier in the uptime (likely during the 03:36-03:44 Administrator activity or before the 05:08 logon); 50972 falls within the 05:08 logon burst (near PowerShell PID 3316, 05:08:37-43).

Analysis

203.78.103.109 is the only non-Microsoft, non-CDN, non-internal peer and has two concurrent long-lived HTTPS sessions: candidate C2 channel. Port-order inference suggests one session may relate to the coreupdater/spoolsv activity and the other to the PowerShell 3316 injected payload (which owns a socket handle, F-007), but owner attribution is NOT established by any record. No disk sources (Sysmon 3, SRUM, firewall log, browser history, DNS cache) are available to corroborate. Confidence: possible.

Evidence (6)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#110	
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#114	
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115",	volatility/ windows.netstat.NetStat.	

UTC	Artifact	Detail	Reference	Original source
		"LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	jsonl#12	
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netstat.NetStat. jsonl#15	
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "10.42.85.10", "ForeignPort": "445", "LocalAddr": "10.42.85.115", "LocalPort": "50957", "Offset": "209519101090720", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netstat.NetStat. jsonl#2	
2020-09-19 01:24:09	Memory: network connections (netstat)	network endpoint created (memory, netstat) svchost.exe Proto=TCPv4 | LocalAddr=0.0.0.0 | LocalPort=3389 | ForeignAddr=0.0.0.0 | ForeignPort=0 | State=LISTENING | PID=520	volatility/ windows.netstat.NetStat. jsonl#23	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

4.5 Persistence

Persistence review based on memory only (Volatility 3 svcscan/svclist, registry scheduled_tasks, filescan, mftscan, handles, psscan, cmdline).
(1) F-012 (high, probable): service "coreupdater" is AUTO_START, WIN32_OWN_PROCESS and was STOPPED at capture. Its SCM Order is 662. The boot-time services are numbered in alphabetical order up to 623, and the per-user services of the 05:08 ricksanchez logon start at 664. So coreupdater was registered at runtime between boot and 05:08. The likely moment is the coreupdater.exe run by the elevated domain Administrator (03:40:49-03:43:10), which suggests the binary installed itself. This is reboot persistence. Timeline event T-010 added.
(2) F-013 (medium, possible): powershell.exe PID 508 started at 05:08:37, the same second as the explorer-launched Run-key programs (OneDrive, vmtoolsd, SecurityHealthSystray). Its parent, PPID 1380, had already exited, and userinit stayed alive for 26 s. This may point to a logon autostart (Run key or logon script). It cannot be confirmed without the hives.
Ruled out (see note): the scheduled tasks are Microsoft/GPO defaults; Explorer /NOUACCHECK is the standard task created at Administrator logon; ad_driver.sys is the FTK Imager driver; the Startup folders contain only desktop.ini; no service binary sits in a user, Temp or ProgramData path.

F-012: Persistence: Windows service "coreupdater" (AUTO_START, own-process) registered at runtime during the Administrator session, after boot and before the 05:08 ricksanchez logon

Severity	**High**
Confidence	probable
Independent sources	memory
Category	persistence
MITRE ATT&CK	T1543.003, T1569.002
Period (UTC)	2020-09-19 03:40:00 → 2020-09-19 03:40:49

Observed facts

Facts (Volatility 3 svclist/svcscan, memory captured 2020-09-19 05:10:39 UTC):
- svclist record coreupdater: Order=662, Start=SERVICE_AUTO_START, Type=SERVICE_WIN32_OWN_PROCESS, State=SERVICE_STOPPED, Binary empty, no PID (volatility/windows.svclist.SvcList.jsonl#22). The same record appears 8 times in svcscan (rows 20, 679, 2906, 3548, 4795, 8067, 12806, 13443), always with the same values.
- The SCM database loaded at boot (01:24:07) is in alphabetical order. It ends at Order 623 with xinputhid (svclist #23). In that order, CoreMessagingRegistrar is at Order 93 (#553). A service named "coreupdater" that existed at boot would therefore sit near Order 93-95. Its Order of 662 means services.exe (PID 616, started 01:24:08) added it at runtime, after the boot enumeration.
- The per-user services for ricksanchez's logon session (_b3e3d7 suffix, created at the 05:08:15 logon) come right after it: Order 664 AarSvc_b3e3d7 (#21) to 682. FTK Imager's ad_driver is at Order 684 (#1).
- Correlated activity: coreupdater.exe (MFT record 26213, $FN created 03:40:00) ran as PID 8324 (PPID 4008, session 3, C137\Administrator RID 500, SeDebugPrivilege enabled) from 03:40:49 to 03:43:10 UTC (psscan #167). This is the only time window in the image when the coreupdater binary was active.

- The registry ImagePath could not be recovered: the "Binary (Registry)" column is empty for all 43,221 svcscan rows, and filescan/dlllist give no path for coreupdater.exe.

Analysis

Interpretation: the service was created between the boot (01:24) and the ricksanchez logon (05:08:15). The Order sequence places it before the 05:08 per-user services. The coreupdater.exe run (03:40:49-03:43:10) by an elevated Domain Admin is the only activity that matches, so the most likely explanation is that the binary installed itself as a service (e.g. CreateService / sc create). This is a hypothesis: no 7045/4697 event or registry hive is available to prove who created it.
The service is AUTO_START and was STOPPED at capture, so it is set up to run coreupdater (probably as LocalSystem, account not recoverable) at the next reboot. It is a reboot-persistence mechanism (T1543.003). Confidence is "probable" because all the evidence comes from one memory image and the ImagePath is missing. To confirm, review the SYSTEM hive (Services\coreupdater ImagePath, ObjectName, LastWrite) and the System.evtx 7045 events.

Evidence (7)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_svclist	{"TreeDepth": "0", "Display": "coreupdater", "Name": "coreupdater", "Offset": "1402927956048", "Order": "662", "Start": "SERVICE_AUTO_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_WIN32_OWN_PROCESS"}	volatility/windows.svclist.SvcList.jsonl#22	
	raw.vol_svclist	{"TreeDepth": "0", "Name": "xinputhid", "Offset": "1402926051120", "Order": "623", "Start": "SERVICE_DEMAND_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_KERNEL_DRIVER"}	volatility/windows.svclist.SvcList.jsonl#23	
	raw.vol_svclist	{"TreeDepth": "0", "Binary": "C:\\Windows\\system32\\svchost.exe -k LocalServiceNoNetwork -p", "Display": "CoreMessaging", "Name": "CoreMessagingRegistrar", "Offset": "1402925696448", "Order": "93", "PID": "904", "Start": "SERVICE_AUTO_START", "State": "SERVICE_RUNNING", "Type": "SERVICE_WIN32_SHARE_PROCESS"}	volatility/windows.svclist.SvcList.jsonl#553	
	raw.vol_svclist	{"TreeDepth": "0", "Display": "Agent Activation Runtime_b3e3d7", "Name": "AarSvc_b3e3d7", "Offset": "1402927959024", "Order": "664", "Start": "SERVICE_DEMAND_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_WIN32_SHARE_PROCESS"}	volatility/windows.svclist.SvcList.jsonl#21	
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "coreupdater", "Name": "coreupdater", "Offset": "1402927956048", "Order": "662", "Start": "SERVICE_AUTO_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_WIN32_OWN_PROCESS"}	volatility/windows.svcscan.SvcScan.jsonl#20	
2020-09-19 03:40:49	Memory: process scan (psscan)	process created (memory, psscan) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3	volatility/windows.psscan.PsScan.jsonl#167	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=26213 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/windows.mftscan.MFTScan.jsonl#56376	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem

F-013: Possible logon autostart: PowerShell PID 508 started in the same second as the Run-key programs of the ricksanchez logon (05:08:37) by a transient parent (PPID 1380)

Severity	**Medium**
Confidence	possible
Independent sources	memory
Category	persistence
MITRE ATT&CK	T1547.001, T1037.001, T1059.001
Period (UTC)	2020-09-19 03:36:26 → 2020-09-19 05:08:37

Observed facts

Facts (psscan/cmdline):
- winlogon.exe PID 6456 (session 2, started 03:17:00) spawned userinit.exe PID 5204 at 05:08:16. userinit exited at 05:08:42, so it ran 26 s; userinit normally exits right after starting the shell (psscan #161).
- explorer.exe PID 5896 started at 05:08:16 (#57) and then launched the usual Run-key autostarts: SecurityHealthSystray PID 356 at 05:08:33 (#121), vmtoolsd.exe PID 7252 "-n vmusr" at 05:08:34 (#156, cmdline #77) and OneDrive.exe PID 8904 at 05:08:37 (#118).
- powershell.exe PID 508 started at 05:08:37 with PPID 1380 (#101). PID 1380 is not present in pslist/psscan. PID 508 exited at 05:08:43 after spawning powershell.exe PID 3316 and conhost PID 728 (#13, #145). Command lines for 508 and 3316 are empty (cmdline #78, #81).
- No suspicious entries were found in the other autostart locations visible in memory. The Startup folders show only desktop.ini (filescan #4544 ProgramData, #5478 ricksanchez). The scheduled tasks with a Logon trigger are only Explorer.EXE /NOUACCHECK (created 03:36:26, the standard unelevated-Explorer task) and AppxDeploymentClient (registry scheduled_tasks #32, #25).

Analysis

Hypothesis: PowerShell started in the same second as the explorer-launched Run-key programs, during the ricksanchez logon and while userinit was still alive. Its parent was short-lived and had already exited. This pattern fits a per-user logon autostart: an HKCU/HKLM Run value, a UserInitMprLogonScript, or a GPO logon script launched through an intermediate cmd/wscript/gpscript process (PID 1380). The child process later held an injected PE (F-004). Other explanations remain open: interactive execution by the user, or a remote push via WMI/PsExec. The Run keys, Winlogon\Userinit, Environment\UserInitMprLogonScript and GPO scripts.ini are not available because the registry hives were not parsed and there is no disk artefact, so this remains possible only. This entry adds the persistence angle to F-004/F-007.

Evidence (7)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 05:08:16	Memory: process scan (psscan)	process created (memory, psscan) userinit.exe PID=5204 | PPID=6456 | ImageFileName=userinit.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#161	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:37	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe PID=508 | PPID=1380 | ImageFileName=powershell.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#101	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:37	Memory: process scan (psscan)	process created (memory, psscan) OneDrive.exe PID=8904 | PPID=5896 | ImageFileName=OneDrive.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#118	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:33	Memory: process scan (psscan)	process created (memory, psscan) SecurityHealth PID=356 | PPID=5896 | ImageFileName=SecurityHealth | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#121	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:34	Memory: process scan (psscan)	process created (memory, psscan) vmtoolsd.exe PID=7252 | PPID=5896 | ImageFileName=vmtoolsd.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#156	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_cmdline	{"TreeDepth": "0", "PID": "508", "Process": "powershell.exe"}	volatility/ windows.cmdline.CmdLine.jsonl#78	
2020-09-19 03:36:26	Memory: scheduled tasks (registry)	scheduled task created (memory) ; scheduled task last run (memory) C:\Windows\Explorer.EXE Action Arguments=/NOUACCHECK | Principal ID=Author | Enabled=true	volatility/ windows.registry.scheduled_tasks.ScheduledTasks.jsonl#32	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

4.6 Accounts, authentication and privileges

Accounts and authentication were reconstructed from memory only: sessions, getsids token groups, envars, mftscan per-user SID artefacts, hivelist and netscan. Results:
- F-014 (high, probable): C137\Administrator (RID 500, Domain/Enterprise/Schema Admins) logged on over RDP. The S-1-5-14

RemoteInteractive SID is in the tokens of Edge PID 860 and coreupdater PID 8324. This was the account's first-ever logon on the host: the -500 profile artefacts were created at 2020-09-19 03:36:26. The full High IL token was used for the malicious activity in session 3. No RDP source IP survives.
- F-015 (info): account inventory and chronology. defaultuser0 (local RID 1000) and local RID 1001 ("Admin" hive) date from OOBE on 2020-09-18 05:47-05:50. ricksanchez (RID 1106, Domain Admin) first logged on at 22:44:13 on 2020-09-18 and logged on at the console on 2020-09-19 at 05:08:15, with Medium IL tokens. mortysmith (RID 1108, attributed by timing) first logged on at 22:46:45, created Thoughts.lnk/Thoughts.txt, and is linked to the spoolsv injection through its CWD. The session 1 user (03:16-03:17) is undetermined.
- Timeline events T-011 to T-014 added. IOCs I-007 (Administrator SID) and I-008 (mortysmith) added.

F-014: Domain Administrator (RID 500) used via RDP (Remote Interactive, logon type 10): first-ever logon on the host at 03:36:26, session 3, elevated Domain/Enterprise/Schema Admin token

Severity	**High**
Confidence	probable
Independent sources	memory
Category	initial access / lateral movement
MITRE ATT&CK	T1078.002, T1021.001
Period (UTC)	2020-09-19 01:24:09 → 2020-09-19 03:39:37

Observed facts

Facts (memory image only):
- The tokens of MicrosoftEdge.exe PID 860 (session 3, 03:36:40-03:44:52) and coreupdater.exe PID 8324 (session 3, 03:40:49-03:43:10) belong to C137\Administrator, SID S-1-5-21-2232410529-1445159330-2725690660-500. Both tokens contain the well-known SID S-1-5-14 "Remote Interactive Logon" (getsids rows #606 and #625), together with S-1-5-4 Interactive and BUILTIN\Administrators S-1-5-32-544. They also carry Domain Admins (-512), Schema Admins (-518, #614), Enterprise Admins (-519, #615) and Group Policy Creator Owners (-520, #612).
- coreupdater.exe ran at High Mandatory Level S-1-16-12288 (#637), which is a full, unfiltered admin token.
- Per-user artefacts named after SID -500 show this account's first activity on the host. The MFT directory "S-1-5-21-2232410529-1445159330-2725690660-500" was created 2020-09-19 03:36:26 UTC (mftscan #107218), and the .pckgdep files were created 03:36:27 and last written 03:40:45 (#102754). No earlier -500 profile artefact exists apart from the system-wide MergedResources .pri file at 03:16:51, which is generated for every profile. The profile hives C:\Users\Administrator\ntuser.dat and UsrClass.dat were still loaded at capture (hivelist #14, #15).
- In contrast, the tokens of ricksanchez processes (explorer PID 5896, powershell PID 508/3316) contain S-1-5-4 Interactive and S-1-2-0 Local but no S-1-5-14, and their SESSIONNAME is "Console" (envars #657). ricksanchez therefore logged on at the console (type 2).
- Network: the RDP listener on TCP 3389 belongs to svchost PID 520 (netscan #5). No inbound 3389 connection survives in netscan/netstat, so the RDP client IP cannot be recovered. The only internal peer at capture is the DC 10.42.85.10:445 (netscan #88).

Analysis

The S-1-5-14 group is added to a token only for RemoteInteractive logons (Terminal Services/RDP, logon type 10). This establishes that the built-in domain Administrator account reached DESKTOP-SDN1RPT over RDP at about 03:36:23-03:36:26 UTC. It was the first time this account had ever used the host, since the profile was created at that moment. During that session Defender NIS was unloaded (F-009), coreupdater.exe was dropped and run with SeDebugPrivilege (F-005/F-006/F-010), a service was registered (F-012) and loot.lnk was created at 03:46:18. These facts strongly suggest the Administrator credentials were compromised and used for hands-on-keyboard lateral movement, most likely from another internal host (source IP unknown). An administrator's legitimate RDP session is the alternative explanation, but it fits poorly with the defense evasion and the malware drop. Severity is high. Confidence stays probable because every artefact comes from the single memory image: there are no Security 4624/4778, TerminalServices 21/25 or RCM 1149 logs to confirm it.

Evidence (11)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Remote Interactive Logon", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-14"}	volatility/windows.getsids.GetSIDs.jsonl#606	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Remote Interactive Logon", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-14"}	volatility/windows.getsids.GetSIDs.jsonl#625	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "High Mandatory Level", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-16-12288"}	volatility/windows.getsids.GetSIDs.jsonl#637	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Schema Admins", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-21-2232410529-1445159330-2725690660-518"}	volatility/windows.getsids.GetSIDs.jsonl#614	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Enterprise Admins", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-21-2232410529-1445159330-2725690660-519"}	volatility/windows.getsids.GetSIDs.jsonl#615	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Administrator", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-21-2232410529-1445159330-2725690660-500"}	volatility/windows.getsids.GetSIDs.jsonl#600	
2020-09-19 03:39:37	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-2232410529-1445159330-2725690660-500 Record Number=82944 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/windows.mftscan.MFTScan.jsonl#107218	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-19 03:36:30	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-2232410529-1445159330-2725690660-500.pckgdep Record Number=22129 | Attribute Type=FILE_NAME | MFT Type=File	volatility/windows.mftscan.MFTScan.jsonl#102754	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
	raw.vol_hivelist	{"TreeDepth": "0", "File output": "Disabled", "FileFullPath": "\\??\\C:\\Users\\Administrator\\ntuser.dat", "Offset": "227618280730624"}	volatility/windows.registry.hivelist.HiveList.jsonl#14	
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x761c00", "PID": "5896", "Process": "explorer.exe", "Value": "Console", "Variable": "SESSIONNAME"}	volatility/windows.envars.Envars.jsonl#657	
2020-09-19 01:24:09	Memory: network connections (netscan)	network endpoint created (memory) svchost.exe Proto=TCPv4 | LocalAddr=0.0.0.0 | LocalPort=3389 | ForeignAddr=0.0.0.0 | ForeignPort=0 | State=LISTENING | PID=520	volatility/windows.netscan.NetScan.jsonl#5	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem

F-015: Account inventory and logon chronology reconstructed from memory (per-user SID artefacts, sessions, tokens)

Severity	Informational
Confidence	probable
Independent sources	memory
Category	context / accounts
MITRE ATT&CK	T1078.002, T1087
Period (UTC)	2020-09-18 05:47:23 → 2020-09-19 05:08:16

Observed facts

Per-account summary. Dates are the first and last per-user SID artefacts in mftscan, plus sessions and tokens:
- Local SID S-1-5-21-41211245-796119838-3940169921. RID 1000 = defaultuser0 (OOBE): L.defaultuser0.cdp and -1000.pckgdep created 2020-09-18 05:47:17 (#103228). RID 1001: first .pckgdep at 2020-09-18 05:49:51, last write 2020-09-19 03:16:52 (#104464); its SID directory was created 05:50:04 (#117835). This is probably the local "Admin" account, whose hive C:\Users\Admin\ntuser.dat is loaded (hivelist #13).
- C137\ricksanchez, RID 1106, Domain Admins. First logon 2020-09-18 22:44:11-22:44:21: profile folder (#10758), FontCache and SID directory created 22:44:13 (#102070), L.ricksanchez.cdp created 22:44:21. Current console logon in session 2 at 2020-09-19 05:08:15 (sihost #53, explorer SESSIONNAME Console #57). LOGONSERVER is \\CITADEL-DC01 (#602). Tokens are Medium IL with Administrators (UAC filtered).
- C137 RID 1108, attributed to mortysmith. Per-user artefacts start 2020-09-18 22:46:45 (-1108 SID directory #28040, .pckgdep #100724, last write 23:01:20). L.mortysmith.cdp was created 22:46:46 (#26401) and Thoughts.lnk

22:47:34 (#79896). The file \Users\mortysmith\Desktop\Thoughts.txt is in filescan (#5309).
- C137\Administrator, RID 500 (Domain/Enterprise/Schema Admins). First logon 2020-09-19 03:36:26, RemoteInteractive, session 3 (F-014). Last process activity is MicrosoftEdge PID 860, which exited 03:44:52.
- Session 1 (MicrosoftEdge PID 4592, 03:16:05-03:17:00): the user is not recoverable (the process had exited and has no token in getsids). Session 2 (csrss/winlogon/dwm) was created at 03:17:00, the same second session 1 ended, and had no user until ricksanchez logged on at 05:08:15.
- Service/machine contexts: DESKTOP-SDN1RPT$, LOCAL SERVICE, NETWORK SERVICE, SYSTEM.

Analysis

No new account creation shows up in memory. All interactive users are either domain accounts with domain-joined profiles created on 2020-09-18, or OOBE/local accounts. Three of the accounts that logged on interactively are highly privileged: ricksanchez is in Domain Admins, and Administrator is in Domain, Enterprise and Schema Admins. Their credential material (NTLM hashes and Kerberos tickets) was therefore cached in lsass.exe. lsass was opened by the injected code in spoolsv.exe (F-003), and by coreupdater.exe, which had SeDebugPrivilege (F-010). This makes domain-wide credential theft plausible. The spoolsv.exe CWD is C:\Users\mortysmith\Desktop (handles #14195), which links the injected spooler to the mortysmith context. The likely scenario is a payload started from morty's desktop that injected into spoolsv, but this is a hypothesis. Mapping RID 1108 to mortysmith rests on matching creation times (22:46:45 vs 22:46:46), not on a SAM/AD lookup. Logon types other than RemoteInteractive vs Console, logon failures, 4648 explicit credentials, account creation and group changes could not be checked: there are no EVTX logs, and no registry plugins were run against the SAM/SECURITY hives that sit in memory (hivelist #7, #8).

Evidence (18)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 05:47:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-41211245-796119838-3940169921-1000.pckgdep Record Number=83741 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#103228	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 21:41:53	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-41211245-796119838-3940169921-1001.pckgdep Record Number=87543 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#104464	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 05:50:04	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-41211245-796119838-3940169921-1001 Record Number=84914 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#117835	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_hivelist	{"TreeDepth": "0", "File output": "Disabled", "FileFullPath": "\\??\\C:\\Users\\Admin\\ntuser.dat", "Offset": "227618253631488"}	volatility/ windows.registry.hivelist.HiveList.jsonl#13	
2020-09-18 22:44:11	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) ricksanchez Record Number=87848 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#10758	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 22:53:47	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-2232410529-1445159330-2725690660-1106 Record Number=92893 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#102070	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:15	Memory: logon sessions	process in session (memory) sihost.exe Session ID=2 | Process ID=3232 | User Name=C137/ricksanchez	volatility/ windows.sessions.Sessions.jsonl#53	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:16	Memory: logon sessions	process in session (memory) explorer.exe Session ID=2 | Session Type=Console | Process ID=5896 | User Name=C137/ricksanchez	volatility/ windows.sessions.Sessions.jsonl#57	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x1b9f82033f0", "PID": "1172", "Process": "svchost.exe", "Value": "\\\\CITADEL-DC01", "Variable": "LOGONSERVER"}	volatility/ windows.envars.Envars.jsonl#602	
2020-09-18 22:46:45	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record)	volatility/ windows.mftscan.MFTSca	C:\Sleuthline\cases\ test02\memory\

UTC	Artifact	Detail	Reference	Original source
		S-1-5-21-2232410529-1445159330-2725690660-1108 Record Number=90542 | Attribute Type=FILE_NAME | MFT Type=DirInUse	n.jsonl#28040	DESKTOP-SDN1RPT.mem
2020-09-18 22:46:45	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-2232410529-1445159330-2725690660-1108.pckgdep Record Number=90533 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#100724	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 22:46:46	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) L.mortysmith.cdp Record Number=90592 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#26401	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 22:47:34	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Thoughts.lnk Record Number=90986 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#79896	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\Users\\mortysmith\\ Desktop\\Thoughts.txt", "Offset": "209519130648096"}	volatility/ windows.filescan.FileScan.jsonl#5309	
2020-09-19 03:16:05	Memory: process scan (psscan)	process created (memory, psscan) MicrosoftEdge. PID=4592 | PPID=764 | ImageFileName=MicrosoftEdge. | SessionId=1	volatility/ windows.psscan.PsScan.jsonl#71	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:17:00	Memory: process scan (psscan)	process created (memory, psscan) winlogon.exe PID=6456 | PPID=4084 | ImageFileName=winlogon.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#132	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1048608", "HandleValue": "1912", "Name": "\\Device\\ HarddiskVolume3\\Users\\mortysmith\\Desktop", "Offset": "209519093754240", "PID": "2188", "Process": "spoolsv.exe", "Type": "File"}	volatility/ windows.handles.Handles.jsonl#14195	
	raw.vol_hivelist	{"TreeDepth": "0", "File output": "Disabled", "FileFullPath": "\\SystemRoot\\System32\\Config\\SAM", "Offset": "227618137903104"}	volatility/ windows.registry.hivelist.HiveList.jsonl#8	

4.7 Remote access and lateral movement

Remote access and lateral movement were assessed from the memory image only, using netscan, svcscan, psscan, getsids, handles, filescan, mftscan, dlllist and envars.

(1) F-016 (high/probable): there was an inbound RDP session at 03:36:23 UTC, the first ever on the host. This is shown by RDP device artefacts created on first use (rdpidd.PNF, termkbd.PNF, IddCx0102, RdpIdd, TerminalServices-PnPDevices Admin/Operational evtx at 03:36:23-24) and by S-1-5-14 in the session 3 tokens of C137\Administrator (Edge PID 860, coreupdater PID 8324). The 3389 listener is svchost PID 520 (TermService). No live 3389 connection remained at capture, and the source IP is not recoverable. Timeline T-015.

(2) F-017 (medium/possible): drive Z: was mapped to \\CITADEL-DC01\FileShare (LUID 0xb3ce1f, svchost PID 1096) with an ESTABLISHED SMB session to 10.42.85.10:445. ntlanman/davclnt were loaded in the ricksanchez explorer at 05:09:52. The share is a possible staging location (IOC I-009).

(3) Ruled out with the available artefacts: PsExec and equivalents, WinRM/remote PowerShell (WinRM stopped, no wsmprovhost), RMM tools (AnyDesk, TeamViewer, ScreenConnect, RustDesk, Splashtop, Atera), outbound RDP (no mstsc/mstscax/.rdp), and pivoting to other internal hosts (the only internal peer is the DC). The time zone was derived as UTC-7 (PDT) from the ETL file name.

F-016: Inbound RDP session of C137\Administrator at 03:36:23 UTC: first-ever RDP device installation (rdpidd/termkbd PNF, RdpIdd, TerminalServices-PnPDevices logs) + RemoteInteractive tokens; source IP not recoverable

Severity	**High**
Confidence	probable
Independent sources	memory
Category	lateral movement
MITRE ATT&CK	T1021.001, T1078.002
Period (UTC)	2020-09-19 01:24:09 → 2020-09-19 03:36:24

Observed facts

MFT records resident in memory (windows.mftscan) show a burst of RDP-specific first-use artefacts at 2020-09-19 03:36:22-03:36:24 UTC, all with low, recycled MFT record numbers (252-263): C:\Windows\INF\termkbd.PNF (record 255, created 03:36:23 - RDP keyboard device), user-not-present-trace-2020-09-18-20-36-23.etl (record 256, 03:36:23), IddCx0102 (record 257), rdpidd.PNF (record 259, 03:36:23 - Remote Desktop Indirect Display driver), RdpIdd folder (record 260, 03:36:23), Microsoft-Windows-TerminalServices-PnPDevices%4Admin.evtx (record 262) and %4Operational.evtx (record 263), both created 03:36:24. These are the first-ever instances on a host installed on 2020-09-18 ~06:35, which means this was the first RDP session ever established to the machine. The same second matches the session 3 start (03:36:23) and the creation of the Administrator (RID 500) profile (03:36:26, F-014). Tokens of MicrosoftEdge.exe PID 860 (session 3, 03:36:40-03:44:52) and coreupdater.exe PID 8324 (03:40:49-03:43:10) carry S-1-5-14 "Remote Interactive Logon". The RDP listener is TCP 0.0.0.0:3389 / [::]:3389, owned by svchost.exe PID 520 (TermService). TermService is SERVICE_RUNNING with SERVICE_DEMAND_START, and UmRdpService and SessionEnv are running. At capture time (05:10:39) netscan shows no ESTABLISHED connection on 3389, so the session had ended (Edge exited 03:44:52). The session 3 csrss/winlogon are no longer in psscan. Handles in svchost PID 988 point to Microsoft-Windows-RemoteDesktopServices-RdpCoreTS%4Operational.evtx, which would hold the client IP (EID 131), but no EVTX was collected. The local timestamp embedded in the ETL name (2020-09-18 20:36:23 for 03:36:23 UTC) gives a machine time zone of UTC-7 (PDT).

Analysis

Fact: an RDP session opened at 03:36:23 UTC. RDP device artefacts and RemoteInteractive token SIDs show this. In that session the domain Administrator dropped and ran coreupdater.exe and created the coreupdater service. Interpretation: the host was taken over remotely over RDP with domain Administrator credentials (lateral movement INTO this host, T1021.001 with valid domain accounts T1078.002). The credentials were most likely stolen beforehand, which fits the lsass-reading code injected into spoolsv (F-003). The RDP source (an internal host or a pivot) cannot be determined from memory: there is no surviving TCP endpoint and no RdpCoreTS/TerminalServices-LocalSessionManager logs. Every reference comes from the single memory image, so this stays 'probable' until the Security 4624 LT10 / RdpCoreTS 131 / LSM 21-25 events are obtained from disk.

Evidence (9)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:36:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) rdpidd.PNF Record Number=259 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#128240	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) termkbd.PNF Record Number=255 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#69180	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) RdpIdd Record Number=260 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#23934	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:24	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Microsoft-Windows-TerminalServices-PnPDevices %4Admin.evtx Record Number=262 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#23939	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) user-not-present-trace-2020-09-18-20-36-23.etl Record Number=256 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#128233	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Remote Interactive Logon", "PID": "860", "Process": "MicrosoftEdge.", "SID": "S-1-5-14"}	volatility/ windows.getsids.GetSIDs.jsonl#606	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Remote Interactive Logon", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-	volatility/ windows.getsids.GetSIDs.	

UTC	Artifact	Detail	Reference	Original source
		14"}	jsonl#625	
2020-09-19 01:24:09	Memory: network connections (netscan)	network endpoint created (memory) svchost.exe Proto=TCPv4 | LocalAddr=0.0.0.0 | LocalPort=3389 | ForeignAddr=0.0.0.0 | ForeignPort=0 | State=LISTENING | PID=520	volatility/ windows.netscan.NetScan.jsonl#5	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "Remote Desktop Services", "Name": "TermService", "Offset": "1402925935040", "Order": "457", "PID": "200", "Start": "SERVICE_DEMAND_START", "State": "SERVICE_RUNNING", "Type": "SERVICE_WIN32_SHARE_PROCESS"}	volatility/ windows.svcscan.SvcScan.jsonl#187	

F-017: Mapped network drive Z: -> \\CITADEL-DC01\FileShare and ESTABLISHED SMB session to the DC (10.42.85.10:445) at capture

Severity	**Medium**
Confidence	possible
Independent sources	memory
Category	lateral movement
MITRE ATT&CK	T1021.002, T1039
Period (UTC)	2020-09-19 05:09:52 → 2020-09-19 05:09:52

Observed facts

The Workstation service host svchost.exe PID 1096 holds handle 0x9b0 (offset 209519096092816) to \Device\Mup\;Z:0000000000b3ce1f\CITADEL-DC01\FileShare. filescan recovers the same FILE_OBJECT (\;Z:0000000000b3ce1f\CITADEL-DC01\FileShare). This is a drive letter Z: mapped in logon session LUID 0xb3ce1f to the share FileShare on the domain controller CITADEL-DC01. netscan shows TCP 10.42.85.115:50957 -> 10.42.85.10:445 ESTABLISHED, with no owner or creation time (kernel SMB redirector). In the session of C137\ricksanchez (console, session 2), explorer.exe PID 5896 loaded ntlanman.dll and davclnt.dll at 05:09:52 UTC (MPR.dll at 05:09:09), which is network-provider activity 4 s before FTK Imager started (05:09:56). LOGONSERVER=\\CITADEL-DC01 in the ricksanchez environment. No other internal host appears in netscan/netstat. There is no outbound 3389, 5985/5986 or 445 connection to any host other than the DC.

Analysis

A drive mapped to a DC share is common in a domain (GPO/logon script). The LUID cannot be mapped to a user with the available plugins: there is no logon-session-to-LUID table. The explorer network-provider DLL loads tie the use of the share to ricksanchez around 05:09:52, so it may be responder or user activity just before acquisition. Given loot.lnk (03:46:18) and the Domain Admin compromise, the share is a possible collection/staging location (T1039) and should be reviewed on the DC (5140/5145 from 10.42.85.115, FileShare contents, loot files). There is no evidence that this host was used as a pivot towards other internal machines.

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1048576", "HandleValue": "2480", "Name": "\\Device\\Mup\\;Z:0000000000b3ce1f\\CITADEL-DC01\\FileShare", "Offset": "209519096092816", "PID": "1096", "Process": "svchost.exe", "Type": "File"}	volatility/ windows.handles.Handles.jsonl#11901	
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\;Z:0000000000b3ce1f\\CITADEL-DC01\\FileShare", "Offset": "209519096092816"}	volatility/ windows.filescan.FileScan.jsonl#3433	
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "10.42.85.10", "ForeignPort": "445", "LocalAddr": "10.42.85.115", "LocalPort": "50957", "Offset": "209519101090720", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#88	
2020-09-19 05:09:52	Memory: loaded DLLs	DLL loaded (memory) C:\Windows\System32\ntlanman.dll PID=5896 | Process=explorer.exe	volatility/ windows.dlllist.DllList.jsonl#936	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

UTC	Artifact	Detail	Reference	Original source
	raw.vol_envars	{"TreeDepth": "0", "Block": "0x761c00", "PID": "5896", "Process": "explorer.exe", "Value": "\\\\CITADEL-DC01", "Variable": "LOGONSERVER"}	volatility/windows.envars.Envars.json#641	

4.8 User activity on files and folders

I rebuilt user file and folder activity from memory only: windows.mftscan (FILE_NAME and STANDARD_INFORMATION per record, no parent path), mftscan.ADS (Zone.Identifier), filescan and handles.

New findings:
*- **F-018 (high, probable, collection).** In the Administrator RDP session (session 3), I attributed the activity through the Administrator Recent directory (record 520, created 03:36:24, modified 03:47:39):*
- 03:45:02: shell folders and desktop.ini of all user profiles (ricksanchez, mortysmith, OOBE local) were accessed in the same second.
- 03:45:53: Portal_gun.png was opened in Photos.
- 03:46:18: loot.lnk was created (MFT 87445; the record was reused from the deleted pckgdep file of RID 1001).
- 03:47:34: a mortysmith-profile desktop.ini was accessed.
- 03:47:39: Desktop.lnk was created and the Explorer jump lists were updated.
- This activity shows the session was still interactive until at least 03:47:39, after Edge exited (03:44:52) and coreupdater.exe exited (03:43:10).
- The target of "loot" cannot be resolved in memory.
*- **F-019 (medium, probable).** The PowerShell CustomDestinations jump list 590aee7bdd69b59b was committed at 05:08:37 (temp file JNKXKDDMCSDYIYL32PRB.temp renamed). This shows powershell.exe PID 508 ran with a visible window in the ricksanchez desktop, which adds to F-007 and F-013.*
*- **F-020 (info).** Baseline user activity on 2020-09-18 between 22:44 and 23:08:*
- Thoughts.lnk was created, pointing to mortysmith\Desktop\Thoughts.txt.
- Plans.lnk was created.
- Edge downloads Jessica.jpg and Portal_gun.png carry ZoneId=3.

I added timeline events T-016 to T-018 and IOC I-010 (Desktop.lnk). The responder artefacts (Incident_drive (E).lnk, sysinfo.txt, autorunsc MOTW) are recorded as noise in the note.

F-018: Administrator RDP session file/folder activity 03:45-03:47: profile folders enumerated, Portal_gun.png opened, "loot" item opened (loot.lnk) and Desktop folder opened (Desktop.lnk)

Severity	**High**
Confidence	probable
Independent sources	memory
Category	collection
MITRE ATT&CK	T1005, T1074.001, T1083
Period (UTC)	2019-12-07 09:14:52 → 2020-09-19 03:47:39

Observed facts

This sequence was reconstructed from memory-resident MFT records (windows.mftscan). Its $SI and $FN times agree, so there is no sign of timestomping. All times are UTC.
- 03:45:02: $SI Accessed is updated at the same second on desktop.ini files and the shell folders 3D Objects, Downloads, Music, Saved Games, Searches and Links. The records belong to three different profile generations: records 87861/87863/88092/88263/88316, created 22:44:11-16 (ricksanchez profile); records 89697-89727, created 22:46:44 (mortysmith profile, RID 1108); and records 84486/84487, created 2020-09-18 05:49:47 (OOBE-era local profile). Examples: rows 36769 (Downloads, rec 1527), 126444 (Music, rec 87863), 97439 (3D Objects, rec 89698), 76144 (desktop.ini, rec 84486).
- 03:45:53: Portal_gun.png (MFT record 96326) gets $SI Accessed = 03:45:53. The file was created on 2020-09-18 at 23:07:44 and carries a Zone.Identifier ADS (ZoneId=3, LastWriterPackageFamilyName=Microsoft.MicrosoftEdge). In the same second and the next, mspaint.exe.mui and Blank_PhotosSplashWideTile.png are accessed (03:45:53), then

PhtoMDL2.ttf (Photos app font, rec 102649, 03:46:02) and iconcache/AutomaticDestinations (03:45:54). This fits the image being opened in the Photos app.
- 03:46:16: Windows Search catalog CiST0000.001 (rec 83350) accessed.
- 03:46:18: loot.lnk is created (MFT record 87445, $SI=$FN=03:46:18, link count 1). The record was reused: it previously held S-1-5-21-41211245-796119838-3940169921-1001.pckgdep, the AppRepository package-dependency file of the OOBE local account RID 1001, which was therefore deleted before 03:46:18.
- 03:47:34: desktop.ini rec 89652 accessed. It was created 2020-09-18 22:46:43, at the start of the mortysmith profile creation, and sits next to the AppData record 89655.
- 03:47:39: Desktop.lnk is created (rec 87493). At the same second the Recent directory rec 520 has $SI M/A updated to 03:47:39. This directory was created at 03:36:24, during the Administrator first logon, so it is the Administrator profile's Recent folder. The Explorer jump lists 5f7b5f1e01b83767 and f01b4d95cf55d32a.automaticDestinations-ms (recs 23462/23433, created 03:36:33-43 in the Administrator profile) are also modified at 03:47:39.
- The memory holds no MFT entry or filescan object named "loot" other than loot.lnk, so the LNK target (folder or file named "loot") cannot be resolved.

Analysis

Facts vs. interpretation:
- Recent*.lnk files are created by the shell when a user opens a file or folder. loot.lnk and Desktop.lnk were created at 03:46:18 and 03:47:39. The Recent directory that changed at 03:47:39 was created during the first logon of C137\Administrator, so both LNKs are attributed to the Administrator RDP session (session 3). Attribution is probable: mftscan does not give parent references.
- Just before that, desktop.ini and shell folders of every local profile were accessed in the same second. This is consistent with the intruder browsing C:\Users or searching across profiles, including the mortysmith profile. The spoolsv injection (F-003) also has its CWD in mortysmith\Desktop, where Thoughts.txt resides. A downloaded image (Portal_gun.png) was then opened, an item called "loot" was opened, and a Desktop folder was opened. The "loot" name, which matches offensive-tooling naming, and the timing (3 minutes after coreupdater.exe exited at 03:43:10) suggest local data collection or staging (T1005/T1074.001). The content, location and possible exfiltration of "loot" are not established. Its target could be on the local disk or on the mapped share (Z: -> \\CITADEL-DC01\FileShare).
- The Edge process of the session (PID 860) exited at 03:44:52. Activity after that time is Explorer/Photos and shows the session was still interactive until at least 03:47:39, later than the 03:44 window estimated in step 1.
- Limits: memory-only. There is no LNK body (target path, volume serial, MAC), no ShellBags, no RecentDocs and no $UsnJrnl to confirm the target or any deletions. Confidence is therefore probable.

Evidence (15)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:46:18	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=87445 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#78567	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:46:18	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) loot.lnk Record Number=87445 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#78568	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 21:41:48	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-21-41211245-796119838-3940169921-1001.pckgdep Record Number=87445 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#15020	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:47:39	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=87493 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#82924	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:47:39	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Desktop.lnk Record Number=87493 | Attribute Type=FILE_NAME | MFT	volatility/ windows.mftscan.MFTScan.jsonl#82925	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

UTC	Artifact	Detail	Reference	Original source
		Type=File		
2020-09-19 03:36:24	Memory: MFT records in memory	file created (memory, MFT record) Record Number=520 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#64444	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:24	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Recent Record Number=520 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#64445	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 23:07:44	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=96326 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#49436	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 01:31:22	Memory: MFT records in memory	file created (memory, MFT record) Record Number=102649 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#84896	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2019-12-07 09:14:52	Memory: MFT records in memory	file created (memory, MFT record) Record Number=1527 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#36769	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 22:46:44	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=89698 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#97439	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 22:44:11	Memory: MFT records in memory	file created (memory, MFT record) Record Number=87863 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#126444	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 22:46:43	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=89652 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#89564	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:43	Memory: MFT records in memory	file created (memory, MFT record) Record Number=23462 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#117560	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_ads	{"TreeDepth": "0", "ADS Filename": "Zone.Identifier", "Filename": "Portal_gun.png", "Hexdump": "5b 5a 6f 6e 65 54 72 61 6e 73 66 65 72 5d 0d 0a 4c 61 73 74 57 72 69 74 65 72 50 61 63 6b 61 67 65 46 61 6d 69 6c 79 4e 61 6d 65 3d 4d 69 63 72 6f 73 6f 66 74 2e 4d 69 63 72 6f 73 6f 66 74 45 64 67 65 5f 38 77 65 6b 79 62 33 64 38 62 62 77 65 0d 0a 5a 6f	volatility/ windows.mftscan.ADS.jsonl#12	

F-019: PowerShell custom jump list (590aee7bdd69b59b.customDestinations-ms) written at 05:08:37 - interactive PowerShell window at ricksanchez logon

Severity	**Medium**
Confidence	probable
Independent sources	memory
Category	execution
MITRE ATT&CK	T1059.001
Period (UTC)	2020-09-19 05:08:37 → 2020-09-19 05:08:37

Observed facts

MFT record 91449 was created at 2020-09-19 05:08:37 UTC as JNKXKDDMCSDYIYL32PRB.temp (rows 74873-74875). It was then renamed 590aee7bdd69b59b.customDestinations-ms (rows 133783-133785). $SI: Created and Modified 05:08:37, MFT-changed 05:08:38, Accessed 05:09:51. $FN: 05:08:37. Writing a .temp file and renaming it is the normal pattern the shell uses to commit a CustomDestinations jump list. AppID 590aee7bdd69b59b is the Windows PowerShell AppID. The time matches the start of powershell.exe PID 508 (05:08:37, psscan) and its child PID 3316 (05:08:43). Both run in the console session of C137\ricksanchez, which logged on at 05:08:15.

Analysis

This adds detail to F-007 and F-013. The shell creates a PowerShell jump list only when powershell.exe registers with the taskbar, which means it ran with a visible window in an interactive desktop (ricksanchez session 2). This does not fit a fully hidden service or WMI launch. A visible window would be expected with a Run-key, Startup or logon-script launch, or a

manual one, so the jump list alone does not separate a benign from a malicious launch. The injected PE in PID 3316 (F-004) is still the malicious indicator. The attribution of record 91449 to the ricksanchez profile is inferred from timing, since mftscan gives no parent directory. The jump list content (tasks) is not in memory.

Evidence (4)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 05:08:37	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) JNKXKDDMCSDYIYL32PRB.temp Record Number=91449 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#74875	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:37	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=91449 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#133783	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:37	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) 590aee7bdd69b59b.customDestinations-ms Record Number=91449 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#133785	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:37	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe PID=508 | PPID=1380 | ImageFileName=powershell.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#101	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

F-020: Baseline user file activity 2020-09-18 22:44-23:08 (ricksanchez / mortysmith): Thoughts.txt, Plans, Edge image downloads (Jessica.jpg, Portal_gun.png)

Severity	**Informational**
Confidence	probable
Independent sources	memory
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2020-09-18 22:47:34 → 2020-09-18 23:07:44

Observed facts

The mftscan and filescan records for user activity before the intrusion show the following (UTC):
- 22:47:02: Microsoft Edge.lnk is created in a new profile (rec 90987).
- 22:47:34: Thoughts.lnk is created (rec 90986, Recent item). Its target is \Users\mortysmith\Desktop\Thoughts.txt, a FILE_OBJECT still present in filescan; the MFT record of Thoughts.txt is not memory-resident.
- 22:48:06: Plans.lnk is created (rec 91266). Its target "Plans" is not found in memory.
- 23:00:21-23:07:34: Edge/Bing image cache entries (th*.jpg, OIP*.jpg, Portal_gun*.png/jpg, Portal_Gun[1].htm 23:05:00).
- 23:01:11: Jessica.jpg is saved (rec 94877, ADS Zone.Identifier ZoneId=3, LastWriterPackageFamilyName=Microsoft.MicrosoftEdge). It was opened at 23:01:11-46, and the AutomaticDestinations jump list cb05cc8c5a282971 was created at 23:01:11.
- 23:07:44: Portal_gun.png is saved (rec 96326, Zone.Identifier ZoneId=3 from Edge) and Portal_gun.lnk is created (rec 96327). The Administrator session accessed it again at 03:45:53 (see F-018).

Analysis

This is normal user behaviour: browsing, saving pictures, and editing text notes on a user Desktop. It gives the baseline against which the 03:36-03:47 Administrator activity stands out. Thoughts.txt and Plans may hold content relevant to the case (memory only keeps their names), so they should be recovered from disk. The LNK files are attributed to mortysmith or ricksanchez from profile creation timing (ricksanchez profile 22:44:11, mortysmith 22:46:43-44). This attribution is probable, not certain.

Evidence (7)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 22:47:34	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Thoughts.lnk Record Number=90986 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#79896	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\Users\\mortysmith\\ Desktop\\Thoughts.txt", "Offset": "209519130648096"}	volatility/ windows.filescan.FileScan.jsonl#5309	
2020-09-18 22:48:06	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Plans.lnk Record Number=91266 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#42892	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 23:01:11	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Jessica.jpg Record Number=94877 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#541	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_ads	{"TreeDepth": "0", "ADS Filename": "Zone.Identifier", "Filename": "Jessica.jpg", "Hexdump": "5b 5a 6f 6e 65 54 72 61 6e 73 66 65 72 5d 0d 0a 4c 61 73 74 57 72 69 74 65 72 50 61 63 6b 61 67 65 46 61 6d 69 6c 79 4e 61 6d 65 3d 4d 69 63 72 6f 73 6f 66 74 2e 4d 69 63 72 6f 73 6f 66 74 45 64 67 65 5f 38 77 65 6b 79 62 33 64 38 62 62 77 65 0d 0a 5a 6f 6e	volatility/ windows.mftscan.ADS.jsonl#1	
2020-09-18 23:07:44	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Portal_gun.png Record Number=96326 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#49438	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 23:07:44	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Portal_gun.lnk Record Number=96327 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#49441	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

4.9 Web browsing, downloads and e-mail

Web browsing and download origin were reconstructed from memory only (mftscan DOMStore/INetCache names, the Edge DownloadHistory and Downloads directory $SI times, vol_ads Zone.Identifier, psscan, filescan, handles, netscan).

Key result, F-021 (high, probable): in the RDP session of C137\Administrator, Edge Legacy PID 860 (03:36:40-03:44:52) visited a host by raw IP 194.61.24.x at 03:39:26 (DOMStore 194.61.24[1].xml, MFT 25584). The Edge DownloadHistory directory (MFT 24179) was updated at 03:39:57. coreupdater.exe (MFT 26213) was created at 03:40:00 and finalised at 03:40:04 (FN changed). The Administrator Downloads directory (MFT 426) was modified at 03:40:42, and coreupdater.exe was executed at 03:40:49. Taken together this points to a browser download of the payload, done right after the Defender NIS driver was unloaded (F-009).

Recorded in this step: timeline events T-019 and T-020, and IOC I-011 (194.61.24.0/24). Baseline browsing on 2020-09-18 and the responder tools are listed in a note. There is no e-mail client, no other browser, and no download from another user's session.

F-021: coreupdater.exe most probably downloaded with Microsoft Edge (Legacy) by C137\Administrator from a web host at 194.61.24.x (03:39:26-03:40:04 UTC)

Severity	**High**
Confidence	probable
Independent sources	memory
Category	initial access / ingress tool transfer
MITRE ATT&CK	T1105, T1204.002, T1078.002, T1021.001
Period (UTC)	2020-09-19 03:36:24 → 2020-09-19 03:40:49

Observed facts

There are no browser databases in this case (no chromium/firefox/WebCache tables). The browser activity was rebuilt from windows.mftscan, psscan, filescan and the Zone.Identifier ADS.
Sequence in the Administrator RDP session (session 3), with Edge Legacy PID 860 (PPID 764, alive 03:36:40-03:44:52, psscan row 158):
- 03:39:04: the Edge profile was initialised and the MSN start page was loaded. The DOMStore item www.msn[1].xml was created (MFT 24781).
- 03:39:26: DOMStore item "194.61.24[1].xml" was created (MFT 25584, short name 194612~1.XML, mftscan rows 51833/112170/114790). In the same second, ntp[1].htm (MFT 25585) and container "3" (MFT 25586, 03:39:27) were created. Edge Legacy DOMStore names are the host name without its last label, so this matches a site on an IP address 194.61.24.<x>. The last octet is not recoverable.
- 03:39:57: the $SI modified/changed/accessed times of the Edge DownloadHistory directory (MFT 24179, created 03:36:41, mftscan row 109281) moved to 03:39:57. Edge Legacy writes a download-history entry when a download starts.
- 03:40:00: coreupdater.exe was created (MFT 26213, $FN Created/Modified 03:40:00, $FN Accessed 03:40:01, $FN Updated 03:40:04, row 56378). The 4 s gap between FN creation and the FN change is consistent with a download being written and then finalised or renamed.
- 03:40:42: the $SI modified time of the Administrator Downloads directory moved (MFT 426, created 03:36:24 with the -500 profile, row 5594).
- 03:40:49: coreupdater.exe ran (PID 8324, PPID 4008, psscan row 167), 7 s after the Downloads directory changed.
These are the only DOMStore hosts on the host: www.bing, update10, settings-tipset, microsoftedgewelcome.microsoft, login.live, rickandmorty.fandom, discord, public.servenobid, imasdk.googleapis (2020-09-18 22:44-23:08, ricksanchez/mortysmith), www.msn and 194.61.24 (2020-09-19 03:39). 194.61.24.x is the only raw-IP host and the only non-default site visited in the Administrator session. vol_ads has no Zone.Identifier for coreupdater.exe; ADS content survives in memory only for resident or cached records, so HostUrl/ReferrerUrl cannot be read. 194.61.24.x is not in netscan/netstat (at capture the only external non-Microsoft peer was 203.78.103.109:443).

Analysis

Interpretation (probable): the domain Administrator, in the RDP session, opened Edge and browsed by IP to a host in 194.61.24.0/24. From there a download started (03:39:57), coreupdater.exe was written (03:40:00-03:40:04), most likely into C:\Users\Administrator\Downloads (directory modified 03:40:42), and it was executed 7 s later from Explorer/SmartScreen (03:40:49, ChxSmartScreen pf). Before this, at 03:39:36-03:39:45, Windows Security was opened and the Defender NIS driver was unloaded (F-009). This pattern (disable protection, download a payload from a raw IP over HTTP, run it elevated) is typical of hands-on-keyboard ingress tool transfer through a compromised privileged RDP account.
Caveats: all the evidence comes from the memory image (one source). The parent folder of MFT 26213 is not given by mftscan, so its location in Downloads is inferred from timing. The full URL, the last octet and the scheme cannot be recovered without WebCacheV01.dat / spartan.edb of the Administrator profile (filescan row 4452 shows WebCacheV01.dat was open) or the Zone.Identifier on disk. An alternative is that the file was copied over the RDP clipboard or drive redirection while the web visit is a coincidence. The DownloadHistory change 3 s before file creation makes a browser download more likely.

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:39:26	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) 194.61.24[1].xml Record Number=25584 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#51833	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:39:26	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) 194612~1.XML Record Number=25584 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#112169	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:39:26	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) ntp[1].htm Record Number=25585 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#51835	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:41	Memory: MFT records in memory	file created (memory, MFT record) Record Number=24179 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#109281	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) coreupdater.exe Record Number=26213 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#56378	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:24	Memory: MFT records in memory	file created (memory, MFT record) Record Number=426 | Attribute Type=STANDARD_INFORMATION | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#5594	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:40	Memory: process scan (psscan)	process created (memory, psscan) MicrosoftEdge. PID=860 | PPID=764 | ImageFileName=MicrosoftEdge. | SessionId=3	volatility/ windows.psscan.PsScan.jsonl#158	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:40:49	Memory: process scan (psscan)	process created (memory, psscan) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3	volatility/ windows.psscan.PsScan.jsonl#167	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\Users\\Administrator\\ AppData\\Local\\Microsoft\\Windows\\WebCache\\ WebCacheV01.dat", "Offset": "209519123245328"}	volatility/ windows.filescan.FileScan.jsonl#4452	
2020-09-19 03:39:04	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) www.msn[1].xml Record Number=24781 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#29019	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem

4.10 Defense evasion and anti-forensics

I reviewed defense evasion and anti-forensics using the memory image only (Volatility 3).
New findings:
- F-022 (info/probable): no sign of log clearing or logging being disabled. The EventLog service (svchost PID 988) was running with handles open on Security, System, Windows PowerShell and the RDP/SMB channels. The cached evtx $SI M times run up to 03:52, 04:01 and 05:09-05:10. The PowerShell Operational evtx was first created at 05:08:40, so PowerShell logging was active during the 05:08 PowerShell chain and the disk EVTX is a priority. No wiping or LOLBin anti-forensic tool was found (sdelete, cipher /w, wevtutil, vssadmin, CCleaner, BleachBit, MpCmdRun). The Recycle Bin holds only desktop.ini.
- F-023 (medium/probable): Defender state at capture. WinDefend (MsMpEng PID 2404) and WdFilter were running, as were SecurityHealthService, wscsvc and mpssvc. WdNisSvc and WdNisDrv were stopped (complements F-009). Sense (MDE) was stopped, so there is no EDR telemetry.
- F-024 (info/probable): no timestomping. coreupdater.exe MFT 26213 has $SI equal to $FN. Every SI<FN case is explained by package files copied with their original timestamps or by MFT record reuse. Records 227-262 on the responder volume are noise. The SSDT is clean (all ntoskrnl), unhooked_system_calls shows no divergence, the direct-syscall hits are false positives (signed DLL ranges), and no DKOM-hidden process was found. The memory evasion remains the reflective injection already reported in F-003 and F-004.
- Note: possible missing coreupdater prefetch and a possible RDP end around 03:52, both unconfirmed.

F-023: Microsoft Defender state at capture: WinDefend/WdFilter running, only Network Inspection (WdNisSvc/WdNisDrv) stopped; no MDE (Sense stopped)

Severity	**Medium**
Confidence	probable
Independent sources	memory
Category	defense evasion
MITRE ATT&CK	T1562.001
Period (UTC)	2020-09-19 01:24:09 → 2020-09-19 03:39:45

Observed facts

svcscan state of the security services at capture time:
- WinDefend: SERVICE_RUNNING, AUTO_START, PID 2404, binary "C:\ProgramData\Microsoft\Windows Defender\platform\4.18.2009.2-0\MsMpEng.exe" (row 59).
- WdFilter: RUNNING, BOOT_START (row 75). WdBoot: STOPPED (row 77; normal once boot completes).
- WdNisSvc: STOPPED, DEMAND_START (row 69). WdNisDrv: STOPPED (row 70). WdNisDrv.sys was unloaded at 03:39:45 (F-009).
- Sense (Defender for Endpoint): STOPPED, DEMAND_START (row 247).
- SecurityHealthService: RUNNING, PID 5664 (row 250). wscsvc: RUNNING. mpssvc: RUNNING, PID 1408 (row 390). SgrmBroker: RUNNING.
MsMpEng.exe PID 2404 (created 01:24:09, 9 threads) is in pslist. WdFilter still had 25 kernel callbacks and a timer registered (callbacks row 1). No Defender exclusion or policy registry data is available, because no hives were parsed.

Analysis

Real-time protection (MsMpEng plus the WdFilter minifilter) was still in place at capture. The defense impairment is limited to the network inspection component, which was stopped while the Administrator had Windows Security open, right before the download of coreupdater.exe (F-009). This fits a manual change in the Windows Security UI over the RDP session, for example toggling real-time or cloud protection, which also stops NIS. A full Defender disable that was later re-enabled, or path exclusions added for coreupdater.exe, cannot be excluded: the registry keys (HKLM\SOFTWARE\Microsoft\Windows Defender\Exclusions, Real-Time Protection) and the Defender Operational log (5001/5007/5004) are needed. Sense being stopped means there is no EDR telemetry to fall back on.

Evidence (8)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_svcscan	{"TreeDepth": "0", "Binary": "\"C:\\ProgramData\\Microsoft\\Windows Defender\\platform\\4.18.2009.2-0\\MsMpEng.exe\"", "Display": "Microsoft Defender Antivirus Service", "Name": "WinDefend", "Offset": "1402926031872", "Order": "585", "PID": "2404", "Start": "SERVICE_AUTO_START", "State": "SERVICE_RUNNING", "Type": "SERVICE_WIN32_OWN_PROCESS"}	volatility/windows.svcscan.SvcScan.jsonl#59	
	raw.vol_svcscan	{"TreeDepth": "0", "Binary": "\\FileSystem\\WdFilter", "Name": "WdFilter", "Offset": "1402926014720", "Order": "569", "Start": "SERVICE_BOOT_START", "State": "SERVICE_RUNNING", "Type": "SERVICE_FILE_SYSTEM_DRIVER"}	volatility/windows.svcscan.SvcScan.jsonl#75	
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "獀摜鼻獀摜鼻蔑案俣秸助戳獡捩獟牴犊扭笞躡粵挤慨麤牴榆獹躡獀摜鼻ㄖ殼瞟Ζ氈軷戳獡捩潟瑽敲淳躡粵挤慨麤牴榆獹躡獀摜鼻獀摜鼻蔑案址彈漣湩瑩慅楊歐瞙眸也婀璩圀慢楳彣癮牴犊瞭坟粵挤慨麤牴榆獹殼瞶瑽瞙鼻瑽瞙鐽秸晾ㄑ蔑啜圀鼻Ζ氈軷戳獡捩潟瑽敲哈圀档牡瑟慲瑩瞬坟獀摜鼻獀摜鼻蔑案俣秸助戳獡捩獟牴犊扭笞殼哈圀档牡瑟慲瑩瞬坟獀摜鼻ㄖ殼瞟Ζ氈軷戳獡捩潟瑽敲淳殼哈圀档牡瑟慲瑩瞬坟獀摜鼻獀摜鼻蔑案址彈漣湩瑩慅楊歐瞙眸也婀璩圀慢楳彣瑽敲淳畢瞺啄圀档牡瑟慲瑩瞬眸瑽瞙鼻瑽瞙鐽秸晾蕖嚐圀鼻Ζ氈軷戳獡捩獟牴犊扭笞臏粵挤慨麤牴榆獹臏獀摜鼻", "Name": "WdNisSvc", "Offset": "1402926015184", "Order": "575	volatility/windows.svcscan.SvcScan.jsonl#69	
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "Microsoft Defender Antivirus Network Inspection System Driver", "Name": "WdNisDrv", "Offset": "1402926010544", "Order": "574", "Start": "SERVICE_DEMAND_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_KERNEL_DRIVER"}	volatility/windows.svcscan.SvcScan.jsonl#70	
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "瑶躎啄黔獢慴整瞳鼻瑽瞙兀秸晾蕖嚐鯃捯潼潦ㄖ殼瞙Ζ氈軷挤摯捥瑶躎啄黔獢慴整瞳鼻瑽瞙兀秸晾饃婀瑽圀潤敷癣瞳卟啄黔獢慴整瞳鼻瑽瞙兀秸晾蕖嚐鯃捯潼潦ㄖ齜偋冥摯捥瑶潎摯瞹眸饃婀瑽圀潤敷癣瞳卟啄黔獢慴整瞳鼻瑽瞙兀秸晾蕖嚐鯃捯潼潦ㄖ殼瞙Ζ氈軷挤摯捥瑶殼葟录才瑽聊瑙鼻獀摜鼻蔑案瞙Ζ氈軷挤摯捥瑶殼鼠录才瑽聊瑙鼻獀摜鼻蔑案矧銥豪滢楣昔瞼眸坋彈淆敷癣彷潭敷ㄖ殼瞙Ζ氈軷挤摯捥瑶殼鼠录才瑽聊瑙鼻獀摜鼻蔑案矧銥豪滢楣昔瞼眸饃婀瑽圀潤敷癣瞳啟啄黔獢慴整瞳鼻瑽瞙兀秸晾蕖嚐鯃捯潼潦ㄖ殼瞙Z", "Name": "Sense", "Offset": "1402925916784",	volatility/windows.svcscan.SvcScan.jsonl#247	

UTC	Artifact	Detail	Reference	Original source
		"Order": "397",		
2020-09-19 01:24:09	Memory: process list (pslist)	process created (memory, pslist) MsMpEng.exe PID=2404 | PPID=616 | ImageFileName=MsMpEng.exe | SessionId=0 | Threads=9	volatility/windows.pslist.PsList.jsonl#32	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
	raw.vol_callbacks	{"TreeDepth": "0", "Callback": "272684865443760", "Module": "WdFilter", "Type": "PspLoadImageNotifyRoutine"}	volatility/windows.callbacks.Callbacks.jsonl#1	
2020-09-19 03:39:45	Memory: unloaded kernel modules	kernel module unloaded (memory) WdNisDrv.sys 272684914507776	volatility/windows.unloadedmodules.UnloadedModules.jsonl#7	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem

F-024: Timestomping and kernel/user hooking review: no timestomp on coreupdater.exe or attacker files; SSDT and ntdll stubs clean

Severity	**Informational**
Confidence	probable
Independent sources	memory
Category	anti-forensics
MITRE ATT&CK	T1070.006, T1014
Period (UTC)	2019-07-16 17:11:36 → 2020-09-19 05:09:26

Observed facts

Timestomping. I compared $SI and $FN for every cached MFT record with $FN created on or after 2020-09-18 22:00.
- coreupdater.exe (MFT 26213): $SI C/M = 03:40:00 (SI Updated/Accessed 03:40:49) and $FN C/M = 03:40:00 (FN Updated 03:40:04). They are consistent, so SI is not earlier than FN.
- All SI<FN cases are benign:
  - Cortana/Search package data files (apps.csg, settings.csg, *synonyms.txt, *globals.txt, *conversions.txt; $SI dated 2019-07/10, $FN 22:44:30, 22:46:53 and 03:36:41). These are copied at each first user logon with their source timestamps preserved.
  - Records whose SI rows were cached from an earlier use of the record: loot.lnk MFT 87445 reused from the S-1-5-~1.PCK file of 21:41:48, python3.exe MFT 104672/88750, adtz_globals.dll.
- Records 227/260/261/262 hold sysinfo.txt (05:03:02), DC01 (04:38:49), Protected (05:04:24) and "New folder", renamed DESKTOP-SDN1RPT (05:09:26). They share record numbers with C: RDP artefacts (rdpidd.PNF 259, RdpIdd 260, Data 261, PnPDevices evtx 262) from 03:36:23-24. This is a different volume, consistent with the responder's evidence drive E:.
Hooks and hiding:
- vol_ssdt: all 781 entries resolve to ntoskrnl.
- vol_unhooked_system_calls: 45 functions, none with more than one distinct implementation.
- vol_direct_system_calls hits are in SearchIndexer PID 3708 (254), svchost PID 4312 (128), svchost PID 520 (2), svchost 904, ctfmon and WmiPrvSE. They are attributed to signed DLL ranges (iertutil.dll, FirewallAPI.dll, clbcatq.dll) and contain standard ntdll stubs.
- 56 callbacks with no module show null or implausible addresses.
- psxview: no live process is missing from pslist.

Analysis

The attacker did not alter the timestamps of the dropped payload, so the 03:40:00-03:40:49 MFT chronology can be trusted. The direct-syscall hits are attributed to legitimate system DLL ranges and are textbook ntdll stubs, most likely page-sharing or VAD attribution artefacts (false positives). The module-less callbacks look like parsing smear. No rootkit-level evasion (SSDT or inline ntdll hooks, DKOM-hidden processes) was found. The only memory evasion is the user-mode reflective injection already reported (spoolsv.exe PID 2188 F-003, powershell.exe PID 3316 F-004). The conclusion depends on the partial MFT cache: files whose records were not resident in memory could not be checked.

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=26213 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#56376	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) coreupdater.exe Record Number=26213 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#56378	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2019-07-16 17:11:36	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Record Number=24160 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#31242	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-18 21:41:48	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) S-1-5-~1.PCK Record Number=87445 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#15019	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:03:02	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) sysinfo.txt Record Number=260 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#114358	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:23	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) RdpIdd Record Number=260 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#23934	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:09:26	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) DESKTOP-SDN1RPT Record Number=262 | Attribute Type=FILE_NAME | MFT Type=DirInUse	volatility/ windows.mftscan.MFTScan.jsonl#114362	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_ssdt	{"TreeDepth": "0", "Address": "272684835679072", "Index": "0", "Module": "ntoskrnl", "Symbol": "NtAccessCheck"}	volatility/ windows.ssdt.SSDT.jsonl#1	
	raw.vol_direct_system_calls	{"TreeDepth": "0", "Address": "140726349991939", "Disasm": "0x7ffd681a1003: add eax, 0xc32ecdc3; 0x7ffd681a1008: nop dword ptr [rax + rax]; 0x7ffd681a1010: mov r10, rcx; 0x7ffd681a1013: mov eax, 0x13ff; 0x7ffd681a1018: test byte ptr [0x7ffe0308], 1; 0x7ffd681a1020: jne 0x7ffd681a1025; 0x7ffd681a1022: syscall ; 0x7ffd681a1024: ret ; ", "PID": "3708"	volatility/ windows.malware.direct_system_calls.DirectSystemCalls.jsonl#4	
	raw.vol_direct_system_calls	{"TreeDepth": "0", "Address": "140726497199904", "Disasm": "0x7ffd70e04720: mov r10, rcx; 0x7ffd70e04723: mov eax, 0; 0x7ffd70e04728: test byte ptr [0x7ffe0308], 1; 0x7ffd70e04730: jne 0x7ffd70e04735; 0x7ffd70e04732: syscall ; 0x7ffd70e04734: ret ; ", "PID": "4312", "Process": "svchost.exe", "Range": "\\\\Windows\\\\System32\\\\FirewallAPI.dll"}	volatility/ windows.malware.direct_system_calls.DirectSystemCalls.jsonl#258	

F-022: Anti-forensics review: no evidence of event log clearing/disabling, logging channels still open by EventLog service at capture

Severity	**Informational**
Confidence	probable
Independent sources	memory
Category	anti-forensics
MITRE ATT&CK	T1070.001, T1562.002, T1562.003
Period (UTC)	2020-09-18 05:41:34 → 2020-09-19 05:08:40

Observed facts

There are no EVTX contents in this memory-only case, so 1102, 104, 4616 and Defender 5001/5007 cannot be queried. What memory shows:
(1) The EventLog service (svchost.exe PID 988, created 01:24:08, 18 threads) was SERVICE_RUNNING (svcscan row 508). It held open File handles on \Windows\System32\winevt\Logs\System.evtx (handle 0x524, handles row 10924),

Security.evtx (0x530, row 10927), Windows PowerShell.evtx (0x544, row 10932), TerminalServices-LocalSessionManager Operational/Admin, RemoteConnectionManager Operational, SmbClient/SMBServer Security and TerminalServices-PnPDevices Admin/Operational (rows 11034-11265). Security.evtx and System.evtx are also in filescan (rows 1143, 1117).
(2) Cached $SI modification times show channels kept being written through the incident window:
- RdpCoreTS Operational: 03:52:14 (MFT 89157)
- System.evtx: 04:01:28 (MFT 27975)
- Windows PowerShell.evtx: 05:09:40
- TerminalServices-PnPDevices Operational: 05:10:38
Microsoft-Windows-PowerShell%4Operational.evtx (MFT 93203) was created at 05:08:40, 3 s after powershell.exe PID 508 started. This means PowerShell operational logging was active (not disabled) at the time of the 05:08 PowerShell chain.
(3) No wiping or anti-forensic tool appears in psscan, mftscan or filescan: no sdelete, cipher /w, CCleaner, BleachBit, wevtutil, vssadmin, fsutil, auditpol or MpCmdRun processes. The cipher.exe hits are OS component files from 2020-09-18 06:35-06:37. The only $Recycle.Bin content seen is S-1-5-21-...-1106\desktop.ini (filescan row 5448). No $I/$R files are present.
(4) Defender Operational evtx (MFT 81794): the cached $SI M time is 01:25:11 and nothing is newer. It was not among the files handled by PID 988.

Analysis

At capture time, event logging (Security, System, PowerShell, RDP channels) was working, and nothing in memory points to a log clear or audit-policy tampering. The PowerShell Operational log was created at 05:08:40 by the 05:08 activity, which strongly suggests the disk copy holds 4103/4104/40961 events for PIDs 508/3316. That makes the disk EVTX a priority acquisition.
A log clear followed by continued logging (1102/104) cannot be excluded without the EVTX files. The cached MFT records are partial snapshots, and $SI M times of open evtx files update lazily. So the 01:25:11 date on Defender Operational is not proof that no Defender events (for example 5001/5010 or WdNis stop events around 03:39:45) were written. Absence of evidence here reflects limited artifacts, not a confirmed absence.

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1180063", "HandleValue": "1328", "Name": "\\Device\\HarddiskVolume3\\Windows\\System32\\winevt\\Logs\\Security.evtx", "Offset": "209519066688144", "PID": "988", "Process": "svchost.exe", "Type": "File"}	volatility/windows.handles.Handles.jsonl#10927	
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1180063", "HandleValue": "1316", "Name": "\\Device\\HarddiskVolume3\\Windows\\System32\\winevt\\Logs\\System.evtx", "Offset": "209519066674944", "PID": "988", "Process": "svchost.exe", "Type": "File"}	volatility/windows.handles.Handles.jsonl#10924	
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1180063", "HandleValue": "1348", "Name": "\\Device\\HarddiskVolume3\\Windows\\System32\\winevt\\Logs\\Windows PowerShell.evtx", "Offset": "209519066691744", "PID": "988", "Process": "svchost.exe", "Type": "File"}	volatility/windows.handles.Handles.jsonl#10932	
	raw.vol_svcscan	{"TreeDepth": "0", "Display": "Windows Event Log", "Name": "EventLog", "Offset": "1402925743360", "Order": "136", "PID": "988", "Start": "SERVICE_AUTO_START", "State": "SERVICE_RUNNING", "Type": "SERVICE_WIN32_SHARE_PROCESS"}	volatility/windows.svcscan.SvcScan.jsonl#508	
2020-09-19 05:08:40	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) Microsoft-Windows-PowerShell%4Operational.evtx Record Number=93203 | Attribute Type=FILE_NAME | MFT Type=File	volatility/windows.mftscan.MFTScan.jsonl#131149	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-18 05:41:34	Memory: MFT records in memory	file created (memory, MFT record) Record Number=27975 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/windows.mftscan.MFTScan.jsonl#143841	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem
2020-09-18 22:53:08	Memory: MFT records in memory	file created (memory, MFT record) Record Number=89157 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/windows.mftscan.MFTScan.jsonl#141846	C:\Sleuthline\cases\test02\memory\DESKTOP-SDN1RPT.mem

UTC	Artifact	Detail	Reference	Original source
2020-09-18 05:41:39	Memory: MFT records in memory	file created (memory, MFT record) Record Number=81794 | Attribute Type=STANDARD_INFORMATION | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#136684	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\Windows\\System32\\ winevt\\Logs\\Security.evtx", "Offset": "209519066688144"}	volatility/ windows.filescan.FileScan .jsonl#1143	
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\$Recycle.Bin\\S-1-5-21-2232410529-1445159330-2725690660-1106\\desktop.ini", "Offset": "209519130828096"}	volatility/ windows.filescan.FileScan .jsonl#5448	

4.11 Collection, exfiltration and communications

I assessed collection, exfiltration and communications using the memory image only. I searched mftscan and filescan for archives and found none: the only hit was loot.lnk (F-018), plus benign driver .cab files. No compression or transfer tool was found in psscan or in the prefetch names (7z, rar, rclone, curl, bitsadmin, certutil, robocopy, ftp, WinSCP). OneDrive setup ran at 03:37 for the Administrator, and OneDrive.exe ran for ricksanchez from 05:08, with no sign of transfer use. No BITS job artefacts were found. netscan and netstat give no owner for any connection. External non-Microsoft peers are limited to 203.78.103.109:443, on local ports 50875 and 50972. From the sequential port ordering and the AFD endpoint handles in the injected processes, port 50972 is probably the powershell.exe PID 3316 channel (05:08:43) and port 50875 is possibly the injected spoolsv PID 2188. I recorded this as F-025 (medium/possible): a probable C2 / possible exfil channel, with no evidence of staging or bulk transfer.

F-025: Collection/exfiltration review: no archive, compression or transfer tool found; 203.78.103.109:443 sessions are the only candidate C2/exfil channel (ephemeral port ordering points to the 05:08 PowerShell chain and spoolsv injection)

Severity	**Medium**
Confidence	possible
Independent sources	memory
Category	collection / exfiltration / command and control
MITRE ATT&CK	T1071.001, T1041, T1573, T1560, T1005, T1039
Period (UTC)	2020-09-19 03:46:18 → 2020-09-19 03:46:18

Observed facts

Facts (memory image only, Volatility 3):
- Archives: windows.mftscan, searched for .zip/.7z/.rar/.tar/.tgz/.vhd/.dmp/.kirbi/ntds/rclone/7z/winrar/megasync names created after 2020-09-18 20:00 UTC, returned only loot.lnk (MFT 87445, FILE_NAME C/M/E/A 2020-09-19 03:46:18, row 78568), which is already covered by F-018. The only .cab files are printer-driver package cabs from 2020-09-18 21:41:50 (prnms011, MFT 87491). windows.filescan contains no archive or "loot" file object.
- Tools: Prefetch names in mftscan created from 03:30 onwards are WUDFHOST, ONEDRIVESETUP (03:37:29), FILESYNCCONFIG (03:37:34), SECURITYHEALTHHOST (03:39:47), CHXSMARTSCREEN (03:40:49), RUNTIMEBROKER, POWERSHELL (05:08:44), MSINFO32, DLLHOST, FTK IMAGER and BACKGROUNDTRANSFERHOST (05:10:43). There is no 7z/WinRAR/rar/rclone/curl/bitsadmin/certutil/robocopy/ftp/makecab/tar/WinSCP prefetch, and none of these images appears in psscan. BITS: only qmgr.dll is mapped. No BITS job artefact is recoverable.
- OneDrive: OneDrive.exe was running for ricksanchez from the 05:08 logon. The files SyncEngine-2020-09-19.0508.8904.1.aodl and Downloader_2020-09-19_050839 log exist (filescan rows 5010, 5092). OneDriveSetup/FileSyncConfig prefetch at 03:37:29-34 corresponds to OneDrive per-user setup during the Administrator's first logon. Nothing shows OneDrive sync used for data transfer.
- Network (netscan/netstat): every connection has an empty PID/Owner. External peers are 203.78.103.109:443 from local ports 50875 and 50972 (both ESTABLISHED; netscan rows 110/114, netstat rows 12/15), plus Microsoft/Akamai/Edgecast peers (13.107.x, 204.79.197.x, 131.253.33.254, 52.x, 23.x, 72.21.91.29:80, 104.92.247.90:80). Internal: 10.42.85.10:445 from local port 50957 (SMB to DC, Z: = \\CITADEL-DC01\FileShare, F-017).
- Socket handles: spoolsv.exe PID 2188 (F-003, reflectively injected) holds \Device\Afd and four \Device\Afd\Endpoint handles (handles rows 13880, 13887, 13892, 13893, 14292). powershell.exe PID 3316 (F-004, injected) holds one \Device\

Afd\Endpoint (row 28137). coreupdater.exe exited at 03:43:10 and no socket survives for it.
- Ephemeral port ordering (Windows assigns these sequentially): 50875 < 50957 (SMB at the 05:08 logon) < 50965/50966 (logon-time Microsoft connections) < 50972 < 50976-50991 (Edge/OneDrive/Store at 05:08-05:10).

Analysis

I found no evidence of staging (no archive), of compression tools or of bulk transfer tools in memory. Absence is not proof, because only files whose MFT records were cached in RAM are visible, and SRUM, EVTX (5156, Sysmon 3/22, BITS 3/59/60), $UsnJrnl and the browser upload history are not in the scope of this case. The "loot" item opened at 03:46:18 by the domain Administrator (F-018) is the strongest collection indicator; its target and contents are unknown. Data could also have been moved over the RDP clipboard/drive redirection or to \\CITADEL-DC01\FileShare, and neither path is visible in memory.
Port 50972 to 203.78.103.109 was allocated right after the 05:08:3x logon-time connections, which matches the 05:08:43 start of injected powershell.exe PID 3316. That process holds exactly one AFD endpoint, so 50972 is probably its C2 channel. Port 50875 is older (before the 05:08 logon). Its most plausible owner is the injected spoolsv.exe PID 2188, which holds several AFD endpoints. spoolsv normally holds few or no outbound sockets, but the attribution is inferred, not observed (owner fields are null). Overall assessment: 203.78.103.109 is very likely an active command-and-control and possible exfiltration endpoint (HTTPS 443) at capture time. Its volume is unknown.

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#110	
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#114	
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netstat.NetStat.jsonl#12	
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netstat.NetStat.jsonl#15	
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1442207", "HandleValue": "572", "Name": "\\Device\\Afd", "Offset": "209519069517776", "PID": "2188", "Process": "spoolsv.exe", "Type": "File"}	volatility/ windows.handles.Handles.jsonl#13880	
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1442207", "HandleValue": "600", "Name": "\\Device\\Afd\\Endpoint", "Offset": "209519069533376", "PID": "2188", "Process": "spoolsv.exe", "Type": "File"}	volatility/ windows.handles.Handles.jsonl#13887	
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1442207", "HandleValue": "2368", "Name": "\\Device\\Afd\\ Endpoint", "Offset": "209519091203248", "PID": "3316", "Process": "powershell.exe", "Type": "File"}	volatility/ windows.handles.Handles.jsonl#28137	
2020-09-19 03:46:18	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) loot.lnk Record Number=87445 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#78568	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_filescan	{"TreeDepth": "0", "Name": "\\Users\\ricksanchez\\ AppData\\Local\\Microsoft\\OneDrive\\logs\\Personal\\ SyncEngine-2020-09-19.0508.8904.1.aodl", "Offset": "209519128394784"}	volatility/ windows.filescan.FileScan.jsonl#5010	
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "10.42.85.10", "ForeignPort": "445", "LocalAddr": "10.42.85.115", "LocalPort": "50957", "Offset": "209519101090720", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#88	

4.12 Incident timeline and correlation

I assembled the incident timeline from memory artefacts only (24 events, T-001 to T-024). Four events were added in this step: T-021 (session 1 Edge PID 4592, 03:16:05-03:17:00, user undetermined), T-022 (coreupdater.exe PID 8324 exits at 03:43:10 after running from 03:40:49, with SeDebugPrivilege), T-023 (Administrator Edge PID 860 exits at 03:44:52) and T-024 (state at capture, 05:10:39: injected PE at 0x1840000 in spoolsv PID 2188 with a handle on lsass, plus 2 ESTABLISHED sessions to 203.78.103.109:443).

Chain of events (UTC): the domain Administrator logged on over inbound RDP at 03:36:23. Edge visited 194.61.24.x at 03:39:26. WdNisDrv was unloaded at 03:39:45. coreupdater.exe was dropped at 03:40:00 and executed at 03:40:49; the AUTO_START service "coreupdater" was most probably installed while it ran. Profiles were enumerated and loot.lnk was created between 03:45 and 03:47:39. At ricksanchez's logon, the injected PowerShell chain 508 -> 3316 started at 05:08:37 and 05:08:43. Live C2 sessions to 203.78.103.109 were open at capture.

Time consistency (F-026): EPROCESS times, $SI/$FN times, unloaded-module times, the jump list and prefetch all agree to the second. coreupdater.exe has $SI = $FN, so there is no timestomping. The time zone is derived as UTC-7, and meta gives no time bias.

New IOCs I-012 to I-017: the spoolsv injected region, the PowerShell injected region and thread, the DOMStore 194.61.24[1].xml file, C137\Administrator, C137\ricksanchez, and the coreupdater.exe MFT 26213 timestamps.

F-026: Incident chronology and cross-source time consistency (memory-only)

Severity	**Informational**
Confidence	probable
Independent sources	memory
Category	context
MITRE ATT&CK	T1021.001, T1105, T1204.002, T1543.003, T1562.001, T1055.001, T1059.001
Period (UTC)	2020-09-19 03:36:40 → 2020-09-19 05:08:37

Observed facts

Consolidated sequence (UTC): 01:24:07 boot; 01:24:09 spoolsv.exe PID 2188 start (injected later, time unknown); 03:16:05-03:17:00 session 1 Edge PID 4592 (user undetermined); 03:36:23 inbound RDP, C137\Administrator RID 500 session 3 (first logon, profile 03:36:26); 03:36:40 Edge PID 860; 03:39:26 visit to 194.61.24.x (DOMStore); 03:39:36 Windows Security opened; 03:39:45 WdNisDrv.sys unloaded; 03:39:57 DownloadHistory updated; 03:40:00 coreupdater.exe written (MFT 26213, FN finalised 03:40:04); 03:40:49 coreupdater.exe PID 8324 run (High IL, SeDebugPrivilege enabled), service "coreupdater" AUTO_START registered (SCM order 662) in this window; 03:43:10 coreupdater exit; 03:44:52 Edge exit; 03:45:02-03:47:39 profile enumeration, Portal_gun.png, loot.lnk (03:46:18), Desktop.lnk (last activity of session 3); 05:08:15 ricksanchez console logon; 05:08:37 powershell PID 508 -> 05:08:43 PID 3316 with injected PE; 05:08:44 first PowerShell prefetch; 05:09:56 FTK Imager; 05:10:39 capture with 2 ESTABLISHED sessions to 203.78.103.109:443. Time consistency: process create/exit times (EPROCESS), $SI/$FN times (mftscan) and unloaded-module times are mutually coherent to the second (e.g. DOMStore 03:39:26 < WdNisDrv unload 03:39:45 < file creation 03:40:00 < process start 03:40:49; jump list commit 05:08:37 = powershell PID 508 start; prefetch 05:08:44 one second after PID 508 exit). coreupdater.exe $SI = $FN (no timestomp). Local time zone derived as UTC-7 (PDT); meta machine_active_time_bias_min is None.

Analysis

No clock skew or time-zone inconsistency is observable between memory sources; all of them derive from the same kernel clock, so this is not an independent validation against an external clock (no EVTX/disk). Two gaps remain: (1) initial access vector to the Administrator credentials and the RDP source IP are not recoverable; (2) the injection time into spoolsv PID 2188 and the link between the 03:40 coreupdater activity and the 05:08 PowerShell chain are inferred (persistence via service/logon autostart is the most probable bridge) but not proven. The 03:16 session 1 may be a precursor but its user is unknown.

Evidence (8)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:40:49	Memory: process scan (psscan)	process created (memory, psscan) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3	volatility/ windows.psscan.PsScan.jsonl#167	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:36:40	Memory: process scan (psscan)	process created (memory, psscan) MicrosoftEdge. PID=860 | PPID=764 | ImageFileName=MicrosoftEdge. | SessionId=3	volatility/ windows.psscan.PsScan.jsonl#158	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:40:00	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) coreupdater.exe Record Number=26213 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#56378	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:39:45	Memory: unloaded kernel modules	kernel module unloaded (memory) WdNisDrv.sys 272684914507776	volatility/ windows.unloadedmodules.UnloadedModules.jsonl#7	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:39:26	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) 194.61.24[1].xml Record Number=25584 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#51833	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:37	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe PID=508 | PPID=1380 | ImageFileName=powershell.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#101	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 05:08:37	Memory: MFT records in memory	file created (memory, MFT record) ; file modified (memory, MFT record) JNKXKDDMCSDYIYL32PRB.temp Record Number=91449 | Attribute Type=FILE_NAME | MFT Type=File	volatility/ windows.mftscan.MFTScan.jsonl#74875	C:\Sleuthline\cases\ test02\memory\ DESKTOP-SDN1RPT.mem
	raw.vol_info	{"TreeDepth": "0", "Value": "15.19041", "Variable": "Major/Minor"}	volatility/ windows.info.Info.jsonl#9	

5. Indicators of compromise

ID	Type	Value	Context	First seen (UTC)	Last seen (UTC)	References
I-001	service	coreupdater	Auto-start own-process service; matching coreupdater.exe run as domain Administrator PID 8324 at 03:40:49 UTC	2020-09-19 03:40:49	2020-09-19 03:40:49	volatility/ windows.svcscan.SvcScan.jsonl#20 volatility/windows.pslist.PsList.jsonl#50
I-002	file	coreupdater.exe	Suspicious binary, MFT record 26213, created 2020-09-19 03:40:00 UTC	2020-09-19 03:40:00	2020-09-19 03:40:00	volatility/ windows.mftscan.MFTScan.jsonl#56378
I-003	ip	203.78.103.109	Two ESTABLISHED TCP sessions to port 443 (local 50875, 50972), owner not resolved; candidate C2 (to be confirmed)			volatility/ windows.netscan.NetScan.jsonl#110 volatility/windows.netscan.NetScan.jsonl#114
I-004	file	loot.lnk	LNK created 2020-09-19 03:46:18 UTC (MFT 87445) during domain Administrator session after coreupdater.exe execution; possible staging folder "loot".	2020-09-19 03:46:18	2020-09-19 03:46:18	volatility/ windows.mftscan.MFTScan.jsonl#78568
I-005	service	WdNisDrv/WdNisSvc stopped (Defender NIS) at 2020-09-19 03:39:45	Defender Network Inspection driver unloaded 15 s before coreupdater.exe drop; behavioural indicator of Defender tampering (F-009).	2020-09-19 03:39:45	2020-09-19 03:39:45	volatility/ windows.unloadedmodules.UnloadedModules.jsonl#7 volatility/windows.svcscan.SvcScan.jsonl#70
I-006	command	e2vgmtzqt3xcbq.mca (mutant name)	Random-looking named mutant in mutantscan with no owning handle recovered; unexplained, low-confidence lead to compare against threat intel.			volatility/ windows.mutantscan.MutantScan.jsonl#440
I-007	sid	S-1-5-21-2232410529-1445159330-2725690660-500	The built-in domain Administrator (C137) was used over RDP (S-1-5-14) at 03:36-03:44 UTC on 2020-09-19. This session dropped and ran coreupdater.exe. Treat the account as compromised.	2020-09-19 03:39:37	2020-09-19 03:39:37	volatility/ windows.getsids.GetSIDs.jsonl#625 volatility/windows.mftscan.MFTScan.jsonl#107218

ID	Type	Value	Context	First seen (UTC)	Last seen (UTC)	References
I-008	user	C137\mortysmith (RID 1108)	Interactive logon on 2020-09-18 at 22:46:45. The injected spoolsv.exe (PID 2188) had C:\Users\mortysmith\Desktop as its CWD, so this user context may be the origin of the injection.	2020-09-18 22:46:46	2020-09-18 22:46:46	volatility/windows.mftscan.MFTScan.jsonl#26401 volatility/windows.handles.Handles.jsonl#14195
I-009	path	\\CITADEL-DC01\FileShare (mapped as Z:, LUID 0xb3ce1f)	Network share on the DC mapped at capture time. Possible collection/staging location; review its access logs and contents on the DC.			volatility/windows.handles.Handles.jsonl#11901 volatility/windows.filescan.FileScan.jsonl#3433
I-010	file	Desktop.lnk (Administrator Recent, MFT 87493, 2020-09-19 03:47:39)	Recent item created during the Administrator RDP session, 81 s after loot.lnk. It points to a Desktop folder that was opened, possibly the mortysmith Desktop.	2020-09-19 03:47:39	2020-09-19 03:47:39	volatility/windows.mftscan.MFTScan.jsonl#82925
I-011	ip	194.61.24.0/24 (194.61.24.x, last octet not recoverable)	Web host the domain Administrator visited by IP in Edge Legacy at 2020-09-19 03:39:26 UTC (DOMStore 194.61.24[1].xml, MFT 25584). This is the probable download source of coreupdater.exe (written 03:40:00). Search proxy/firewall/DNS logs for 194.61.24.* around 03:39-03:41 UTC.	2020-09-19 03:39:26	2020-09-19 03:39:26	volatility/windows.mftscan.MFTScan.jsonl#51833
I-012	command	spoolsv.exe PID 2188 – injected PE, RWX VadS region 0x1840000 (MZ), unlinked from loader lists, handle to lsass.exe PID 656	Memory IOC (F-003): reflective DLL/PE injection in print spooler with credential-access handle on LSASS; dump region for YARA/hashing.			volatility/windows.malware.malfind.Malfind.jsonl#1 volatility/windows.malware.ldrmodules.LdrModules.jsonl#32 volatility/windows.handles.Handles.jsonl#14206
I-013	command	powershell.exe PID 508 -> PID 3316 (ricksanchez, session 2) – injected PE RWX region 0x10c6c070000, RWX-started thread TID 3892	Memory IOC (F-004/F-007): PowerShell chain started 05:08:37/05:08:43 UTC at ricksanchez logon, parent PPID 1380 exited; probable C2 stager (port 50972 -> 203.78.103.109:443 probable).	2020-09-19 05:08:37	2020-09-19 05:08:43	volatility/windows.malware.malfind.Malfind.jsonl#6 volatility/windows.malware.suspicious_threads.SuspiciousThreads.jsonl#44 volatility/windows.psscan.PsScan.jsonl#101 volatility/windows.psscan.PsScan.jsonl#154
I-014	file	194.61.24[1].xml (Edge Legacy DOMStore, Administrator profile, MFT 25584, created 03:39:26 UTC)	Browser artefact of the visit to the raw-IP web host 194.61.24.x that most probably served coreupdater.exe (F-021).	2020-09-19 03:39:26	2020-09-19 03:39:26	volatility/windows.mftscan.MFTScan.jsonl#51833
I-015	user	C137\Administrator (domain RID 500, Domain/Enterprise/Schema Admins)	Account used via inbound RDP (session 3, 03:36:23-03:47:39 UTC) to download and run coreupdater.exe; treat credentials as compromised (F-014/F-016).			volatility/windows.getsids.GetSIDs.jsonl#619 volatility/windows.getsids.GetSIDs.jsonl#606
I-016	user	C137\ricksanchez (RID 1106, Domain Admins)	Session in which the injected PowerShell chain ran at logon 05:08:37 (F-004/F-007/F-013); account potentially compromised / targeted by logon autostart.	2020-09-19 05:08:37	2020-09-19 05:08:43	volatility/windows.psscan.PsScan.jsonl#101 volatility/windows.pslist.PsList.jsonl#81
I-017	file	coreupdater.exe MFT record 26213 ($FN C/M 2020-09-19 03:40:00, $FN Updated 03:40:04, Accessed 03:40:01 UTC; parent directory not resolvable in mftscan)	Payload file timestamps for disk pivot ($MFT/$J/Amcache SHA1 to be retrieved on disk image).	2020-09-19 03:40:00	2020-09-19 03:40:00	volatility/windows.mftscan.MFTScan.jsonl#56378

6. MITRE ATT&CK mapping

Technique	Findings / events
T1003.001	F-003, F-010, T-024
T1005	F-018, F-025, T-018
T1014	F-024
T1021.001	F-006, F-011, F-014, F-016, F-021, F-026, T-005, T-013, T-015
T1021.002	F-017
T1036.005	F-005
T1037.001	F-013
T1039	F-017, F-025

Technique	Findings / events
T1041	F-025
T1055	F-004, F-007, T-004
T1055.001	F-003, F-026, T-024
T1059.001	F-004, F-007, F-013, F-019, F-026, T-004, T-008
T1070.001	F-022
T1070.006	F-024
T1071.001	F-011, F-025, T-024
T1074.001	F-018, T-007
T1078.002	F-006, F-014, F-015, F-016, F-021, T-005, T-013
T1083	F-018, T-016
T1087	F-015
T1105	F-021, F-026, T-019, T-020, T-023
T1134	F-010
T1204.002	F-005, F-006, F-021, F-026, T-006, T-022
T1543.003	F-005, F-006, F-012, F-026, T-003, T-010, T-022
T1546.003	F-007
T1547.001	F-007, F-013
T1560	F-025
T1562.001	F-009, F-023, F-026, T-009
T1562.002	F-022
T1562.003	F-022
T1569.002	F-012
T1573	F-025
T1620	F-003, F-004, T-024

7. Answers to the questions

(no question asked)

8. Recommendations

Immediate actions (containment)

- Isolate DESKTOP-SDN1RPT (10.42.85.115) from the network while preserving its state. Block 203.78.103.109 and 194.61.24.0/24 at the perimeter firewall and proxy, and search for other internal hosts talking to them.
- Treat C137\Administrator (RID 500) and C137\ricksanchez (RID 1106) as compromised. Reset their passwords, invalidate their sessions and tickets, and review every Domain/Enterprise/Schema Admin membership.
- Because Domain/Enterprise/Schema Admin tokens were used and lsass was probably accessed: plan a double krbtgt password reset and reset the service, admin and cached credentials used on this host (including mortysmith).
- Acquire a full disk image or KAPE triage of DESKTOP-SDN1RPT (EVTX, registry hives, $MFT/$J, Prefetch, Amcache, SRUM, Edge WebCache), and memory plus logs of CITADEL-DC01, before any remediation.

Eradication and recovery

- Remove the 'coreupdater' service (HKLM\SYSTEM\CurrentControlSet\Services\coreupdater) and the coreupdater.exe binary (MFT 26213), after hashing and preserving them.
- Identify and remove the user-context autostart that launched powershell.exe at the ricksanchez logon: check Run/RunOnce, logon scripts, GPO scripts, WMI subscriptions and the PowerShell profiles.
- Because system processes are injected (spoolsv.exe) and Domain Admin rights were used, rebuild the workstation from known-good media rather than cleaning it.
- Re-enable Microsoft Defender Network Inspection (WdNisSvc/WdNisDrv) and check that Defender Tamper Protection is on. Deploy or re-enable MDE (Sense).
- Review \\CITADEL-DC01\FileShare and the DC for malicious files, new accounts, GPO changes and new services.

Hardening

- Prohibit interactive and RDP logons by the built-in domain Administrator and other Tier-0 accounts on workstations: use a tiered administration model, the 'Deny log on through Remote Desktop Services' right and the Protected Users group.
- Restrict inbound RDP to workstations (firewall plus Network Level Authentication, from jump hosts only) and require MFA for RDP.
- Enable LSA Protection (RunAsPPL) and Credential Guard. Deploy LAPS for local administrator accounts.
- Enable Defender Tamper Protection, ASR rules (block executables from the web unless they meet prevalence/age criteria; block credential stealing from lsass) and SmartScreen enforcement.
- Enable PowerShell Script Block Logging, Module Logging and Transcription, and Constrained Language Mode or AppLocker/WDAC for non-admin users.
- Disable the Print Spooler service on workstations that do not need it.

Detection and monitoring

- Alert on new service installation (System 7045 / Security 4697), especially when the binary is in a user Downloads or Temp folder or the service name is 'coreupdater'.
- Alert on RDP logons (4624 LogonType 10) by RID-500 or Domain/Enterprise Admin accounts to workstations, and on first-time RDP to a host (TerminalServices-RemoteConnectionManager 1149, LocalSessionManager 21/25).
- Alert on Defender component changes: WdNisDrv unload/stop, Defender events 5001/5010/5012/5013, and 7036 events for WdNisSvc.
- Sysmon/EDR: ProcessAccess on lsass.exe from spoolsv.exe or powershell.exe (GrantedAccess including 0x10/0x1410); CreateRemoteThread into spoolsv.exe; powershell.exe spawned by userinit or with no explorer parent at logon.
- Network: alert on connections to 203.78.103.109 and 194.61.24.0/24; alert on browser downloads from raw-IP URLs; alert on a long-lived TLS beacon from spoolsv.exe or powershell.exe.
- YARA/memory hunting: unbacked MZ RWX private regions in spoolsv.exe/powershell.exe; mutant 'e2vgmtzqt3xcbq.mca'.

Further investigation

- Disk triage of DESKTOP-SDN1RPT:
  - Security 4624/4625/4672/4688;
  - TerminalServices logs, to recover the RDP source IP;
  - System 7045 for the coreupdater ImagePath and install time;
  - PowerShell Operational 4103/4104 from 05:08:37, to recover the script content;
  - Defender Operational;
  - Edge WebCache/DownloadHistory, for the full 194.61.24.x URL and the Zone.Identifier of coreupdater.exe;
  - Amcache, for the SHA1 of coreupdater.exe;
  - the target of loot.lnk.
- Extract coreupdater.exe and the injected regions (spoolsv 0x1840000, powershell 0x10c6c070000) with Volatility dumpfiles/malfind --dump, and run reverse engineering, sandbox analysis and config extraction (C2 profile, beacon type).
- Domain controller CITADEL-DC01: authentication logs for C137\Administrator before 03:36 UTC (initial credential compromise), DCSync/replication events (4662), GPO and logon-script changes, file access on FileShare, and new accounts or services.
- Identify the host that originated the RDP session at 03:36:23 (DC 4624/4768/4769 and firewall/NetFlow logs) and investigate it as the probable previous foothold.
- Enterprise-wide hunt for: the coreupdater service/file, connections to 203.78.103.109 and 194.61.24.0/24, the mutant e2vgmtzqt3xcbq.mca, and Administrator RID 500 logons on other workstations.
- Check perimeter, proxy and NetFlow logs for the volume of data sent to 203.78.103.109, to confirm or exclude exfiltration (the 'loot' item).

Appendix A: Summary of the analysis steps

Step 1: System context and scope

Case = single memory image DESKTOP-SDN1RPT.mem (2 GiB, SHA256 AC6EE0E8...67C3), Volatility 3 only. Host DESKTOP-SDN1RPT, Windows 10 x64 build 19041, domain C137.LOCAL (DC CITADEL-DC01, SMB session to 10.42.85.10), IP 10.42.85.115, boot 2020-09-19 01:24:07 UTC, capture 05:10:39 UTC. Users: C137\ricksanchez (RID 1106, Domain Admins, console session 2 from 05:08:15), domain Administrator (RID 500) in session 3 (03:36-03:44) running MicrosoftEdge and suspicious coreupdater.exe (PID 8324, PPID 4008, 03:40:49-03:43:10), Admin profile hive loaded. OS install approx 2020-09-18 06:35 UTC (C:\Users $FN creation). Collection tool FTK Imager (E:\, 05:09:56) noted as noise. Findings F-001 (system description) and F-002 (scope limitations) recorded; timeline T-001, T-002.

Not verified: No disk artifacts (registry, EVTX, Prefetch, Amcache, MFT/USN from disk), no detections; time zone unknown; no Wi-Fi/USB/volume data; userassist/amcache plugins empty; hashdump/lsadump/dumpfiles not run so no binary hashes. All evidence is one source (memory) so high findings cannot reach 'confirmed' by independent corroboration.

Step 2: Triage of automatic detections

No automatic detections exist (detections=0, hash_hits/hashset=0; no EVTX in this memory-only case). Memory anomaly plugins were triaged as substitute alerts. True positives: F-003 (critical, probable) reflective PE in spoolsv.exe PID 2188 (malfind MZ RWX 0x1840000, ldrmodules unlinked) holding a 0x1410 handle on lsass.exe PID 656 and cwd C:\Users\mortysmith\Desktop -> injection + credential access; F-004 (high, probable) powershell.exe 508->3316 (ricksanchez, Domain Admin) at 05:08:37-43 after logon, unlinked MZ RWX region 0x10c6c070000 and RWX-started thread TID 3892; F-005 (high, probable) service "coreupdater" AUTO_START + coreupdater.exe (MFT 26213 created 03:40:00) run as domain Administrator PID 8324 03:40:49-03:43:10. IOCs: coreupdater service/file, 203.78.103.109:443 (2 established sessions, owner unresolved). FPs listed in note (MsMpEng malfind, csrss/WmiPrvSE/audiodg pebmasquerade, generic suspicious_threads, FTK/autorunsc responder tools).

Not verified: All evidence is from one memory image, so high/critical findings remain 'probable'. Command lines of PowerShell PIDs 508/3316 and coreupdater are not resident; service ImagePath and coreupdater.exe path/hash not recovered; netscan Owner/PID empty, so C2 attribution to a process is impossible; parent PIDs 1380 and 4008 absent from pslist/psscan.

Step 3: Program execution

Program execution reconstructed from memory only. Malicious/suspicious executions: (1) coreupdater.exe PID 8324 (F-006, high/probable) dropped 03:40:00 (MFT 26213, SI=FN, no timestomp), executed 03:40:49 ($SI access + ChxSmartScreen pf same second -> Explorer launch) in session 3 by C137\Administrator RID 500 with High IL and Domain/Enterprise/Schema Admins, exited 03:43:10; auto-start service "coreupdater" (F-005). Administrator session 3 started 03:36:23 with RDP display driver artefacts (RDP likely). loot.lnk created 03:46:18 (IOC). (2) PowerShell chain 508->3316 for ricksanchez at 05:08:37-43 (F-007, high/probable): parent PPID 1380 missing, no command line / console history, first-ever POWERSHELL pf created 05:08:44, socket handle; complements F-004 (injected PE). (3) spoolsv injection already F-003. Context F-008: msinfo32, FTK Imager (E:), autorunsc (Zone.Identifier referrer James\Downloads\SysinternalsSuite.zip) = responder tools. Timeline T-005..T-008 added.

Not verified: No disk artefacts (Prefetch run counts, Amcache SHA1, registry ShimCache/BAM/UserAssist, SRUM, EVTX 4688/4104/Sysmon) - all conclusions rest on a single source (memory) and remain 'probable'. coreupdater.exe full path, hash and command line unrecoverable (exited); powershell command lines unreadable; parent processes 4008 and 1380 not carved. hash_hits empty.

Step 4: Memory (RAM)

I reviewed the memory image (Volatility 3 only) for this step. It covered network, services, scheduled tasks, drivers and unloaded modules, DLLs, privileges, mutants, ADS, console history, psxview, suspicious threads and direct syscalls. New findings: F-009 (high/probable, defense evasion): Defender's network inspection driver WdNisDrv.sys was unloaded at 03:39:45 UTC. Nine seconds earlier, at 03:39:36, the Administrator (RID 500) opened the Windows Security app; the

SECURITYHEALTHHOST prefetch file appeared at 03:39:47. At capture time WdNisSvc and WdNisDrv were STOPPED. coreupdater.exe was written to disk 15 s after the unload, at 03:40:00. F-010 (high/probable): coreupdater.exe (PID 8324) was the only process with SeDebugPrivilege enabled at runtime (Present,Enabled without the Default flag). F-011 (medium/possible): there were two ESTABLISHED sessions to 203.78.103.109:443 from local ports 50875 and 50972; netscan and netstat give no owning process. The host also had an ESTABLISHED SMB session to the DC on 10.42.85.10:445 and an RDP listener on 3389 (svchost PID 520). The other external peers are Microsoft/CDN. Added timeline event T-009, IOC I-005 (WdNisDrv stopped) and IOC I-006 (mutant e2vgmtzqt3xcbq.mca, no owner found). Checked and ruled out: scheduled tasks are benign (sc config upnphost, usoclient, Explorer /NOUACCHECK, gpupdate). No service runs a binary from a user or Temp folder; the coreupdater service has an empty binary path in svcscan. ad_driver.sys in ricksanchez\AppData\Local\Temp is most likely the FTK Imager driver (responder tool). The odd psxview rows (pslist false / psscan true) are stale duplicate copies of live processes. The direct-syscall and non-file-backed thread hits are in system and browser processes and look benign. No cmd history was recovered (conhost 728 has CommandCount 0). Zone.Identifier ADS exist only for Edge downloads and the Sysinternals tools; none for coreupdater.exe. The DLL lists contain no unexpected DLL from a user folder (only OneDrive and FTK Imager from E:).

Not verified: There are no disk artifacts (EVTX, Prefetch files, Amcache, registry, SRUM, Defender logs), so nothing in memory could be checked against the disk. Every high finding is therefore single-source and stays "probable". I could not determine which process owns the 203.78.103.109 connections. The coreupdater service binary path is empty, and coreupdater.exe has no Zone.Identifier and no handle that would show its full path or how it was downloaded. The plugins hollowprocesses, processghosting, etwpatch, svcdiff, userassist, amcache and skeleton_key were not provided. I did not finish checking the callbacks, SSDT and driverirp tables for hooks within this step.

Step 5: Persistence

Persistence review based on memory only (Volatility 3 svcscan/svclist, registry scheduled_tasks, filescan, mftscan, handles, psscan, cmdline).
(1) F-012 (high, probable): service "coreupdater" is AUTO_START, WIN32_OWN_PROCESS and was STOPPED at capture. Its SCM Order is 662. The boot-time services are numbered in alphabetical order up to 623, and the per-user services of the 05:08 ricksanchez logon start at 664. So coreupdater was registered at runtime between boot and 05:08. The likely moment is the coreupdater.exe run by the elevated domain Administrator (03:40:49-03:43:10), which suggests the binary installed itself. This is reboot persistence. Timeline event T-010 added.
(2) F-013 (medium, possible): powershell.exe PID 508 started at 05:08:37, the same second as the explorer-launched Run-key programs (OneDrive, vmtoolsd, SecurityHealthSystray). Its parent, PPID 1380, had already exited, and userinit stayed alive for 26 s. This may point to a logon autostart (Run key or logon script). It cannot be confirmed without the hives.
Ruled out (see note): the scheduled tasks are Microsoft/GPO defaults; Explorer /NOUACCHECK is the standard task created at Administrator logon; ad_driver.sys is the FTK Imager driver; the Startup folders contain only desktop.ini; no service binary sits in a user, Temp or ProgramData path.

Not verified: There are no disk artefacts, no EVTX and no parsed registry hives. As a result, these could not be checked: the ImagePath, ObjectName and LastWrite of Services\coreupdater (the svcscan "Binary (Registry)" column is empty everywhere), events 7045/4697/4698/4702/106/5861, Run/RunOnce, Winlogon Userinit/Shell, IFEO Debugger, AppInit_DLLs, HKCU CLSID COM hijacks, Active Setup, Print monitors/processors, BootExecute, WMI subscriptions (OBJECTS.DATA not parsed), the hosts file content, Office startup and PendingFileRenameOperations. The scheduled_tasks plugin returned many empty rows because of paged-out data. The creation time of the coreupdater service is inferred, not recorded.

Step 6: Accounts, authentication and privileges

Accounts and authentication were reconstructed from memory only: sessions, getsids token groups, envars, mftscan per-user SID artefacts, hivelist and netscan. Results:
- F-014 (high, probable): C137\Administrator (RID 500, Domain/Enterprise/Schema Admins) logged on over RDP. The S-1-5-14 RemoteInteractive SID is in the tokens of Edge PID 860 and coreupdater PID 8324. This was the account's first-ever logon on the host: the -500 profile artefacts were created at 2020-09-19 03:36:26. The full High IL token was used for the malicious activity in session 3. No RDP source IP survives.

- F-015 (info): account inventory and chronology. defaultuser0 (local RID 1000) and local RID 1001 ("Admin" hive) date from OOBE on 2020-09-18 05:47-05:50. ricksanchez (RID 1106, Domain Admin) first logged on at 22:44:13 on 2020-09-18 and logged on at the console on 2020-09-19 at 05:08:15, with Medium IL tokens. mortysmith (RID 1108, attributed by timing) first logged on at 22:46:45, created Thoughts.lnk/Thoughts.txt, and is linked to the spoolsv injection through its CWD. The session 1 user (03:16-03:17) is undetermined.
- Timeline events T-011 to T-014 added. IOCs I-007 (Administrator SID) and I-008 (mortysmith) added.

Not verified: There are no EVTX logs, so the Security/TS/RCM event IDs (4624, 4625, 4648, 4672, 4720, 4732, 1149, 21/25) and logon failures could not be checked. The SAM and SECURITY hives are in memory, but no registry, hashdump or cachedump output is available, so account creation, last logon, password changes and group membership changes in the SAM are undetermined. The RID 1108 to mortysmith mapping is inferred from timestamps only. The RDP client IP and the session 1 user could not be recovered.

Step 7: Remote access and lateral movement

Remote access and lateral movement were assessed from the memory image only, using netscan, svcscan, psscan, getsids, handles, filescan, mftscan, dlllist and envars.
(1) F-016 (high/probable): there was an inbound RDP session at 03:36:23 UTC, the first ever on the host. This is shown by RDP device artefacts created on first use (rdpidd.PNF, termkbd.PNF, IddCx0102, RdpIdd, TerminalServices-PnPDevices Admin/Operational evtx at 03:36:23-24) and by S-1-5-14 in the session 3 tokens of C137\Administrator (Edge PID 860, coreupdater PID 8324). The 3389 listener is svchost PID 520 (TermService). No live 3389 connection remained at capture, and the source IP is not recoverable. Timeline T-015.
(2) F-017 (medium/possible): drive Z: was mapped to \\CITADEL-DC01\FileShare (LUID 0xb3ce1f, svchost PID 1096) with an ESTABLISHED SMB session to 10.42.85.10:445. ntlanman/davclnt were loaded in the ricksanchez explorer at 05:09:52. The share is a possible staging location (IOC I-009).
(3) Ruled out with the available artefacts: PsExec and equivalents, WinRM/remote PowerShell (WinRM stopped, no wsmprovhost), RMM tools (AnyDesk, TeamViewer, ScreenConnect, RustDesk, Splashtop, Atera), outbound RDP (no mstsc/mstscax/.rdp), and pivoting to other internal hosts (the only internal peer is the DC). The time zone was derived as UTC-7 (PDT) from the ETL file name.

Not verified: There are no EVTX, so the following could not be checked: Security 4624 LT10/4648/4778/5140/5145, RdpCoreTS 131, TerminalServices-LSM 21/24/25, WinRM 6/91/168, PowerShell 400, System 7045. There are no registry hives (Terminal Server Client, MountPoints2, Network\Z, ShellBags), no RDP bitmap cache and no full $MFT. The RDP source IP and the user who owns LUID 0xb3ce1f cannot be determined. The 203.78.103.109:443 sessions still have no identified owner process.

Step 8: User activity on files and folders

I rebuilt user file and folder activity from memory only: windows.mftscan (FILE_NAME and STANDARD_INFORMATION per record, no parent path), mftscan.ADS (Zone.Identifier), filescan and handles.

New findings:
- **F-018 (high, probable, collection).** In the Administrator RDP session (session 3), I attributed the activity through the Administrator Recent directory (record 520, created 03:36:24, modified 03:47:39):
  - 03:45:02: shell folders and desktop.ini of all user profiles (ricksanchez, mortysmith, OOBE local) were accessed in the same second.
  - 03:45:53: Portal_gun.png was opened in Photos.
  - 03:46:18: loot.lnk was created (MFT 87445; the record was reused from the deleted pckgdep file of RID 1001).
  - 03:47:34: a mortysmith-profile desktop.ini was accessed.
  - 03:47:39: Desktop.lnk was created and the Explorer jump lists were updated.
  - This activity shows the session was still interactive until at least 03:47:39, after Edge exited (03:44:52) and coreupdater.exe exited (03:43:10).
  - The target of "loot" cannot be resolved in memory.
- **F-019 (medium, probable).** The PowerShell CustomDestinations jump list 590aee7bdd69b59b was committed at 05:08:37 (temp file JNKXKDDMCSDYIYL32PRB.temp renamed). This shows powershell.exe PID 508 ran with a visible

window in the ricksanchez desktop, which adds to F-007 and F-013.
- **F-020 (info).** Baseline user activity on 2020-09-18 between 22:44 and 23:08:
  - Thoughts.lnk was created, pointing to mortysmith\Desktop\Thoughts.txt.
  - Plans.lnk was created.
  - Edge downloads Jessica.jpg and Portal_gun.png carry ZoneId=3.

I added timeline events T-016 to T-018 and IOC I-010 (Desktop.lnk). The responder artefacts (Incident_drive (E).lnk, sysinfo.txt, autorunsc MOTW) are recorded as noise in the note.

Not verified: There is no disk or KAPE collection, so none of these were available: LNK bodies (target, volume serial, MAC), Jump List contents, ShellBags, RecentDocs/OpenSave/LastVisited, WordWheelQuery, Windows Timeline, SIDR, Thumbcache, Notepad tabs, $MFT/$UsnJrnl/$LogFile (creations, renames and deletions beyond memory-resident records), Recycle Bin, or USB/USBSTOR/EMDMgmt/setupapi. The consequences:
- The target, content and possible exfiltration of "loot" are unknown.
- The parent directories of LNK files are inferred from timing.
- MOTW absence on coreupdater.exe is not established.
- Removable-media history, other than the responder E: drive, cannot be assessed.
- Thoughts.txt and Plans contents are not in memory.

Step 9: Web browsing, downloads and e-mail

Web browsing and download origin were reconstructed from memory only (mftscan DOMStore/INetCache names, the Edge DownloadHistory and Downloads directory $SI times, vol_ads Zone.Identifier, psscan, filescan, handles, netscan).

Key result, F-021 (high, probable): in the RDP session of C137\Administrator, Edge Legacy PID 860 (03:36:40-03:44:52) visited a host by raw IP 194.61.24.x at 03:39:26 (DOMStore 194.61.24[1].xml, MFT 25584). The Edge DownloadHistory directory (MFT 24179) was updated at 03:39:57. coreupdater.exe (MFT 26213) was created at 03:40:00 and finalised at 03:40:04 (FN changed). The Administrator Downloads directory (MFT 426) was modified at 03:40:42, and coreupdater.exe was executed at 03:40:49. Taken together this points to a browser download of the payload, done right after the Defender NIS driver was unloaded (F-009).

Recorded in this step: timeline events T-019 and T-020, and IOC I-011 (194.61.24.0/24). Baseline browsing on 2020-09-18 and the responder tools are listed in a note. There is no e-mail client, no other browser, and no download from another user's session.

Not verified: No browser databases were parsed (WebCacheV01.dat/spartan.edb of the Administrator profile are referenced in filescan but were not extracted). Because of that, the full URL, the last octet of 194.61.24.x, typed URLs and search queries, and the Edge download record cannot be recovered. coreupdater.exe has no Zone.Identifier in memory, so HostUrl/ReferrerUrl are unavailable. mftscan gives no parent directory, so the Downloads location is inferred from timing. All the evidence comes from the memory image (a single source), so corroborating with disk artefacts (MFT/$J, Zone.Identifier, WebCache) or proxy/firewall logs is required to confirm.

Step 10: Defense evasion and anti-forensics

I reviewed defense evasion and anti-forensics using the memory image only (Volatility 3).
New findings:
- F-022 (info/probable): no sign of log clearing or logging being disabled. The EventLog service (svchost PID 988) was running with handles open on Security, System, Windows PowerShell and the RDP/SMB channels. The cached evtx $SI M times run up to 03:52, 04:01 and 05:09-05:10. The PowerShell Operational evtx was first created at 05:08:40, so PowerShell logging was active during the 05:08 PowerShell chain and the disk EVTX is a priority. No wiping or LOLBin anti-forensic tool was found (sdelete, cipher /w, wevtutil, vssadmin, CCleaner, BleachBit, MpCmdRun). The Recycle Bin holds only desktop.ini.
- F-023 (medium/probable): Defender state at capture. WinDefend (MsMpEng PID 2404) and WdFilter were running, as were SecurityHealthService, wscsvc and mpssvc. WdNisSvc and WdNisDrv were stopped (complements F-009). Sense (MDE) was stopped, so there is no EDR telemetry.

- F-024 (info/probable): no timestomping. coreupdater.exe MFT 26213 has $SI equal to $FN. Every SI<FN case is explained by package files copied with their original timestamps or by MFT record reuse. Records 227-262 on the responder volume are noise. The SSDT is clean (all ntoskrnl), unhooked_system_calls shows no divergence, the direct-syscall hits are false positives (signed DLL ranges), and no DKOM-hidden process was found. The memory evasion remains the reflective injection already reported in F-003 and F-004.
- Note: possible missing coreupdater prefetch and a possible RDP end around 03:52, both unconfirmed.

Not verified: No EVTX contents, so 1102/104/4616/Kernel-General 1/Defender 5001-5013 could not be checked. No registry hives (Defender exclusions, DisableRealtimeMonitoring, PowerShell logging policy). No full $MFT, $J or $LogFile (mass deletion, prefetch deletion and timestomp could only be checked on records cached in memory). vol_etwpatch not available. The system time change could not be assessed.

Step 11: Collection, exfiltration and communications

I assessed collection, exfiltration and communications using the memory image only. I searched mftscan and filescan for archives and found none: the only hit was loot.lnk (F-018), plus benign driver .cab files. No compression or transfer tool was found in psscan or in the prefetch names (7z, rar, rclone, curl, bitsadmin, certutil, robocopy, ftp, WinSCP). OneDrive setup ran at 03:37 for the Administrator, and OneDrive.exe ran for ricksanchez from 05:08, with no sign of transfer use. No BITS job artefacts were found. netscan and netstat give no owner for any connection. External non-Microsoft peers are limited to 203.78.103.109:443, on local ports 50875 and 50972. From the sequential port ordering and the AFD endpoint handles in the injected processes, port 50972 is probably the powershell.exe PID 3316 channel (05:08:43) and port 50875 is possibly the injected spoolsv PID 2188. I recorded this as F-025 (medium/possible): a probable C2 / possible exfil channel, with no evidence of staging or bulk transfer.

Not verified: The case has no SRUM (network usage per application), no EVTX (5156, Sysmon 3/22, BITS 3/59/60, firewall 2004/2005), no $MFT/$UsnJrnl, no 7-Zip history, no browser upload history and no OneDrive ODL or SyncEngineDatabase parsing. Byte volumes to 203.78.103.109 are unknown. The socket owners are null in netscan, so attribution rests on inference. The target of "loot" and the contents of \\CITADEL-DC01\FileShare could not be recovered. RDP clipboard and drive-redirection transfers cannot be assessed.

Step 12: Incident timeline and correlation

I assembled the incident timeline from memory artefacts only (24 events, T-001 to T-024). Four events were added in this step: T-021 (session 1 Edge PID 4592, 03:16:05-03:17:00, user undetermined), T-022 (coreupdater.exe PID 8324 exits at 03:43:10 after running from 03:40:49, with SeDebugPrivilege), T-023 (Administrator Edge PID 860 exits at 03:44:52) and T-024 (state at capture, 05:10:39: injected PE at 0x1840000 in spoolsv PID 2188 with a handle on lsass, plus 2 ESTABLISHED sessions to 203.78.103.109:443).

Chain of events (UTC): the domain Administrator logged on over inbound RDP at 03:36:23. Edge visited 194.61.24.x at 03:39:26. WdNisDrv was unloaded at 03:39:45. coreupdater.exe was dropped at 03:40:00 and executed at 03:40:49; the AUTO_START service "coreupdater" was most probably installed while it ran. Profiles were enumerated and loot.lnk was created between 03:45 and 03:47:39. At ricksanchez's logon, the injected PowerShell chain 508 -> 3316 started at 05:08:37 and 05:08:43. Live C2 sessions to 203.78.103.109 were open at capture.

Time consistency (F-026): EPROCESS times, $SI/$FN times, unloaded-module times, the jump list and prefetch all agree to the second. coreupdater.exe has $SI = $FN, so there is no timestomping. The time zone is derived as UTC-7, and meta gives no time bias.

New IOCs I-012 to I-017: the spoolsv injected region, the PowerShell injected region and thread, the DOMStore 194.61.24[1].xml file, C137\Administrator, C137\ricksanchez, and the coreupdater.exe MFT 26213 timestamps.

Not verified: Several things cannot be recovered from memory: the initial access vector (how the Administrator credentials were obtained), the RDP source IP, the last octet of 194.61.24.x, the coreupdater.exe hash and full path, the spoolsv injection time, the owner of each 203.78.103.109 connection, the target of loot.lnk, and the PowerShell command line. With memory as the only source, nothing

can be corroborated independently. The disk image (EVTX 4624/4688/7045, Sysmon, Amcache, $MFT/$J, the SYSTEM and NTUSER hives) is needed to confirm the findings.

Step 13: Summary, answers and recommendations

Final report elements are recorded. set_summary holds the verdict, the executive summary (RDP intrusion as the domain Administrator at 03:36:23 UTC, Defender NIS driver unloaded, coreupdater.exe downloaded from 194.61.24.x and run elevated with service persistence, profile enumeration and 'loot', injected PEs in spoolsv.exe (lsass handle) and the PowerShell chain at the ricksanchez logon, live C2 to 203.78.103.109:443), a detailed technical chronology referencing F-001 to F-026, and the limitations. set_recommendations holds the immediate, remediation, hardening, detection and further-investigation items. No guidance questions were provided, so no answer_question calls were needed.

Not verified: Memory-only case: there are no EVTX, hives or disk artefacts, so every critical/high finding stays probable (single source). Not recoverable: the coreupdater.exe hash and ImagePath, the PowerShell command lines, the RDP source IP, the full 194.61.24.x address, the loot.lnk target, the C2 socket owners, initial credential access, and whether data was exfiltrated.

Appendix B: Automatic detections

Engine	Level	Count

Appendix C: Working notes (leads ruled out, false positives)

- [step 1] Collection activity (excluded from findings): FTK Imager.exe PID 6544 launched by explorer.exe (PID 5896) at 2020-09-19 05:09:56 UTC from removable/external path "E:\FTK Imager\FTK Imager.exe" (vol_cmdline row 87, vol_pslist row 87) - this is the memory acquisition tool, ~45 s before the image SystemTime (05:10:39 UTC). msinfo32.exe (PID 7328, 05:09:13) from explorer is likely the responder checking system info. powershell.exe PID 508 (PPID 1380, non-existent) -> PID 3316 at 05:08:37-05:08:43 at ricksanchez logon needs review in later steps (could be logon script).
- [step 2] Step 2 triage: detections table = 0 rows (no Hayabusa/Chainsaw/Zircolite/Defender output: no EVTX/disk artifacts in this memory-only case); hash_hits and hashset = 0 rows (no offline hash list loaded, so no match proves nothing). Memory 'alert' plugins used as pivots instead. False positives: malfind MsMpEng.exe PID 2404 (Defender signature/emulation RWX, no MZ); malfind PowerShell 3316 zero/data RWX regions (JIT) apart from the MZ one; pebmasquerade csrss.exe 424 (garbage PEB path = unreadable/smeared PEB), WmiPrvSE 2548 and audiodg 6276 (case difference wmiprvse.exe / AUDIODG.EXE only); suspicious_threads in svchost, dllhost, WmiPrvSE, SearchApp, Edge, OneDrive, etc. (start addresses in 0x7ffd... ranges of system DLLs/CFG/JIT, typical Win10 noise). ShimCache: FTK Imager and autorunsc64.exe from removable media (SIGN.MEDIA) = responder tools (autorunsc64 modified 2020-09-13).
- [step 3] Step 3 (execution): disk execution artefacts (Prefetch content, Amcache, ShimCache hive, BAM, UserAssist, SRUM, 4688/Sysmon/PowerShell EVTX) are NOT available - memory-only case. Execution was reconstructed from psscan/pslist/pstree, cmdline, getsids, sessions, ShimCache in memory (10 entries, only FTK/autorunsc/system) and Prefetch file creation times in memory-resident MFT records (creation of a .pf ~ first execution, no run count). Command lines for PIDs 508/3316/2188 are empty (PEB paged); coreupdater.exe exited, path/hash unrecoverable. No COREUPDATER pf found in resident MFT (non conclusive). Parents PID 4008 (coreupdater) and 1380 (powershell stager) not recoverable.
- [step 5] Persistence step - ruled out / context: (1) ad_driver kernel service (SvcList #1, Order 684, DEMAND_START, RUNNING) loaded from C:\Users\RICKSA~1\AppData\Local\Temp\ad_driver.sys (modules #193) is the AccessData/FTK Imager acquisition driver loaded at memory capture (05:09-05:10) -> responder tool, not persistence. (2) Scheduled tasks from registry (vol_scheduled_tasks): sc.exe config upnphost start= auto, usoclient StartWork, rundll32 dfdts.dll, AppxDeploymentClient, Explorer.EXE /NOUACCHECK (created 03:36:26 at Administrator logon - standard CreateExplorerShellUnelevatedTask), gpupdate /target:user (GPO) -> all Microsoft/OS defaults. (3) Startup folders:

only desktop.ini (filescan #4544, #5478). (4) Key handles to IFEO/Winlogon/CLSID are normal system handles. (5) No service in svcscan/svclist with a binary in Users/Temp/ProgramData (except Defender MsMpEng). Not verifiable in memory-only case: Run/RunOnce values, IFEO Debugger, AppInit_DLLs, COM hijack, Active Setup, Print monitors, BootExecute, WMI consumers (OBJECTS.DATA not parsed), hosts file content, 7045/4697/4698 events.

- [step 6] Step 6 (accounts) limitations: this is a memory-only case. No Security, TerminalServices or RCM EVTX logs are available, so 4624/4625/4648/4672/4720/4732, 21/22/25 and 1149 cannot be checked. The SAM and SECURITY hives are loaded in memory (hivelist #7/#8), but the printkey, hashdump, lsadump and cachedump plugins were not run or are not in the database. Account creation, password changes and group changes are therefore undetermined. The RDP source IP is not recoverable: no inbound 3389 connection remains in netscan or netstat. The session 1 user (Edge PID 4592, 03:16:05-03:17:00) is undetermined. The MergedResources-*.pri files for SIDs 1001/1106/1108/500 (05:47:56 and 03:16:34-53) are created system-wide and are NOT logon evidence.
- [step 7] Step 7 ruled out (memory only): no PsExec/PAExec/RemCom (no PSEXESVC service, file, pipe or ADMIN$ handle). No WinRM use: the WinRM service is DEMAND_START/STOPPED, no wsmprovhost process (the binary exists from the OS install 2020-09-18 06:37), and no listener on 5985/5986. No remote access tools: no AnyDesk/TeamViewer/ScreenConnect/RustDesk/Splashtop/Atera/ngrok/plink/chisel in filescan, mftscan, svcscan or psscan. No outbound RDP: there is no mstsc process, no mstscax.dll loaded, no .rdp file or Terminal Server Client cache in memory, and mstsc.exe exists only from the OS install. RemoteRegistry is DISABLED. MPR/ntlanman/davclnt loads in explorer 05:08-05:09 are attributed to the ricksanchez session / Z: mapping. Machine time zone derived from the ETL name user-not-present-trace-2020-09-18-20-36-23.etl created 03:36:23 UTC = UTC-7 (PDT). Not available: EVTX (Security 4624/4648/5140/5145, RdpCoreTS 131, LSM 21/24/25, WinRM 6/91/168, PowerShell 400, System 7045), registry hives (Terminal Server Client, MountPoints2, Network\Z), bitmap cache, raw.mft.
- [step 8] Step 8 (user file activity) - memory-only case. raw.lnk, jumplist, shellbags, RecentDocs/OpenSave/LastVisited, wxt_activity, SIDR, thumbcache, Notepad tabs, $MFT/$J/$LogFile, RecycleBin, USB/USBSTOR/EMDMgmt/setupapi are NOT available (no disk collection), so none of these could be checked. The analysis used windows.mftscan (memory-resident MFT records, no parent path), windows.mftscan.ADS (Zone.Identifier), filescan and handles. Ruled out as noise: Incident_drive (E).lnk / Incident_drive (E) (2).lnk / DESKTOP-SDN1RPT.lnk in ricksanchez Recent (05:09:46) = responder FTK Imager USB (E:). sysinfo.txt (rec 260, 05:03:02-05:03:55) = msinfo32 export by the responder. The autorunsc Zone.Identifier with referrer C:\Users\James\Downloads\SysinternalsSuite.zip = responder tools. No Zone.Identifier ADS was recovered for coreupdater.exe (rec 26213), but the ADS plugin only covers memory-resident records, so MOTW absence is not established. No archive (.zip/.7z/.rar) or staging file was found in memory besides loot.lnk. The removable device history (other than E:) cannot be assessed.
- [step 9] Step 9 (web/downloads/e-mail): no browser or e-mail parsers ran (memory-only case: no chromium_*/firefox_*/hindsight/dissect_iexplore tables). Edge Legacy is the only browser present; no chrome.exe, firefox.exe or msedge.exe and no .crdownload/.partial in mftscan. There is no e-mail client (no Outlook process, no .pst/.ost/.eml). The Edge DOMStore hosts from 2020-09-18 22:44-23:08 (rickandmorty.fandom, discord, login.live, servenobid, imasdk.googleapis, bing, msn) are baseline user browsing. The images Jessica.jpg and Portal_gun.png carry Edge Zone.Identifier ZoneId=3 with no HostUrl (F-020 baseline). Autoruns Zone.Identifier ReferrerUrl=C:\Users\James\Downloads\SysinternalsSuite.zip is a responder tool (F-008). Edge PID 4592 in session 1 (03:16:05-03:17:00) was the local RID 1001 profile (pckgdep -1001 artefacts) loading the default start page; nothing suspicious. No typed searches can be recovered without WebCacheV01.dat (Administrator WebCacheV01.dat was open, filescan row 4452, but it was not extracted or parsed).
- [step 10] Step 10 notes.
  (a) No COREUPDATER.EXE-*.pf name was found among the cached MFT records or filescan objects. Other prefetch files of the same session are present: CHXSMARTSCREEN 03:40:49, RUNTIMEBROKER 03:40:56, SECURITYHEALTHHOST 03:39:47. This could mean the prefetch file was deleted, but the MFT cache in memory is partial, so it is NOT reported as a finding. Verify C:\Windows\Prefetch and $J on disk.
  (b) The RdpCoreTS Operational evtx (MFT 89157) was last written at 03:52:14. Unistore/HxStore files were created at 03:52:09 and Windows Update/CryptnetUrlCache files at 03:51:27-28. The session 3 RDP disconnect/logoff may fall around 03:52, but this is an unconfirmed hypothesis.

(c) Low MFT record numbers 227/260/261/262 (DC01 04:38:49, sysinfo.txt 05:03:02, Protected 05:04:24, "New folder" renamed DESKTOP-SDN1RPT 05:09:26) are on a different volume than the C: RDP artefacts with the same numbers. They are consistent with the responder's evidence drive (E:, FTK Imager) and are treated as collection noise. (d) The Volatility vol_etwpatch plugin output is not available, so ETW patching was not assessed. The ntdll stub comparison (unhooked_system_calls) is clean.

- [step 11] Step 11: the network peers 13.107.x, 204.79.197.x, 131.253.33.254, 52.242.211.89, 52.167.250.154, 13.78.149.173, 23.47.52.90, 23.57.32.143, 72.21.91.29:80 and 104.92.247.90:80 are Microsoft, Akamai or Edgecast infrastructure. They match logon-time Edge/OneDrive/Store traffic (05:08-05:10) and are not counted as IOCs. The BACKGROUNDTRANSFERHOST prefetch (05:10:43) is UWP background transfer and is not BITS exfiltration. The printer-driver .cab files (MFT 87491, 2020-09-18 21:41:50) are benign. The OneDrive logs (.aodl/.etl) are present but cannot be parsed from memory.

Appendix E: Artifact coverage

Artifact	Tool	Mapping	Events	First (UTC)	Last (UTC)
Memory: MFT records in memory	Volatility 3	validated	175764	2013-02-04 23:20:16	2020-09-19 05:11:17
Memory: loaded DLLs	Volatility 3	validated	1408	2020-09-19 01:24:08	2020-09-19 05:10:23
Memory: process scan (psscan)	Volatility 3	validated	178	2020-09-19 01:24:04	2020-09-19 05:10:21
Memory: process list (pslist)	Volatility 3	validated	99	2020-09-19 01:24:04	2020-09-19 05:10:51
Memory: logon sessions	Volatility 3	validated	93	2020-09-19 01:24:04	2020-09-19 05:10:09
Memory: network connections (netscan)	Volatility 3	validated	90	2020-09-19 01:24:08	2020-09-19 05:09:15
Memory: network connections (netstat)	Volatility 3	validated	61	2020-09-19 01:24:08	2020-09-19 03:13:28
Memory: unloaded kernel modules	Volatility 3	validated	10	2020-09-19 01:24:07	2020-09-19 05:08:58
Memory: scheduled tasks (registry)	Volatility 3	validated	8	2020-09-18 05:41:37	2020-09-19 03:36:26
Memory: ShimCache	Volatility 3	validated	7	2018-04-02 18:27:36	2020-09-19 01:30:02

Appendix F: Traceability

DuckDB database	C:\Sleuthline\cases\test02\db\case.duckdb SHA-256 67a1d9c96d8b403ef602caa9992aac031649de5914cd079f01c9a71b4514927f
Pipeline version	commit 5bced44bc74f; content sha256 2186C3B0776C5224; deployed 2026-10-20 16:21
Collection integrity	not checked (run the inventory again with a recent version of the pipeline)
Normalization mappings	v2.1 sha256 64cf863da3e02e13b38e9092ef9948938ec097ee8617ce46efa64bf23ae9c158
Model	anthropic / claude-opus-5-5
Agent runs	2026-10-20T14:40:49Z → 2026-10-20T16:01:01Z: anthropic/claude-opus-5-5, playbook sha256 b7acf00c27b74e1f…
Usage	11,148,171 input tokens (of which 10,136,133 read from cache), 155,332 output tokens, 242 calls
Full trace	C:\Sleuthline\cases\test02\analysis\transcript.jsonl: every SQL query, result and model answer
Analysis state	C:\Sleuthline\cases\test02\analysis\findings.json
Exports	iocs.csv, key_timeline.csv, incident_timeline.csv
Reference format	<output relative to parsed>#<record number>: CSV data line (header excluded) or JSONL line
Report generated on	2026-10-20 16:01 UTC